

EVALUATING AND DESIGNING COMPUTING SYSTEMS
FOR THE FUTURE OF WORK

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DOCTOR OF PHILOSOPHY

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Abstract

From collaborative software to generative AI, computing technologies are reshaping communication, collaboration, and productivity in the workplace. Yet with the growing complexities of computing platforms at the workplace, it becomes increasingly challenging to foresee their impacts on workers and organization. This can lead to not only poor user experience but also sometimes problematic applications that mirror and exacerbate societal issues. How can we better understand user behavior over workplace computing platforms? How can we build applications for better future of work that align with our needs and values with emerging computing technologies? Inspired by Herbert Simon’s vision towards building the science of the artificial, this dissertation aims to shed light on these questions through the development of novel empirical measurements, technical methods, and designs for studying workplace computing systems enabled by recent advances in computing technologies. Specifically, this dissertation present three works demonstrating these approaches, including an analysis of remote meeting multitasking behavior through mining millions of online meetings, emails and file edits; the development of an AI algorithm for predicting team fractures; and a design and evaluation study on a generative AI-based scientific feedback system for researchers. These projects exemplify the opportunities to leverage computation to better understand, support and augment work practices.

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Chapter 1

Introduction

What technology would improve human collaboration and decision-making? Throughout history, people have been captivated by the potential of technology to enhance the workplace experience and have imagined various possibilities for the future of work.

For instance, English writer, journalist and educator Arthur Mee, famously wrote in 1898, *“If, as it is said to be not unlikely in the near future, the principle of sight is applied to the telephone as well as that of sound, earth will be in truth a paradise, and distance will lose its enchantment by being abolished altogether.”* He was envisioning some technology similar to what we know to be video conferencing today.

1.1 Computing technologies at the workplace

Throughout history, people have consistently developed and embraced a variety of technologies to enhance productivity and efficiency in the workplace. From the industrial age’s mechanical innovations to today’s digital revolution, each era has seen its own form of technological adoption [42]. Nowadays, the integration of workplace computing technologies has become particularly pervasive.

In their daily routines, many knowledge workers now rely on collaborative software platforms. Tools such as Zoom and Microsoft Teams facilitate virtual meetings and discussions, while workplace instant messaging apps like Slack enable continuous, asynchronous communication [48]. These technologies represent a significant shift from the traditional model of workplace collaboration, which primarily relied on face-to-face interactions [190, 113].

Moreover, cutting-edge technologies like Virtual Reality (VR) and Augmented Reality (AR) are pushing the boundaries of conventional workplace interactions [94]. Products such as Meta’s Oculus and Apple’s VisionPro are redefining collaborative spaces and introducing immersive experiences that were previously unimaginable in a professional setting. These tools allow for a more engaging and interactive form of collaboration, which can further bridge the gap between remote teams spread

across different geographical locations.

Additionally, the advent of generative AI technologies and applications has marked a significant milestone in workplace computing. Tools like ChatGPT and GitHub Copilot leverage artificial intelligence to assist with tasks ranging from coding to customer service, thereby enhancing productivity and sparking creativity. Recent studies have highlighted their potential to revolutionize how tasks are performed, emphasizing their role in not just automating routine tasks but also in providing new ways to approach complex problems [188, 80, 148, 151, 51].

With all these advances in workplace computing technologies, have we created the ideal workplace?

Despite considerable attention, the overall impact of workplace technologies remains ambiguous, as evidenced by polarized sentiments in news articles. On the one hand, positive sentiments emerge with headlines like "Why is Zoom suddenly everywhere? It's easy and fun to use."¹ On the other extreme, negative views are expressed in articles such as The New York Times' "Why Zoom Is Terrible"² This polarization extends to broader debates about the efficacy of computing technologies, with ongoing disputes about whether technology is enhancing or detracting from the workplace experience.

How do we navigate such disagreements and make progress on workplace computing technologies?

This dissertation seeks to explore this question by implementing Herbert A. Simon's pioneering framework from "Sciences of the Artificial" [221], focusing on workplace computing systems. This framework integrates *empirical evidence*, *technical methods*, and *design*. By adopting an interdisciplinary methodology that combines aspects of computer science and social science, this work illustrates how Simon's approach enhances our understanding of computer-mediated work. It provides a deeper understanding of recent developments in workplace computing technologies – specifically, collaboration tools and generative AI – and the novel work experiences these technologies enable.

1.2 Background: prior research on computer mediated work

This dissertation is grounded in scholarly literature on computer-mediated work. The increasing adoption of workplace computing technologies has sparked significant academic interest among scholars in both computer science and social science.

In computer science, a subcommunity within human-computer interaction (HCI) known as Computer Supported Collaborative Work (CSCW) emerged in the 1980s. This group is dedicated to studying how computer systems can support collaborative activities and coordination. For instance, Johansen [121] classifies collaboration technologies based on time and space dimensions, categorizing collaboration technologies into four types: "Same Time, Same Place" collaboration tools such as whiteboards, "Same Time, Different Place" collaboration tools such as video conferencing tools,

¹<https://www.businessinsider.com/zoom-video-everywhere-google-hangouts-skype-2020-3>

²<https://www.nytimes.com/2020/04/29/sunday-review/zoom-video-conference.html>

“Different Time, Same Place” collaboration tools such as digital bulletin board, “Different Time, Different Place” collaboration tools such as email. Olson and Olson [190] discuss why distance collaboration is hard to design, and highlights mechanisms such as lack of common ground. Hollan and Stornetta [113] critiques the then-prevailing notion that the ultimate goal of collaboration technologies was simply to replicate face-to-face interactions, and argued that focusing solely on creating technologies that simulate being physically together (i.e., "being there") might limit the potential of these technologies to enhance human communication in ways that face-to-face interaction cannot. Based on these foundational works, CSCW scholars have studied and designed various computing technologies that enable novel work experience. For instance, there has been great interests in computer-mediated crowd work, or crowdsourcing, where tasks are computationally outsourced to the crowd. Bernstein et al. [30] present a word processing interface that leverage Mechanical Turk workers to help the writer shorten, proofread or edit parts of their documents. Valentine et al. [238] present a system that organizes crowd workers into computationally-represented structures inspired by those used in organizations - roles, teams, and hierarchies - which support emergent and adaptive coordination toward open-ended goals. Rich literature also explores how online groups and crowds lead to the success of Wikipedia [129, 128], and crowdfunding platforms [115].

Meanwhile, social scientists have conducted foundational research in this area, particularly in studying human behavior within workplace computing systems. For instance, rich literature has studied virtual teams and remote work. Gibson and Gibbs [96] discusses the challenges and strategies for effective leadership in virtual teams, emphasizing the role of trust and communication in overcoming geographical dispersion. [111] studies why remote teams experience more conflict. They suggest that distance reduces familiarity and friendship, and that technology could lead to uneven information and coordination difficulties. Walsham further [241] studies how cultural differences affect the implementation and use of information systems in multinational organizations. With regard to workplace communication tool, Daft and Lengel [72] introduced the concept of media richness, which assesses the effectiveness of communication technologies based on their capacity to process rich information. Scholars have further studied the implications of workplace technologies on productivity. While Bloom et al. found from a large-scale randomized experiment at Ctrip that home working led to a 13% performance increase, literature has also documented mixed evidence in other scenarios [20]. Literature has also studied workplace technology adoption and adaptation. Venkatesh and Davis [240] explore the Technology Acceptance Model (TAM), detailing how perceptions of usefulness and ease of use affect the adoption of new technologies in the workplace. Lapointe and Rivard [143] analyzes resistance to information technology in organizations, and identified five basic components of resistance: behaviors, object, subject, threats, and initial conditions. More recently, scholars have studied how algorithms at work reshape organizational control [126], decision making [183], etc.

As increasingly powerful computing technologies, such as collaboration tools and generative AI,

are introduced in the workplace, there is a growing need for research on these novel experiences to complement our existing understanding of computer-mediated work.

1.3 Thesis Contributions

In this dissertation, I apply Herbert A. Simon’s research framework towards building a *science* of workplace computing technology, integrating *empirical evidence*, *technical methods*, and *design*. I use the proposed framework to investigate significant recent developments in workplace computing technologies, specifically collaboration tools and generative AI.

1.3.1 Towards a science of workplace computing system

This particular approach, is motivated by the foundational work, *the science of the artificial* [221], by Herbert A. Simon, one of the founding fathers of modern artificial intelligence and decision-making theories. This book, is themed around how artificial phenomena ought to be categorized, and whether such phenomena belong within the domain of science, in a similar way as natural science subjects such as physics. Specifically, Simon wrote, “*Natural science is knowledge about natural objects and phenomena. We ask whether there cannot also be ‘artificial’ science knowledge about artificial objects and phenomena.*”.

In his book, Simon present a high level vision some important factors that would contribute towards a science of the artificial, namely *empirical evidence*, *technical methods*, and *design*.

- **Empirical Evidence.** Simon argues for the importance of establishing empirical evidence to understand artificial objects and phenomena, e.g. “*The contingency of artificial phenomena has always created doubts as to whether they fall properly within the compass of science. . . . The genuine problem is to show how empirical propositions can be made at all about systems*”.
- **Technical Methods.** Throughout the book, Simon emphasized the importance of technical methods, such as modelling, simulation, etc.
- **Design.** Simon argues that “*It’s important to study not only ‘how things are’ but also ‘how things ought to be’*”.

Following Simon’s vision, this dissertation explores establishing *empirical evidence*, *technical methods*, and *design*, to study computing systems at the workplace. This dissertation aims to provide initial research results towards answering the following central questions.

- **Empirical Evidence.** How can we leverage increasingly available large-scale data from workplace software to study human behavior over computing platforms?

- **Technical Methods.** How can we better model, sense and predict human behavior at the workplace?
- **Design.** How can we design applications that align with our needs and values at workplace with computing technologies?

1.4 Thesis Overview

To demonstrate this approach, the current dissertation assembles three studies dedicated to examining recent computing technologies that are used or have the potential to be used in the workplace.

To exemplify how *empirical evidence* can be used to study workplace computing system, Chapter 2 provides a large scale analysis of multitasking behavior during remote meetings. This analysis focuses on establishing novel measurements and insights about how people work, utilizing the increasingly available data collected from workplace software. In this chapter, we present what we believe is the most comprehensive study of remote meeting multitasking behavior through an analysis of a large-scale telemetry dataset collected from February to May 2020 of U.S. Microsoft employees and a 715-person diary study. The rich dataset records complete worker activities over video conferencing, email and file edits. We focus on studying remote meeting multitasking behavior, as remote meetings are critical for remote work because of the need for synchronous collaboration in the absence of in-person interactions, and that In-meeting multitasking is closely linked to people’s productivity and well-being. However, we currently have limited understanding of multitasking in remote meetings and its potential impact. Through our large-scale data analysis enabled by new empirical evidence, we demonstrate that intrinsic meeting characteristics such as size, length, time, and type, significantly correlate with the extent to which people multitask, and multitasking can lead to both positive and negative outcomes. Our findings suggest important best-practice guidelines for remote meetings (e.g., avoid important meetings in the morning) and design implications for productivity tools (e.g., support positive remote multitasking).

To exemplify how *technical methods* can be used to study how people work, Chapter 3 introduces an AI approach to classify and predict an important dimension of team experience, *i.e.*, team viability. Understanding team viability — a team’s capacity for sustained and future success — is essential for building effective teams. In this study, we aggregate features drawn from the organizational behavior literature to train a viability classification model over a dataset of 669 10-minute text conversations of online teams. We train classifiers to identify teams at the top decile (most viable teams), 50th percentile (above a median split), and bottom decile (least viable teams), then characterize the attributes of teams at each of these viability levels. We find that a lasso regression model achieves an accuracy of .74–.92 AUC ROC under different thresholds of classifying viability scores. From these models, we identify the use of exclusive language such as ‘but’ and ‘except’, and the use of second person pronouns, as the most predictive features for detecting the most viable

teams, suggesting that active engagement with others’ ideas is a crucial signal of a viable team. We further show that only a small fraction of the 10-minute discussion, as little as 70 seconds, is required for predicting the viability of team interaction. This work suggests opportunities for teams to assess, track, and visualize their own viability in real time as they collaborate.

To exemplify how we can envision future workplace computing technologies through *design*, Chapter 4 presents the design and systematic evaluation of a system based on a large language model that generates peer-review-like feedback. This study aims to rethink the traditional ways researchers receive feedback on their work. Expert feedback lays the foundation of rigorous research. However, the rapid growth of scholarly production and intricate knowledge specialization challenge the conventional scientific feedback mechanisms. High-quality peer reviews are increasingly difficult to obtain. Researchers who are more junior or from under-resourced settings have especially hard times getting timely feedback. With the breakthrough of large language models (LLM) such as GPT-4, there is growing interest in using LLMs to generate scientific feedback on research manuscripts. However, the utility of LLM-generated feedback has not been systematically studied. To address this gap, we created an automated pipeline using GPT-4 to provide comments on the full PDFs of scientific papers. We evaluated the quality of GPT-4’s feedback through two large-scale studies. We first quantitatively compared GPT-4’s generated feedback with human peer reviewer feedback in 15 Nature family journals (3,096 papers in total) and the ICLR machine learning conference (1,709 papers). The overlap in the points raised by GPT-4 and by human reviewers (average overlap 30.85% for Nature journals, 39.23% for ICLR) is comparable to the overlap between two human reviewers (average overlap 28.58% for Nature journals, 35.25% for ICLR). The overlap between GPT-4 and human reviewers is larger for the weaker papers. We then conducted a prospective user study with 308 researchers from 110 US institutions in the field of AI and computational biology to understand how researchers perceive feedback generated by our GPT-4 system on their own papers. Overall, more than half (57.4%) of the users found GPT-4 generated feedback helpful/very helpful and 82.4% found it more beneficial than feedback from at least some human reviewers. While our findings show that LLM-generated feedback can help researchers, we also identify several limitations. For example, GPT-4 tends to focus on certain aspects of scientific feedback (e.g., ‘add experiments on more datasets’), and often struggles to provide in-depth critique of method design. Together our results suggest that LLM and human feedback can complement each other. While human expert review is and should continue to be the foundation of rigorous scientific process, LLM feedback could benefit researchers, especially when timely expert feedback is not available and in earlier stages of manuscript preparation before peer-review.

After the presentations of the three studies, Chapter 5 discusses the open challenges that lie within studying computing systems at workplace to inspire further research and innovation.

Some works in this dissertation has been published in academic venues. Chapter 2 is based on Cao et al. [48], which appeared in Proceedings of the 2021 CHI Conference on Human Factors in

Computing Systems. Chapter 3 is based on material from Cao et al. [53], appeared in Proceedings of the ACM on Human-Computer Interaction (CSCW) in 2020. Chapter 4 is based on Liang, Zhang, Cao et al. [155], which is currently under review.

Chapter 2

Large Scale Analysis of Remote Meeting Multitasking Behavior

In this chapter, we focus on leveraging novel digital data sources, such as telemetry data captured by workplace software, to provide large-scale *empirical evidence* on people’s work patterns. Specifically, the chapter analyzes millions of online meetings, emails, and file edits from all Microsoft US employees. This analysis is complemented by a diary study to investigate multitasking behavior during remote meetings.

2.1 Introduction

Remote work is an essential mode of work across industries [35]. Before COVID-19, many companies had already experimented with or implemented various forms of remote work [41], e.g., recruiting remote employees or allowing employees to work from home part-time. During the pandemic, 34.1% of Americans switched to working from home [41], and it is estimated that 37% of jobs in United States can be done remotely [82]. In fact, many large technology companies, such as Quora and Twitter, have announced that they will allow employees to work from home indefinitely [6, 120].

Central to remote work are remote meetings [190, 53], most commonly experienced through video conferencing tools (e.g., Zoom, Microsoft Teams, Google Meet), which help remote team members stay connected, collaborate and function as an organization, despite physical distance. Therefore, it is critical to understand factors that are associated with remote meeting experiences to better support productive and engaging remote collaborations.

In this paper, we focus on one fundamental experience of remote meetings — in-meeting multitasking. Information work is often governed by multiple tasks and activities that an individual

must remember to perform, often in parallel or in rapid succession, a practice that is called *multitasking* [212, 247, 69]. Multitasking behavior is vital, as it is closely linked to one’s productivity and well-being [69, 160] — increasing the numbers of items to be remembered can wreak havoc with task resumption [11], attentional focus [118, 98] and prospective memory [39]. Multitasking during meetings can additionally affect other people and their productivity [117]. Prior work has investigated how people engage in multitasking during collaborative activities such as meetings and video chats, both in-person and online [117, 233, 169, 108]. However, these works are mainly based on small-scale qualitative studies, and to the best of our knowledge, no research to date has reported systematic and comprehensive evidence from large populations. As a result, there is little statistical evidence on the meeting and personal context that are associated with people’s propensity to multitask, the activities remote attendees engage in while multitasking, and how such remote multitasking behavior may affect workers and groups.

Here, we adopt a mixed-methods approach to systematically understand the context, activities, and consequences of multitasking during remote meetings. Specifically, we analyzed a large-scale dataset collected from February to May 2020 of U.S. Microsoft employees. The dataset contains anonymized events from major productivity tools: Microsoft Teams for remote meetings, Microsoft Outlook for email services, and OneDrive and Sharepoint for accessing and editing files remotely. Multitasking can involve activities that are unrelated to the meeting (e.g., email communications) and related (e.g., notes taking) [117]. Therefore we measured emails sent and files edited during remote meetings as a proxy for multitasking and studied the relationship between multitasking and meeting characteristics through controlled regression analysis. Furthermore, we contextualize the evidence from log analysis with verbatims from a 715-person diary study of Microsoft employees globally, run from mid-April to mid August 2020, exploring their remote meeting experiences during COVID-19.

Our results show that: 1) Multitasking is a common behavior in remote meetings with about 30% of meetings involving email multitasking. People also reported that multitasking becomes more frequent as meetings move to remote, 2) Intrinsic meeting characteristics, such as meeting size, length, type, and timing, significantly correlate with the extent to which people multitask, e.g., people are more likely to multitask in recurring meetings than in ad hoc meetings, and 3) In-meeting multitasking during remote meetings can lead to both positive (e.g., improve productivity) and negative (e.g., loss of attention) experiences. Our analysis suggests practical ways people can improve remote meeting experiences. For example, to reduce unnecessary scheduled and recurring meetings, keep meetings short, avoid intensive meetings early in the morning, and allow space for positive multitasking. Our work implies that productivity tools can help people better manage their in-meeting attention and decide which meetings or parts of the meetings to attend.

The contributions of this paper are threefold:

- We present the first large-scale, empirical study of multitasking behavior during remote meetings, accompanied by rich qualitative evidence. This allows us to understand the factors that correlate with remote meeting multitasking and characterize the motivations and potential consequences of such behavior.
- We discover several key issues in current remote meeting configurations and suggest actionable guidelines for how people may schedule effective remote meetings (Section 2.8).
- Our work points to several concrete design implications for future remote collaboration tools, e.g., support meeting “focus mode” and positive multitasking.

2.2 Background and Related Work

Our work is built upon and contributes to the rich HCI literature on remote work and multitasking behavior and its manifestations in meetings. Prior research mostly focused on small-scale in-person studies, whereas our work analyzes a large-scale log dataset accompanied by a large diary study, to systematically reveal multitasking patterns during all-remote meetings.

2.2.1 Remote Work

Remote work has long been an important topic of research across scientific fields. Through a Working-From-Home (WFH) experiment at Ctrip, a 16,000-employee Chinese travel agency, Bloom et al. [35] studied the costs and benefits of remote work, where they showed that WFH led to improved performance but reduced promotion rate. Prior work also investigated other aspects of remote work, such as management of workers [81], organizational design [238, 203], communication challenges [106], team performance [157], well-being [53, 244], and emerging roles [59] and experience [45, 43]. Remote work becomes more ubiquitous after the COVID-19 pandemic [92, 250], and remote meetings have become a central place where people stay connected and collaborate with others [190] — both DeFilippis et al. [79] and Yang et al. [250] demonstrated a significant increase in remote meetings during the pandemic. Here we contribute to the rich literature of remote work by focusing on how in-meeting multitasking, an artifact of traditional in-person meetings, manifested in all-remote settings.

2.2.2 Multitasking in the Workplace

A large body of prior work has focused on how multitasking impacts attention in the workplace, primarily focusing on the distraction caused to an ongoing task that is interrupted by another activity. Czerwinski et al. [69] employed a diary study to show how information workers switch activities due to interruptions in the workplace, focusing on the difficulty of the continuous switching of context.

Iqbal and Horvitz [118] studied how external interruptions cause information workers to enter into a “chain of distraction” where stages of preparation, diversion, resumption and recovery can describe the time away from an ongoing task. Gonzalez and Mark [97] reported on how information workers conceptualize and organize basic units of tasks and how switching occurs across these conceptual units — people were found to spend about 12 minutes in one working unit before switching to another. In the mobile domain, Karlson et al. [125] and Park et al. [197] found that tasks on mobile phones become fragmented across devices and they identified challenges that exist in resuming these tasks. While most studies have looked at multitasking results from distractions, there is a gap in prior work that characterizes interactions corresponding to multitasking during remote meetings, as discussed in Section 2.2.4.

2.2.3 Factors Associated with Multitasking

External [189, 117, 163, 167] and internal [71, 7] interruptions are considered to be the most direct reasons behind multitasking; however there are other indirect factors that are associated with multitasking as well. Past work has shown that personality and organizational environment [71] are associated with multitasking [166]. Additionally, previous work [165] has shown that the time of day is associated with workers’ focus. On average, people were most focused in their work late-morning (11 a.m.) and mid-afternoon (2-3 p.m.), which is known as the “double peak day” of information workers’ rhythm of work. Day of the week also play important role in people’s attention level and multitasking behavior. Mark et al. [165] found a relationship between online activity and Mondays, the day when people report being the most bored, but at the same time also the most focused. However, most of the prior studies are focused on in-person work settings, and it is unclear to what extent these patterns are applicable to remote meetings. Also, the existing evidence is mainly based on small-scale data that may not generalize to large organizations. Our study confirms the associations between time, distractions and multitasking in remote meetings through large-scale log analysis and greatly extends prior knowledge by investigating a broader range of meeting characteristics, such as meeting size, type, and length.

2.2.4 Multitasking during Meetings

While multitasking during one’s own work mostly impacts personal productivity, special consideration of multitasking during meetings is warranted, as this can additionally impact other colleagues and their productivity [117]. Past work has looked at how people engage in multitasking both during in-person meetings and presentations [117, 108, 21], as well as online collaborative activities, such as remote meetings leveraging subjective feedback or perceptions [169, 233]. For example, in educational settings, the use of laptops during a lecture has been shown to have a negative impact on attention, where students tend to engage in activities such as web-surfing or emailing rather than activities related to the lecture [21, 108]. However, in other studies, people who multitask

during in-person meetings report to do so in order to interleave other important activities as they peripherally pay attention to the meeting and engage only when relevant [117]. In online settings, both meeting related and personal multitasking are seen as ways people’s attention could divert from the actual conversation, though multitasking on a single screen is considered more acceptable than when multitasking is happening on a different screen - often presumed to be unrelated to the meeting [169]. Similarly, a study on video-chats among teens found that boredom was the main reason why teens would multitask during a video chat, wherein they would engage in scrolling social feeds or play games [233].

While prior work on meetings and multitasking focus primarily subjective perceptions, no research to the best of our knowledge has looked at large-scale log data accompanied by qualitative evidence on what people are engaging in while attending a meeting and under what conditions people tend to engage in when multitasking. Our analysis of actual interactions can complement subjective perceptions around multitasking motivations, behaviors and potential consequences, and can provide insights into how to conduct meetings for maximal effectiveness.

2.3 Method

To systematically understand multitasking patterns during remote meetings, we proposed the following research questions to guide our research.

Research Question 0 (RQ0) *How much multitasking is happening during remote meetings?*

Research Question 1 (RQ1) *What factors are associated with multitasking during remote meetings?*

Research Question 2 (RQ2) *What do people do when multitasking during remote meetings?*

Research Question 3 (RQ3) *What are the consequences of multitasking during remote meetings?*

As noted above, we adopt a mixed-methods approach to address these questions: analysis of a large-scale anonymized telemetry dataset coupled with a diary study of people’s perceptions and subjective experiences with respect to in-meeting multitasking.

2.3.1 Regression Analysis on Large-Scale Telemetry Dataset

Data Preparation. We collected metadata (without any content information) on remote meetings (Microsoft Teams), email usage (Microsoft Outlook), and file edits (Onedrive/Sharepoint) of US employees from Microsoft. The majority of work and communication in Microsoft are carried out through these platforms. We collected four separate week-long snapshots of data from February to May, 2020: 1) February 24-28, which represents a period of pre-COVID, mostly co-located work,

2) March 23-27, which represents the company’s transition phase from mostly co-located work to remote work, and 3) April 20-24 and 4) May 18-22, to represent fully remote work periods. While we leveraged all four periods to study work rhythm and derive statistics on multitasking behavior, our regression analysis focused on the snapshot from May 18-22, when employees were fully settled into working from home.

Specifically, for each meeting hosted on Microsoft Teams¹, we collected the start and end timestamps of the meeting², meeting size³, and the meeting type (ad hoc, scheduled, recurring or broadcast⁴), and their distributions are presented in Fig. 2.1(a). As shown in Table. 2.2, we discretized continuous meeting attributes in the following ways to ensure robustness of the regression analysis. We grouped meeting duration into four categories — 0-20 mins, 20-40 mins, 40-80 mins, and >80 mins because of the popularity of 30 mins and 60 mins meetings (Fig. 2.1(a)). For meeting size, due to its long-tail nature, we split it into five roughly equal sized bins. We considered morning, afternoon, and after hours as three categories for hour of the day in order to align it with common work rhythms in Microsoft.⁵ Furthermore, we tested the meeting attribute correlation, where we find rather weak correlations ($|r| < 0.15$) among different attribute groups. For instance, the correlation between meeting time (e.g., morning) and meeting size (e.g., >10 attendees) is around 0.07. The weak dependency between meeting attributes motivated us to directly include them in the regression analysis rather than trying to cluster various meeting types and study those clusters in the regression. Finally, we note that the telemetry data only records virtual meetings, so February data (before pandemic in U.S.) does not reflect total volume of meetings people attended, but all meetings after the pandemic are recorded.

We focused on two multitasking behavior during remote meetings that can be measured through available telemetry data: email multitasking and file edit multitasking. To enable the analysis, on Microsoft Outlook, we collected the time when people actively send, respond to, or forward an email. On file platforms, we recorded events related to editing productivity-related files, including Excel, Powerpoint, Word, and PDFs. Note that due to the nature of the data, we are not able to differentiate whether the files are related to the meeting or not, thus the measured multitasking file behavior might be related to the meeting (e.g. notes taking).

To ensure people’s privacy with the telemetry data, we performed pre-processing to de-identify workers before the data was obtained by researchers. Access to the data was strictly limited to authorized members of the research team who went through extensive privacy and ethical training.

¹We only included meetings that are longer than two minutes to filter out data noise. We also filtered out meetings lasting longer than 3 hours, which is likely due to the fact people forget to leave the meeting.

²Meeting start and end timestamps was logged on a per-person basis, i.e, the exact timestamp that each person attended and left a meeting. The average standard deviation of meeting duration among people joining the same meeting is about 2.1% of the maximum meeting duration, indicating that the same meeting generally has similar length across participants.

³Meeting size was logged as the number of all people ever connected to a meeting. Due to aggregated nature of telemetry data, it is impossible to measure the exact maximum concurrent participants.

⁴Meetings that have more than 250 attendees.

⁵We tested different bucketing strategies and it produced robust results.

Regression Analysis. We joined three data sources by unique user identifiers (anonymized), which resulted in 34,524 (meeting, user) records. For each (meeting, user) pair, we labeled it as email multitasking ($Y = 1$) if the user was found to have at least one active email action during the meeting, otherwise a non-multitasking label was assigned ($Y = 0$). File multitasking was labeled similarly. We conducted a regression analysis to understand the relationships between multitasking and meeting characteristics, while controlling for individual variances since the individual tendency to multitask may confound the estimation. We leveraged stratification [178] to group all records by worker and used conditional logistic regression to fit the model. The binary meeting indicators used in the regression model are defined in Table. 2.2. To account for the correlations between the records from the same meetings, we grouped standard errors at the meeting level. We present regression results in terms of estimated odds ratio and their statistical significance in Fig. 2.3. We did not find significant correlations between file-related multitasking and meeting characteristics, which could be attributed to the fact that files edited are often related to the meeting (Section 2.6.1).

To test the robustness of our regression model (Section 2.3.1), we conducted an alternative analysis using Generalized Linear Mixed Effects Models [216]. Specifically, we estimated random intercepts for workers and approximated the posterior using variational Bayes. The alternative results are shown in Table. 2.1, and they are qualitatively similar to those we present in our main analysis.

Effect	Post. Mean	Post. SD
Day:Monday	0.2345	0.0308
Day:Tuesday	0.3154	0.0277
Day:Wednesday	0.2052	0.0276
Day:Thursday	0.1940	0.0287
Type:broadcast	0.8271	0.1710
Type:recurring	0.5125	0.0191
Type:scheduled	0.4235	0.0199
Size:3	0.0591	0.0335
Size:4-5	0.2152	0.0293
Size:6-10	0.3656	0.0311
Size:>10	0.5788	0.0288
Hour:morning	0.6053	0.0186
Hour:afternoon	0.3760	0.0202
Duration:20-40 min	0.7863	0.0203
Duration:40-80 min	1.3049	0.0265
Duration:>80 min	1.9447	0.0699

Table 2.1: Alternative regression results with Generalized Linear Mixed Effects Models [216]. The model includes random intercepts for workers and is approximated using variational Bayes. The results are qualitatively similar to those of our main analysis.

Limitations of Telemetry Data. While the large-scale telemetry data provides us with a lens onto how people behave during meetings, our emphasis on preserving privacy means that we

Table 2.2: Meeting property and discretized bins used for regression analysis. For each meeting property, we create dummy variables and then left one out as baseline. For instance, Friday is left as baseline for the day of the week meeting property, and all effects are relative to meetings on Friday.

Meeting Property	Discretized bins
Meeting duration	0-20 mins meetings (baseline), 20-40 mins meetings, 40-80 mins meetings, >80 mins meetings
Meeting size	2 attendees (baseline), 3 attendees, 4-5 attendees, 6-10 attendees, >10 attendees.
Meeting types	scheduled, recurring, ad hoc (baseline), broadcast
Hour of the day	morning: 7am - noon, afternoon: noon - 7pm, after hours: 7pm - 7am (baseline)
Day of the week	Monday, Tuesday, Wednesday, Thursday, Friday (baseline)

unfortunately do not have access to all behavioral details. The data was collected in anonymized, aggregated form and does not have sensitive attributes that can potentially reveal the identity of an individual employee or a group in a corporation, such as functional labels, job roles, organizational charts or participant social demographics. Email is perceived as stateless communication in the organization we studied; therefore, fine-grained events e.g., email read was not recorded. Similarly, multitasking behavior in 3rd party platforms and non-digital spaces (e.g., house chores), is not measured due to lack of instrumentation. Finally, telemetry data lacks information on the reasons for and consequences of multitasking. We strive to address these limitations by a complementary diary study where we delve deeper into the trends shown in the telemetry data set.

2.3.2 Diary Study

We complement our quantitative analysis with reports drawn from a diary study of employees from Microsoft reporting their experiences of remote meetings during COVID-19. Diary studies in HCI research [205] have been particularly effective in capturing the nuanced experiences of information workers [69, 217].

The diary study data collection ran between mid-April and mid-August 2020. Participants opted-in to the study from bulk email messages sent to internal mailing lists, with rolling recruitment between mid-April and mid-June 2020. Participants were asked to keep a guided diary for two months. Diary reminders were sent as a file via email to participants to set up in their calendars. The diaries consisted of a series of forms embedded in a secure company website. 24 total diary entries were requested, one entry approximately every two working days for two months total. The 24 guided entries were to be filled out in three cycles of eight topics: Physical workspace, Interaction, Productivity, Tools, Multitasking, Types of meetings, Time in meetings, and Approaches to meetings. The three diary entries on Multitasking used the following primary prompt: "What have you noticed about multitasking in recent online meetings?". This was followed by a list of sub-prompts: How and when you multitask; Why you multitask; Managing video and audio; Group expectations around multitasking; Impact of multitasking on productivity; Impact of multitasking on conversations; Features for multitasking; Impact of home life; Suggestions for improvement. Full

methodological details are available in a technical report [206].

849 total participants provided consent and were onboarded, of which 715 completed at least one diary entry. For those who filled out diaries: Gender coverage was 60% Male, 39% Female, and 1% non-binary or preferred not to say. Age coverage was 5% 18-24, 28% 25-34, 30% 35-44, 28% 45-54, 8% Above 55, and 1% Prefer Not To Say. Roles covered were 41% Business & Sales, 30% Development, 11% Creative, Design, and UX, 11% Technical Operations, 5% Administration, and 2% Research. Participants were drawn from almost all regions of the world, primarily 53% North America, 20% Europe, 12% India, 4% China, and 4% South America.

Of 7045 total rows of diary responses, 819 responded specifically to the multitasking prompt⁶ from 413 unique participants. We randomly selected 100 responses (20 per month from April to August). To analyze the responses, we adopted the method of open coding [66]. Five researchers independently analyzed and coded the first 20% of the interview transcriptions and met to discuss the codes until they were in complete agreement on the codes needed. Then, one researcher coded the remaining transcriptions but discussed any questions with the four other researchers so as to guarantee consensus on the codes. When these steps were finished, the whole research team met and thoroughly discussed the extracted content classification. Through sub-categorization and constant comparison [67], we consistently revised the emerging themes and the final themes presented in Sec 4, 5, 6 and 7 were developed.

To ensure people's privacy in the diary study, we de-identified data before analysis (a participant key linked demographics to diary entries, and then all diary entries were scrubbed for names, places, and other identifying referents). Similar to the telemetry data set, only authorized researchers have access to the diary data.

2.4 RQ0: Volume of Multitasking during Remote Meeting

Our analysis suggests that multitasking during remote meetings is ubiquitous, and that people find themselves engaging in more multitasking during remote meetings compared to in-person meetings, possibly as a result of a shift in work rhythms, and the lower cost to get noticed.

Multitasking intensity over time. In our telemetry dataset, from February to May, we find that 31.1%, 30.9%, 29.2%, and 28.9% meetings involve email multitasking, and 23.7%, 23.1%, 24.8%, and 25.5% meetings involve file multitasking. Putting these percentages in the societal context of the cognitive efforts and resources that information work costs [226], our results suggest the importance of understanding and potentially intervening toward such behavior.

More multitasking as a possible result of work rhythm adaptation. Fig. 2.2 illustrates the shift of work rhythms, characterized by the distribution of work related actions (emails sent, files edited and meetings attended) within a day, from Feb 2020 to May 2020. The email and file-related

⁶We did not include responses mentioning "multitasking" under other prompts.

rhythms throughout the day remain roughly unchanged, indicating that people worked in a similar fashion on emails and files as they did in co-located settings. Meanwhile, there is a clear increase in the number of remote meetings, compared to pre-COVID-19 period – note that the telemetry results do not necessarily support the conjecture that people are meeting more since not all in-person meeting frequencies were recorded in the telemetry. However, our diary study results suggest that people do perceive that they have more meetings, and this may be an important underlying cause of multitasking during remote meetings.

"I think this is more of a habit that developed now that folks don't have face to face meetings at all and that the number of meetings has increased so much that it is just more efficient to get the notes and reading out of the way during the meeting than work extra hours end of day or early next day to catch up" (R498)

"There are so many meetings that there is no time to look at email, or get work done in between. I try and work early in the morning and late at night, but as work flows in during the day and needs response, I find myself more and more just multitasking" (R179)

Ease of turning off video and audio may encourage more multitasking. In comparison with traditional face-to-face meetings, remote meetings make it much easier for people to stay in the background by turning video off/muting themselves. Given that multitasking has been culturally associated with impoliteness [201], we assume more multitasking during remote meetings may be caused by the lower probability of getting noticed by others when multitasking. In our diary study responses, we found that turning off the video camera or muting the microphone was closely related to multitasking behavior, as mentioned by many (32% of responses).

"I typically will not multi-task if I have my video on, because people can definitely tell when you're not paying attention. Sometimes I will choose to turn my video on purely so that I am not tempted to multi-task. If I am an optional participant in a meeting, or I am just listening in, unsure if the agenda really calls for my participation, then I am more likely to keep my video off and openly multi-task until someone says my name." (R10)

"In general, I have a feeling that in our group the expectation is that participants do not multitask during meetings, but who knows what you are doing if the camera is off. (So yeah, that's when I turn off my camera too.)" (R14)

2.5 RQ1: WHAT FACTORS ARE ASSOCIATED WITH MULTITASKING

Our analysis suggests that both intrinsic meeting characteristics and external factors are correlated with remote meeting multitasking, as discussed in detail below.

2.5.1 Intrinsic Meeting Characteristics

More multitasking happens in large meetings. As shown in Fig. 2.3(b) and Fig. 2.4(b), larger meetings generally involve more multitasking. The odds of email multitasking in 3 attendee meetings, 4-5 attendee meetings, 6-10 attendee meetings, and >10 attendee meetings are 1.12 ($P = 0.021$), 1.39 ($P < 0.001$), 1.70 ($P < 0.001$) and 2.16 ($P < 0.001$) times the odds of the meetings with only one or two attendees. This could be explained by the fact that participants need to more actively focus on the meeting conversations when the meetings are small. Our empirical finding is also supported by evidence from the diary study, e.g.,

"If it's a large audience, like a webcast or an internal session on some tech topic, I do multitask more." (R4)

"In the big meetings, like town halls, I tend to stop and actually listen when something of interest is being said. the rest of the time, I seem to not focus on work at all. In small meetings, I generally don't multitask at all anymore. It takes all of my brainpower to focus on the conversation." (R182)

More multitasking happens in long meetings. We also observe that more multitasking happens in longer meetings. From telemetry data, we observe that the odds of email multitasking in 20-40 minute meetings, 40-80 minute meetings, and >80 minute meetings are 1.96, 3.22 and 6.21 times the odds of 0-20 minutes meetings ($P < 0.001$), as illustrated in Fig. 2.4(a) and Fig. 2.3(a). As supported by diary study responses, many people mention that they simply cannot concentrate for a long time and thus engage in multitasking during long meetings.

"Additionally, meetings seem to be longer (e.g., I have a three hour brainstorming session with my team today) and I cannot focus on the meeting that long alone. Then I also tend to work on other tasks from now and then." (R21)

Morning meetings involve more multitasking. The time schedule of the meeting also plays an important role in multitasking behavior. Our email multitasking data analysis suggests that morning meetings involve more multitasking compared to afternoon and after hours: the odds of multitasking in the morning are 1.86 times the odds of the after hour meeting baseline ($P < 0.001$), and the odds of email multitasking in the afternoon are 1.54 times the odds of the after hour meeting baseline ($P < 0.001$). We argue that this observation may be closely related to the daily work rhythms of individuals. As demonstrated by prior work [165], in the afternoon people are generally more focused. We also find supporting evidence in the diary study results (6% of responses)

"I will try to glimpse at email more if a meeting is the first thing I do for work in the morning - so that's schedule-related." (R621)

More multitasking happens in recurring and scheduled meetings compared to ad hoc meetings. Next, our results suggest that multitasking is more likely to happen in recurring and scheduled meetings compared to ad hoc meetings. We find significant associations between email

multitasking and meeting types in our telemetry data. Specifically, the odds of email multitasking in recurring and scheduled meetings is 1.59 ($P < 0.001$) and 1.31 ($P = 0.012$) times the odds of multitasking in ad hoc meetings. Ad hoc meetings generally involve a specific focus relevant to the specific attendees, while scheduled meetings, especially recurring meetings, are more likely to involve broader information sharing which does apply equally to each attendee.

"I just came off an online call with my larger design sync, it's a 30 min recurring meeting where we share topics as our design teams are in [two cities]. I didn't need to present, I was a listener, so I found myself responding to Teams messages, emails etc as the call was going on." (R42)

More multitasking happens Monday through Thursday compared to Friday. The telemetry data also demonstrated that Friday has a relatively lower likelihood of multitasking, compared to other days of the week. The odds of email multitasking on Tuesday are 1.35 times the odds of multitasking on Friday ($P < 0.001$), followed by Monday (1.23 times, $P = 0.001$), Wednesday (1.19 times, $P = 0.003$) and Thursday (1.19 times, $P = 0.003$). While this pattern corroborates findings from prior work [165], we note that the result might not generalize broadly, especially for Fridays, since Microsoft encourages fewer meetings on Friday, so the findings might be company-specific.

More multitasking in meetings of lower relevance and engagement. People also frequently mentioned (17 % responses) that they multitask during meetings they find irrelevant or have lack of interest or engagement in, which might be the underlying reason why people multitask more in larger group sizes and longer meetings.

"I tend to multitask more in larger group meetings online. Larger meetings often have topics on the agenda not directly related to what I'm working on day to day so my mind tends to wander when the topic of discussion is irrelevant. I'm definitely aware of trappings of a divided mind. I'm not necessarily productive on these other tasks. So I normally engage in menial work like cleaning and organizing files." (R11)

"I myself frequently have a web page, source code, or build window open in another window, and I divide my attention - most often when the meeting goes to topics that don't concern me, as most of my meetings do." (R15)

"When meetings have a lot of topics or don't apply to me, I start multitasking." (R16)

Sometimes people lose their concentration due to high cognitive load under such meetings of low relevance,

"I've noticed that I only multitask when I am tired and it is difficult for me to focus on the ongoing meeting. And I don't do difficult tasks either, at most I am checking if my PR went through or kick off a build or just look at pictures of cats. " (R14)

"It's really hard to focus, I don't know what people are trying to say or what the actions items after or why we're discussing certain items. When I can't focus or understand what's going on, I tend to check out and look at other things or do something where I feel engaged and useful. " (R346)

2.5.2 Extrinsic Factors

People multitask during meetings to catch up on other work. Another major reason (39 % responses) why people multitask is to catch up on their work. Given the increasing number of meetings compared to the in-person work experience, people find they are having a hard time completing all of their work in time.

"It needs to happen or you cant get all your work done " (R167)

"But these days I am having a lot of meetings, making it hard to find time to get work done. So, feel super tempted to multi task, if not entire day goes in meetings before real work gets started. Another reason to multitask is deadlines..." (R12)

"My team is often quite bad at sticking to an agenda, so I find myself multitasking as a way to feel like I'm still able to "get things done"while I'm 'sitting in' on these meetings. As a designer, that often means that I'm clicking around in Figma. Increasingly, I am also trying to use meeting time to catch up on context for other meetings. This makes paying attention in any meeting very difficult, but with the volume of meetings and the complexity of the context I feel I need to maintain, I often feel like I have no choice." (R9)

"Lately (since COVID) I've been forced to multitask during meetings to meet the deadlines that I've been given (and even then, I still don't always make it)... I don't have enough hours in the day to do all of the work that is required of me." (R346)

People multitask during meetings due to external distractions. We also find people frequently multitask during remote meetings as a result of external distractions - under such situations, people do not purposely multitask, but their attention gets attracted by external sources. Two major classes of distractions are interface design, and the home working environment. As collaboration moves online in remote work, people are interacting with digital tools more than they used to when co-located at work, and people are mentioning that interface designs can be the cause of multitasking behavior, especially pop-ups, e.g.,

"I multi-task in almost every meeting that isn't 1-1 and even in 1-1 meetings it's hard not to multi-task because email and teams chats are popping in. In person for 1-1 i would lock my computer and focus entirely on the person and this is super hard in the WFH setup." (R502)

"There are a lot of people are multitasking as we are using Teams. Teams is prompting up that someone is trying to get hold of us. This lures us in to checking quickly in to who it is an what they want" (R664)

On the other hand, as pointed out by prior qualitative work [10], remote work involves more distractions from the home working environment that could lead people to multitasking, e.g.,

"Since COVID19 the multitasking also includes: - answering children's questions - preparing food for children - Helping children with school work - resolving children's disagreements " (R183)

People multitask during meetings for anxiety relief. Finally, some participants mentioned that anxiety over the COVID-19 pandemic lead them to seek methods for maintaining focus, such as conducting a low cognitive effort non-work activities while monitoring meetings.

"The current situation has increased my general anxiety. This means I have more difficulty focusing on tasks - including meetings. I have been multi-tasking with non-work tasks (i.e. a colouring game on my phone!) quite a bit as I find this actually enables me to focus better on the meeting." (R5)

"Doing some exercises with my shoulders/back during meetings (with video turned off) is great." (R13)

2.6 RQ2: WHAT DO PEOPLE DO WHEN MULTITASKING

Our analysis further shows that people engage in work-related and non-work-related tasks when multitasking during remote meetings.

2.6.1 Work-related tasks

Communication with coworkers is one of the most frequently mentioned multitasking behavior, since people generally consider that communication is quick to complete without much need to focus, or "light weight". In fact, 29 % of diary study participants mention that they engage in email multitasking, which further confirms and strengthens our motivation to analyze email multitasking using the telemetry data set.

"I might use that time to reply to simple emails that don't require much thinking (so I can also pay attention to the meeting)." (R170)

"And there are different levels of multitasking, meaning the kinds of multitasking things I will do during meetings can range from writing down a quick reminder to myself on a different topic (takes a few seconds) to triaging email which takes very short attention shifts and I can come back into the proceedings easily, to more cognitively demanding tasks like writing emails, responding to IM threads etc." (R171)

Meanwhile, several diary study participants mentioned that they also frequently check and test scripts that take time to run during remote meetings, which is also rather lightweight.

"There have been situations where I've multitasked in the sense of responding to email or checking on the results of a long running job while in a meeting that is more focused on consumption of information rather than on my own contributions." (R180)

"Because I have done a lot testing and building scripts. It takes time, so I can work with other stuff while waiting for the result." (R337)

People also mentioned that they engaged in file multitasking, yet these activities are often related to the meeting, e.g., notes taking (R182, R183, R344), checking relevant files (e.g. *"in a meeting discussing aspects of the project I linked to the latest documentation."*, R174), etc., which could be a possible explanation of why there is no significant relationship between file-related multitasking and meeting characteristics.

2.6.2 Non-work-related tasks

Non-work tasks is also an emerging theme in the diary study responses. For instance, checking social media as a break from work.

"I've been multitasking both for personal and professional things - answering emails and chats for work, taking social media and texting breaks for personal." (R168)

Given that working from home environment under COVID-19, people also reported that they engaged in eating, exercise (primarily for anxiety relief and wellness) and other household chores.

"As my kitchen is very close to the desk with my computer, I also get food or drinks from there during the meetings more often." (R20)

"For me personally, I am more likely to do another task that does not require the computer like managing my to do list, writing down notes, cleaning up my office and desk, eating etc." (R171)

"It has been beneficial to walk around my house while on meetings (that don't require me to be in front of my computer). I'm able to put dinner in the oven, feed the dogs, etc." (R668)

2.7 RQ3: THE CONSEQUENCES OF MULTITASKING

Finally, we present our findings on the consequences of multitasking during remote meetings. Although multitasking is typically associated with negative outcomes such as decreased task performance [177, 68], difficulties in decision making [225] and negative affect [254, 19], our participants report that in-meeting multitasking leads to both positive and negative outcomes.

2.7.1 Positive Outcomes

Multitasking may help boost productivity. First, multiple participants (15 %) mentioned that multitasking helps boost their productivity, which echos prior works on the benefits of multitasking on efficiency [70, 161, 204]. Given that, under current remote work settings, there are many more remote meetings compared to the pre-COVID-19 period, people explained how multitasking helps them to get work done. Here's a representative response,

"There are some benefits to multi-tasking. I've been able to get more work done. I've been less frustrated by meetings that weren't very useful to me. I haven't made any significant mess ups in a meeting yet when I've done it." (R330)

Multitasking is at its most productive when workers understand that their own and others' attention in a meeting is a spectrum about which they can make active meeting choices [140].

"I find myself multi-tasking in meetings that do not require my attention, but not in meetings that do. In some ways, I may be more productive given the ease of multi-tasking in remote meetings." (R508)

People also mentioned the positive experience with meeting-related multitasking behavior, such as note taking and searching for information, e.g.,

"The type of multitasking that I feel it has a positive impact on my productivity is when I am taking notes during a meeting or when I am navigating the internet to find some relevant information that is being discuss in the meeting." (R668)

2.7.2 Negative Outcomes

Multitasking leads to loss of attention/engagement. Nevertheless, remote meeting multitasking does cause negative consequences. Among them, the most frequently (36 %) mentioned negative aspect is loss of attention/engagement, where people lose track of the meeting content (which sometimes is important) due to multitasking activities, as demonstrated by the following cases,

"Its easy to get distracted by multitasking and miss something in the meeting." (R2)

"I have to channel my concentration on 1 primary task – whether that be the meeting itself or the side work I'm doing while listening in." (R346)

"If you leave the meeting window to open a deck or other files, it's hard to get back to the meeting window. Same with chats - if you leave the meeting window to send chats to other people, it's hard to get back to the meeting." (R344)

These observations are well-aligned with previous conclusions on the impact of multitasking behavior on people's attention and task resumption [19, 11].

Multitasking leads to mental fatigue. Moreover, we find that remote meeting multitasking behavior does have an impact on well-being: some participants reported that they feel tired after multitasking during remote meetings.

"I tire a bit more with so many meetings and multitasking." (R72)

Multitasking can be disrespectful. One final downside of multitasking is that it has been sometimes regarded as an inappropriate behavior during remote meetings. Some participants explicitly associate multitasking with being impolite.

"I tend to do it less while on video, so they can't tell I'm being rude." (R173)

"I rarely multitask when on camera—it seems rude." (R673)

"People are becoming a bit more brazen; it's sometimes unbelievably clear that they aren't paying close attention to the discussion and with the current WFH situation it's hard not to notice eyes

straying, backlight (windows) changing, etc. It adds another dimension of rudeness but I also do it to others, sometimes not realizing when I follow the stray notification." (R343)

"I have gotten caught off guard a few times though. Someone will ask me a question and I'll have to ask them to repeat it. They don't seem upset, but I'm still embarrassed/ashamed when it happens. This is mostly due to the fact that I know I have slightly negative feelings toward other people when they get caught in the same situation, or when they ask a question that had been asked and answered recently. I don't want to be one of those people or be thought of as one of those people who is not paying attention." (R330)

2.8 Best-practice guidelines for remote meetings

Remote meetings have become the primary way that people connect and collaborate while working from home. As the number and duration of remote meetings has increased, people appear to have been left with less time to focus on their work and thus have gotten into the habit of multitasking to catch up. This draws participants' attention away from the meeting, and can lead to mental fatigue and disrespectful behaviors. Based on our findings, we propose several practical remote meeting best practices that meeting organizers can use to help people attending the meeting actually attend to the meeting. We note, however, the importance to consider specific worker and corporation contexts when applying these guidelines.

Avoid important meetings in the morning. As demonstrated by Fig. 2.2, people still adhere to a similar "double peak" daily work rhythm [172] as they did in pre-COVID, co-located work settings. Through our regression analysis, we find that email multitasking behavior occurs most often in the morning, which coincides with the fact that email actions peak in the morning. Prior evidence also suggests that people are most focused in mid afternoons [165]. Therefore, our results suggest that meeting organizers might avoid scheduling important meetings in the morning, when it is harder for people to concentrate. However, multitasking in the morning is not always bad, as confirmed by our diary study participants (e.g., "balancing home and work"). In fact, scheduling light-weight meetings in the morning may actually help people smoothly transition from "home" to "work" mode under remote work settings [248, 119].

Reduce the number of unnecessary meetings. Many participants stated that they multitasked because they found that the content of some meetings did not apply to them, especially for information sharing and daily stand up meetings. In the telemetry data analysis, we also found that there was much more email multitasking in recurring and scheduled meetings compared to ad hoc meetings. Given there are so many meetings people now need to attend, meeting organizers should reconsider the necessity of numerous meetings, or the frequency of such meetings, so as to help people better focus. The organizers may also consider sharing information asynchronously. For example, sending out a recording of a presentation for attendees to watch on their own, and then

only use the meeting for discussion.

Shorten meeting duration and insert breaks. In our telemetry analysis, we found that longer meetings are associated with more multitasking behavior, which is also verified by the qualitative evidence. As suggested by prior literature, humans have an upper time limit where they can fully engage and pay attention [164], thus we suggest that meeting organizers should shorten the duration of meetings, or insert breaks when meetings have to run long.

Encourage active contribution from the appropriate number of attendees. Finally, many participants mentioned that they multitasked because they don't have anything to actively contribute to the meeting discussion. They generally muted themselves and turned off their video in such scenarios. If organizers want the full attention of participants in an important meeting, they should encourage participants to actively engage through stimulating interactions, especially if it is a large meeting with a variety of attendees. For meetings where active engagement is particularly important, then the invitee list should be as small as necessary to achieve the right level of engagement from attendees.

Allow space for positive multitasking. Our findings suggest that multitasking can be positive under remote environment. Therefore, meeting organizers could consider creating personalized meeting agenda so that people are aware of the timing when relevant agenda items come up. Organizers can also implement a convention where video-on implies full attention, and video-off signals multitasking.

2.9 Design implications for better support of remote meetings

We have seen that multitasking in remote meetings is a complex behavior, with both positive and negative aspects. It can help people be more productive, but may also reduce attention, increase fatigue, and appear impolite. While culturally the term multitasking may have a negative connotation [201], our study adds to a growing body of work that reevaluates differential attention and multitasking in remote work contexts [140] and for users of all abilities [75]. Based on our findings, we argue that given the complex nature of remote meeting multitasking, it is important to encourage its positive aspects while reducing its negative implications. In this section we discuss several ways productivity tools might do this.

Support a 'focus mode' for remote meetings. Our analysis shows that pop-ups in current software interfaces during remote meetings distract people from the meeting itself. To alleviate such distractions, we envision future collaboration platforms having a remote meeting 'focus mode'. After people choose to enter the mode (e.g., for a very important meeting), the tool could block all standard pop-up messages, emails, etc., so as to help them concentrate on the meeting itself. The focus mode could also employ a multitasking alert feature. People could give permission to the app to track their behavior in other applications or even their other devices. For non-meeting focused behavior

(speaking, screensharing, in meeting parallel chat, etc.), the app could alert people about their multitasking and reflect back the reason (e.g., not being able to absorb all of the information within the meeting). As such, the feature could help people engage in the meeting and avoid unintended loss of attention.

Support other types of engagement during remote meetings. Some positive multitasking fits the meeting purpose, but technically requires moving outside the meeting window which may increase the risk of being distracted from the meeting. For instance, people often need to work outside the actual meeting window or platform to take notes or work on files, switching back and forth between this work and the meeting. The ability to have more windows, split views, or even a more broadly-defined meeting space could not only reduce the potential for distraction but also improve the shared use of these resources. Given the various reasons for remote meeting multitasking, it is also important to not arbitrarily consider it unacceptable. Meetings could enable attendance along a scale of high to low engagement, to help set attendees' expectations and more closely match actual behaviour [140].

Help people decide which meetings to attend. Our study suggests that over-multitasking or negative multitasking is associated with the increasing number of meetings during remote work. Apart from organizational leaders actively reducing the number of meetings, future remote meeting tools could develop a feature for people to self-rank the importance of each meeting, or recommend an importance level assigned to each meeting for each individual based on meeting characteristics (e.g., content, size, attendees, etc.), and add the importance level of each meeting on the person's calendar apps and video conferencing platforms. As such, people will have a better idea for each day which meeting is important and when they should pay special attention in order to not miss key information (thus avoiding multitasking). For less important meetings, the system could recommend ways to catch-up later, alternative ways to attend, or help notify the meeting organizer.

Help people skip some parts of the meeting. As suggested by our findings, even within the same meeting, certain parts of the meeting may not be so important as other parts for a specific attendee: the attendee may only be interested in a particular section, and works on other things except for that section. If meetings have agendas, one solution would be to for organizers and attendees to flag expected attention per item, and add increasing and decreasing visible attention when relevant. Future tools can also help better support personalized meeting content importance tracking for attendees. For instance, the system could make use of real time transcriptions of meeting and compare the topic similarity to those that the attendee might be interested in.

2.10 Limitations and generalizability

While our findings build on rich telemetry and diary study data to extend what was previously know about multitasking in remote meetings, it is important to consider them in the context of several

limitations. For one, the data are drawn from one global information technology company (and the telemetry data focus on US workers only), with most participants being information workers. As a result, our findings may not generalize to other worker types under remote settings, or to other cultures. Further, the analyzed data were collected during the COVID-19 pandemic era, but we were not able to distinguish remote work effects from COVID-19 effects (e.g., the impact of remote working from home on people’s mental wellbeing) on multitasking behavior. Additionally, while telemetry data is valuable in providing a large-scale, realistic picture of behavior, it does not provide the motivation motivating that behavior [87], we are not able to distinguish whether the email and file actions we observed were related to the meeting or not. It is likely that some of the positive multitasking observations may be false positives. Finally, the current diary study does not cover every aspect with regard to remote meeting multitasking, which can be addressed in future work, e.g., references to side-channels, such as Whatsapp, Facebook or other apps that cannot be instrumented through telemetry data analysis and people’s perception of multitasking. Nevertheless, we believe that this work presents the most comprehensive analysis of remote meeting multitasking behavior currently available, and could extend to scenarios beyond the narrow-sense of workspace meetings (e.g., real-time distance education that leverages remote meeting tools [57]). We leave the above mentioned limitations as future work.

2.11 Conclusion

In this paper, we presented a large-scale and mixed-methods study of multitasking behavior during remote meetings. We analyzed a large-scale telemetry dataset and conducted a longitudinal diary study during COVID-19 period. Our analysis leads to practical guidelines for remote meeting attendees and design implications for productivity tools, both of which can improve remote meeting experiences. Our results point to the importance of multitasking in the consideration of remote meetings, both with respect to their social and technical components. On the one hand, how remote meetings are scheduled and structured are significantly associated with when and to what extent people divide their attentions. On the other hand, multitasking could imply both positive and negative outcomes for individual worker and work group. More future research efforts are needed to address remote meeting experience as it becomes mainstream, and our work provides a foundational and timely understanding of such.

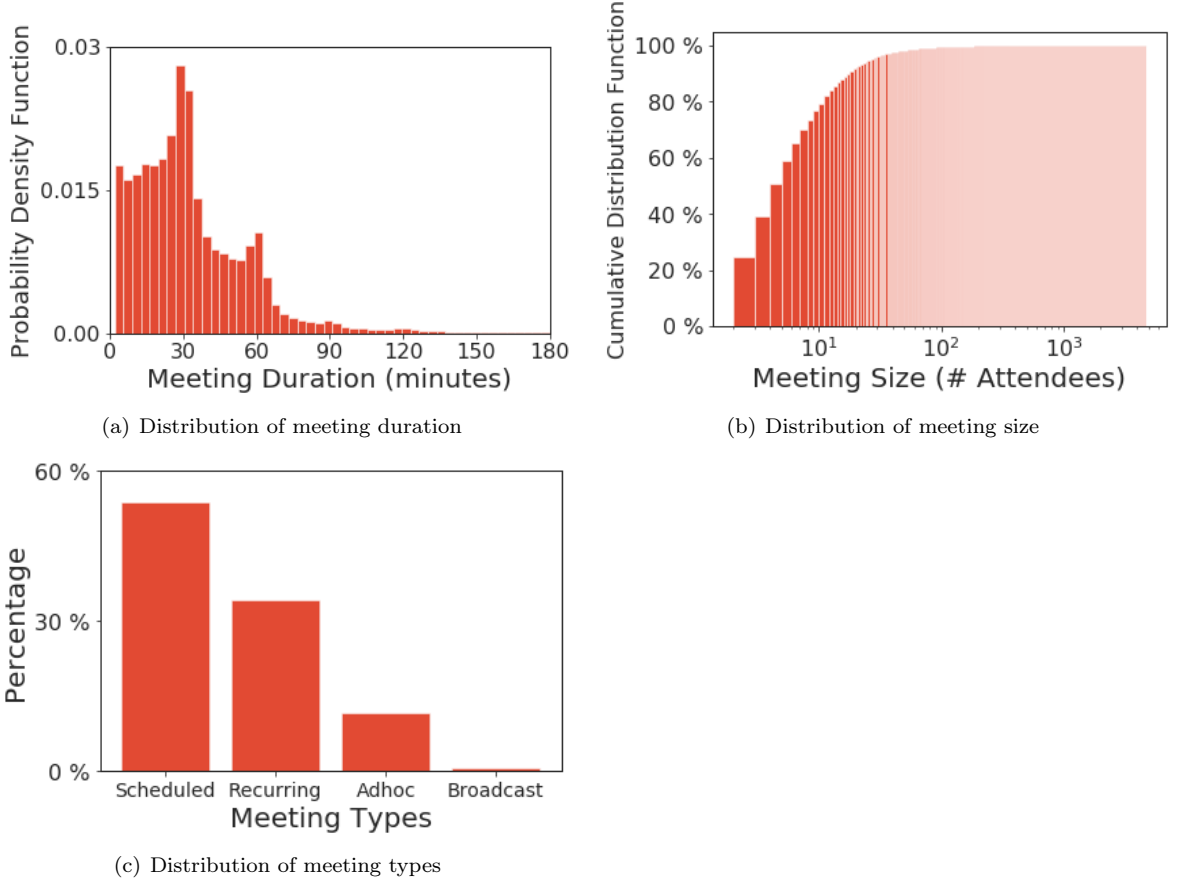
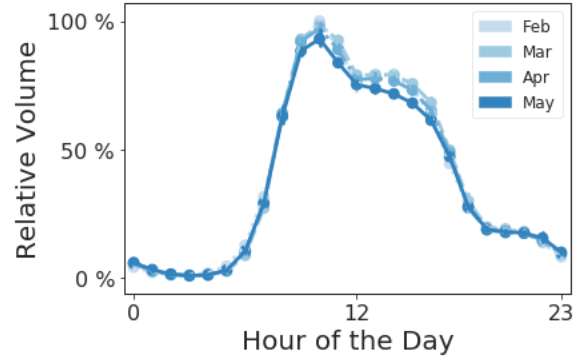
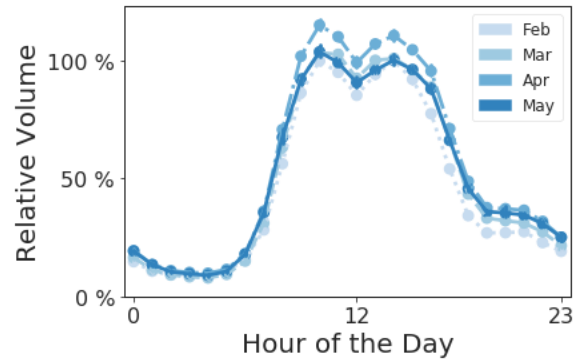


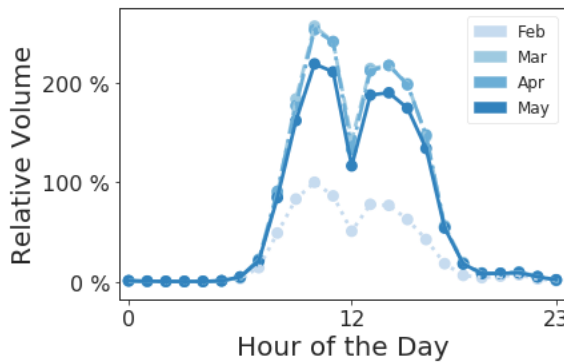
Figure 2.1: Distributions of virtual meeting duration, meeting size, and meeting types measured through our collected telemetry data. We observe two clear peaks surrounding 30 mins and 60 mins in distribution of meeting duration, indicating the popularity of scheduling 30 mins and 60 mins meetings. We observe a long tail distribution of meeting size, where over 20% of all meetings are one on one meetings. Finally, we observe the majority of meetings are scheduled meetings, followed by recurring meetings, adhoc meetings and broadcast meetings. The data distributions motivate us to discretize features into bins specified in Table 1.



(a) Email Action Daily Rhythm



(b) File Action Daily Rhythm



(c) Remote Meeting Daily Rhythm

Figure 2.2: Transition of worker work rhythms (i.e., distribution of work related actions over time within a day) from Feb. 2020 (pre-COVID-19) to May 2020 on email, file editing actions and remote meetings. While email and file usage remains stable, there is a clear increase in the volume (over twice as much) of remote meetings after the breakout of the pandemic. Results are normalized using maximum volume in February.

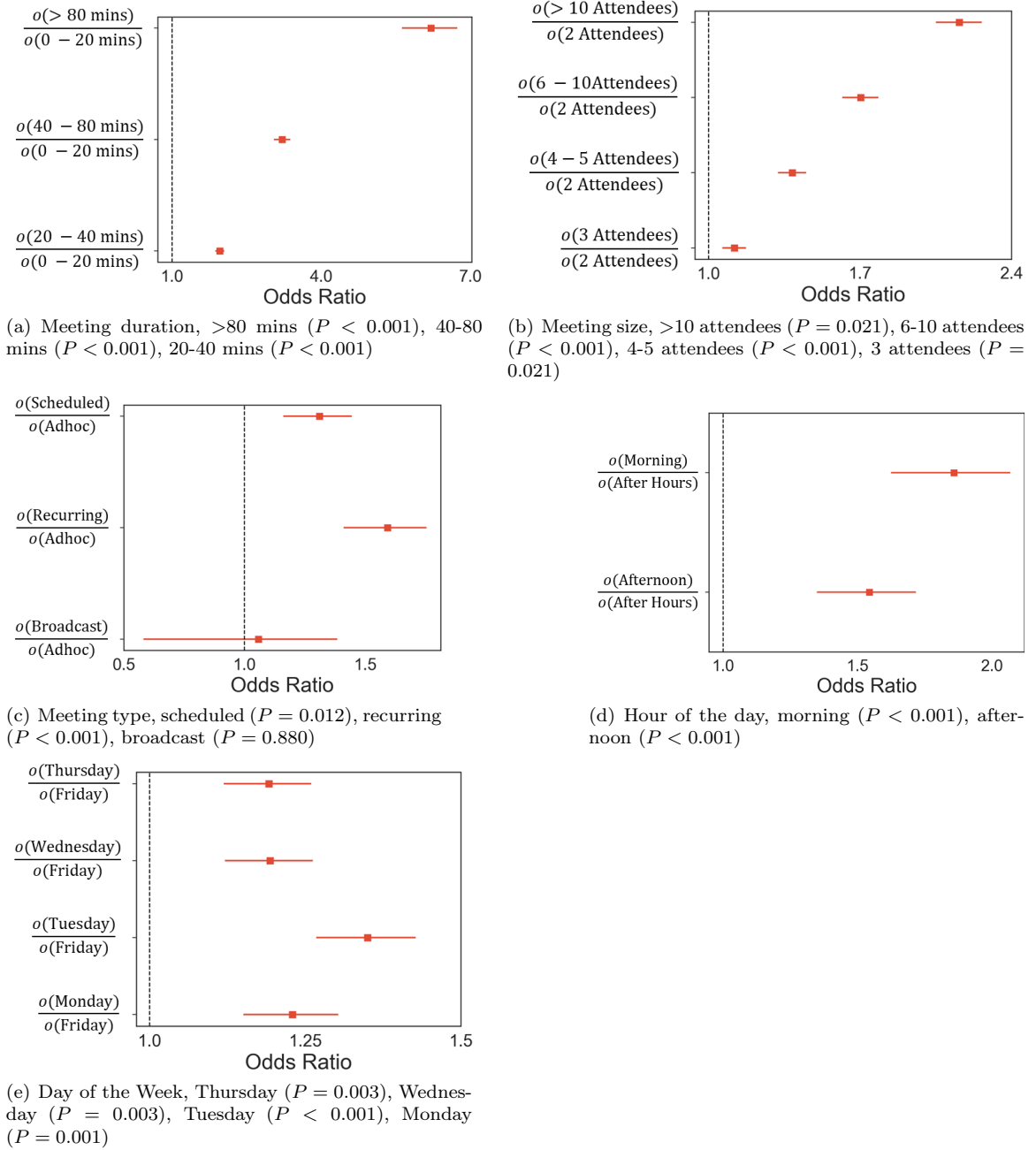


Figure 2.3: Conditional logistic regression results on the relationship between email multitasking and remote meeting characteristics. We find significant associations between email multitasking and meeting duration, meeting size, meeting types, hour of the day and day of the week.

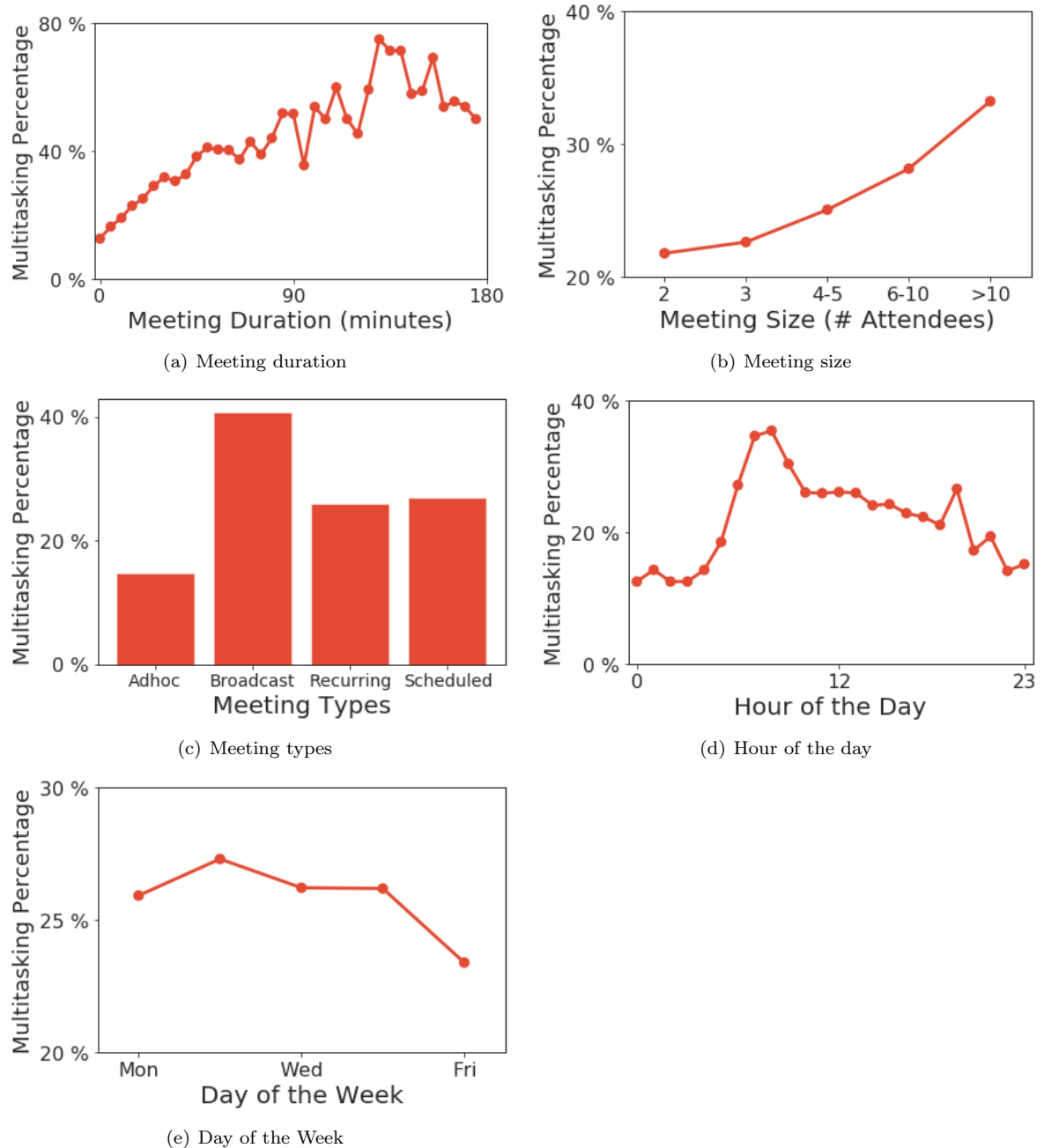


Figure 2.4: Proportion of user-meeting pairs with email multitasking versus meeting characteristics measured by telemetry data. People multitask more in longer meetings, larger meetings, recurring/scheduled meetings, morning meetings and Tuesdays.

Chapter 3

AI predictions of team viability

In the previous chapter, we demonstrated how novel data sources, such as telemetry data collected from workplace software, could be used to better measure and understand work patterns of people at the workplace. In this chapter, we further show computing algorithms, such as natural language processing and AI, can be used as *technical methods* to analyze work practices. On one hand, these methods contribute to a deeper understanding of human behavioral patterns. On the other hand, they provide the technical foundation for developing applications that enhance the work experience. Here, we present a focal study using natural language processing and AI to classify and predict team viability.

3.1 Introduction

Even high performing teams can be miserable. Whether or not a team succeeds at their goals, their team environment can be joyless, toxic, or even downright malicious. *Team performance* is a widely studied metric of successful collaboration [174, 229, 28, 243], but it is only one piece of the picture. *Team viability* — the capacity of a team for sustainable growth and future success [27] — is a less studied yet equally critical component of a sustainable, successful team [171]. A team with high viability can adapt to internal and external changes [55]. A team which suffers a loss of viability will see its members resist, resent, or even actively undermine their teams [142, 179, 227, 239]. A loss of viability can be so disastrous that the members fracture and do not want to work together again [244]. Consequently, it is critical for teams to evaluate and sustain their viability in order to promote continued engagement and future success. In particular, remote teams are more susceptible to antisocial behavior and developing conflicts [110]. As remote collaboration becomes increasingly prevalent and necessary, it becomes even more crucial to study team viability, which reflects both the members' satisfaction with their teammates and the members' behavioral intent to remain in the team [103]. Since team viability is not necessarily skill-based or task-specific, not just remote

teams but any team can strive to improve their viability.

Team viability is typically measured through self-reported questionnaires [64], limiting its effectiveness to after-the-fact instances of low viability. Assessing team viability at early stages of team interaction can offer actionable insights for improving ongoing collaborative efforts, as opposed to collecting retrospective lessons and regrets in the case of post-hoc evaluation. Therefore we ask: can we differentiate more viable and less viable teams based on a glimpse of their interaction? What attributes of team interaction are indicative of high viability? If this were possible, we could aid teams in diagnosing problematic starts and aid them in righting themselves.

To address these questions, we study a dataset of 669 online, 10-minute long, remote team interactions [244, 245], in which each team completed a collaborative task and a post-hoc team viability instrument [64]. We then use this data to judge the efficacy of particular signals in classifying whether a given interaction has a viability above or below the viability thresholds of top decile, median, or bottom decile. Since we are interested in which features differentiate the behavior of more viable and less viable interactions rather than the particular viability score they achieve, we focus on the task of classification.

We design features drawn from organizational behavior literature and compare multiple machine learning models for classification. For instance, we measure cosine similarity of TF-IDF vectors based on unigrams and bigrams to characterize topic alignment in the conversation, and analyze subjectivity and sentiment of the messages to characterize linguistic output using LIWC (Linguistic Inquiry Word Count) [224, 234]. Furthermore, features that cannot be easily detected by algorithms, such as psychological safety [89, 38, 86, 38], were coded by Amazon Mechanical Turk workers. We employ a lasso (L1 regularized) regression model to identify the most predictive features, achieving the highest accuracy of .92 AUC ROC for the classification of the top decile viability teams, .75 AUC ROC for the task with a median split, and .74 AUC ROC for the classification of the lowest decile viability teams. These results remain robust even when controlling for team performance.

Given these models, we highlight a few behavior patterns that we found as being important signals for viability. First, we find that use of exclusive language, such as ‘but’, ‘either’, ‘except’, and addressing via the second person, such as ‘you’ and ‘your’, are significant predictors of higher viability across all three tasks, which suggest teams whose members actively respond to and engage with the ideas of others are more likely to be viable. Sarcastic humor, anxiety terms, and higher average reading level were additionally significant in distinguishing the top decile teams from the rest, suggesting that the most viable teams are engaging in nuanced and complex interactions, in which context and focus matter. On the other hand, the use of words indicating sadness as defined by LIWC and the negativity text sentiment of the most negative individual were associated with the lowest decile viability teams.

Many of the significant features in our models, such as those reflecting word choice and sentiment, were computationally derived rather than human-coded. This result hints at the possibility of

enabling teams to conduct real-time assessment of their viability. Furthermore, we demonstrate that the relationship between viability and team interaction is consistent over time, yet the pattern differs between the first time a team convenes and when it meets again even if they do not recognize their teammates [244], which suggests that viability is a dynamic measure that could be monitored continuously. Finally, we find that models using features computationally derived from the first 70 seconds of an interaction can achieve a comparable level of performance to the models utilizing full 10 minute long interaction, implying that signals of team viability arise even in the first few moments of interaction.

While previous works have focused on predicting and understanding the determinants of team performance [174, 28], team viability is an equally critical goal for promoting effective collaboration. Evaluating viability is especially important in remote teams because they lack a shared context [190], are more likely to have difficulty developing mutual understanding [95], are less cohesive than face-to-face groups [230], and exhibit more competitive behavior [202], which render them more prone to developing conflicts [110]. In this paper, we present models for assessing viability via classification of online team interaction and explore potential signals of team viability in text interaction data. We make our classification algorithm available for others to use.¹

3.2 Background and Related work

Teams are complex dynamic systems that develop as members interact and respond to the unfolding of situational demands over time in a given context [135]. On one hand, a positive team experience can help members come together to produce a whole that is greater than the sum of its parts. On the other hand, a negative team experience can sabotage success to the point where members vow to never work together again. Given the critical role that teams and teamwork play in our economy and society, how might we better understand the viability of any given team?

While team viability and team performance are correlated with a strong and positive relationship for teams completing routine activities [27], teams that perform well can still have disastrously low viability, as exemplified by the eventual fracture between Kahneman and Tversky [150]. Moreover, viability and performance also tend to have a weaker relationship when the team encounters dynamic situations that require different strategies and skills [134]. Thus, team viability should be considered and studied as a distinct feature from team performance.

3.2.1 Team Viability

When team viability is absent or lost to such an extent that the team chooses not to work together in the future, a phenomenon called team fracture occurs [244]. Literature shows that interventions for improving the psychological state of a team and its members can be effective [141]. If we can

¹<https://viability.stanford.edu>

recognize that a team has low viability or is dropping in viability, it would be possible to try to help them before they fail [112]. Thus, there exists an important need for being able to recognize that a team needs help.

Currently, there are two ways that this need may be addressed, and both of them have significant drawbacks. The first mechanism for recognizing that a team needs help is through self-reporting from individual team members or the team as a collective. For example, an individual may raise complaints about their team to the team manager or other parties. This mechanism encounters three problems. First, self-reporting depends on the *ability* of an individual to recognize low or falling team viability. People are often remarkably bad at recognizing their own state [174], and may not realize the decline of team viability in the moment. Second, the *responsibility* of speaking up about problems with team viability entirely falls on the reporting individual. Social loafing and diffusion of responsibility are obstacles to self-reporting even if an individual recognizes problems, particularly when a task is unattractive. Third, *potential consequences* present significant disincentives for the reporting party. Self-reporting may potentially lead to interventions and improvement [141], but it could also lead to total disintegration of collaboration or the ostracization of the reporting individual from the team and even the entire organization [73]. Indeed, even in the best case scenario where the overall viability of a team increases as a result of an individual's report, the social status of that individual is still likely to suffer [195]. Because of all of these reasons, individuals often may not act to prevent the impending failure of a team.

The second existing mechanism for recognizing that a team needs help is through external involvement, often taking the form of third-party actors, such as managers, coaches, and counselors, being called to step in via processes such as quarterly team assessments [170]. External involvement allows for the external evaluation of a team, and third-party actors often have incentive to report and act upon problems with a team's viability [29]. These factors account for some of the difficulties of relying on self-reporting as a solution. However, external involvement has its own drawbacks. First, introducing an evaluating party who is not directly in the team introduces potential inaccuracies in their evaluation [171]. When this party is related to the team, e.g. in the case of a team manager [22], they may bring biases and misaligned incentives of their own about their team in competition with other teams, or the characteristics of individual members of their team [29]. When this evaluating party is entirely separate from the team, e.g. in the case of a counsellor or a team-building professional [17], they may not be able to understand the complete context in which the team lives, and their evaluations may be distorted as well. Second, even if we disregard whether an accurate evaluation of the team is possible, an external human evaluation of the team costs resources, which presents a barrier to entry for smaller teams or organizations. Finally, by the time a third-party actor is summoned to deal with team issues, it may already be too late to matter [136].

Current ways of recognizing that a team needs help with viability are not ideal — a gap exists in the literature, with regards to a systematic investigation of the features of team viability and its

possible predictability. Team interactions have been the subject of countless rounds of observation and analysis. One point of consensus in prior research is that predicting team behaviour is difficult, often because whether the team dynamics are effective is not obvious even to the team members themselves [174]. In this work, we set out to answer a related research question:

Research Question 4 (RQ4) *Can we differentiate high and low viability teams by analyzing team interactions?*

3.2.2 Antecedents of team viability

Existing literature explores predicting team fracture and team performance, but not team viability, using field studies and crowdsourcing. Certain features emerge as strong predictors of team performance in field studies, such as team minimum agreeableness and team mean conscientiousness, openness to experience, collectivism, and preference for teamwork [28]. Interesting insights derived from prior research are that linguistic cues and conversational patterns extracted from the first 20 seconds of a team discussion are predictive of whether it will be a productive discussion [186], and that the type of task matters for the consistency of team success or fracture [244]. Notably, existing literature has focused on a number of broad feature categories, where there is an overlap between the categories. In this paper, we do not aim to identify patterns on the level of the categories, but we do draw upon trends in these categories to select our features.

Work Pattern Features

From proposing ideas to implementing decisions, equal participation is a key aspect of constructive teams [187]. Well-balanced participation is a challenge in virtual, chat-based discussions because they lack in-person cues. For example, in offline interactions, individuals can demonstrate that they are listening to a person speak by nodding their head in agreement, or maintaining eye contact. However, in online interactions, individuals can only convey their listening by sending written responses [137]. This limitation opens the door to many problematic scenarios as it can be exploited by internet trolls who take pleasure in upsetting others, as well as socially-insensitive individuals who simply neglect to respond [127]. When a person does not see evidence that they are being heard, they feel dissatisfied and frustrated. Such frustration makes people less willing to participate and contribute, sometimes disengaging completely.

A team’s performance is also an antecedent to its viability [159, 23], but the relationship is complicated by the stresses of achieving performance in some situations [29]. Producing high quantities of ideas can be equated to high performance since teams that are diverse, insightful, and creative are better able to cultivate a rich flow of ideas. Another way to evaluate a team’s performance is its ability to reach conclusions efficiently, as cohesive teams tend to collectively process ideas and converge to a final decision more quickly than non-cohesive teams [83].

Semantic Features

Constructive communication has an ameliorative effect on groups, reducing conflict and increasing cooperation [24]. For example, positive interactions induce a pleasant overall mood, which is associated with greater cognitive effort and more constructive methods of dealing with disagreement. The foundations of a healthy team atmosphere are also commonly established during a team's first interactions, since initial communication and interaction behaviors set the standard for team culture [232]. In particular, friendly introductions and greetings between group members create social bonds, which lay the foundation for benevolence and trust [102]. To sustain this trust, team members often engage in supportive communication in accomplishing a task (e.g. celebrating achievements, expressing appreciation of each other's contributions). Additionally, since online discussions do not convey the same richness of emotion and reaction that face-to-face communication enables, they are susceptible to misunderstandings when utterances contain sarcasm or irony [180]. Due to the use of sarcasm, what might look like a negative utterance is, in fact, a positive one (and vice versa). Consequently, sarcastic or ironic utterances online can affect team members' perceptions of positivity and interaction behaviors.

Rudeness and aggression can quickly sabotage a healthy conversation (e.g., ad hominem attacks, name-calling, knee-jerk contradiction) [84]. Instead, healthy team dynamics center on agreeableness, respect, and support. From softening the perceived force of a message to allowing all parties to save face, politeness can shape the course of online interactions for the better [252]. Inter-member civility also encourages social connection and rapport, fostering amicable conversation between members and helping groups maintain cohesion throughout a discussion.

Topic Features

Some topical contexts naturally lend themselves to antisocial behavior and discord. Specifically, topics like politics, religion, and ideology are breeding grounds for conflict [252]. Religion and philosophy account for over half of the conflict on Wikipedia [130], while 14% of Americans say they have been harassed online due to their political views [86]. Moreover, people tend to feel more strongly about ideological issues and take matters personally [13]. Such emotional investment makes individuals more susceptible to friction and hostility, with decision-making becoming more emotional than logical.

Word Choice Features

Psychological safety refers to a shared belief held by teammates that a team is safe for interpersonal risk taking [89]. Task conflicts that occur in a psychologically safe environment have been found to improve creativity and decision making without damaging interpersonal relations or interactions between teammates [38]. In this way, a highly productive team may be highly critical but also highly

psychologically safe. In the opposite direction, a highly critical team that is not psychologically safe can transform how teammates respond to challenges. A multitude of factors can result in teams not being psychologically safe, for example name-calling, physical threats, dismissive remarks, and social punishments associated with rejecting the group norms [86, 38]. In 2017, 41% of Americans said they have suffered online harassment, 27% said they have been called offensive names online, and 22% say that someone intentionally embarrassed them online [86]. Victims of such behaviors often experience mental or emotional stress, and are unwilling to work with their perpetrators again. Such stress is particularly common in online mediums, where anonymity emboldens perpetrators to use words that they likely do not use in in-person interactions. These situations motivate another research question:

Research Question 5 (RQ5) *Which behaviors are most strongly associated with high vs. low viability teams?*

3.2.3 The durability of team viability antecedents

Team interactions evolve over time [104] — what indicates a positive relationship early in a team interaction, might indicate a problematic one if the team is expected to be very familiar due to months of close collaboration. Though rare, studies of longitudinal interactions in teams show compounding effects of low viability [144]. And thus the question of how repeated episodes of work might expose unexpected qualities of team viability. Intriguingly, the dataset analyzed in this work has the unique property of involving repeated interactions between the same team without them knowing it, as well as interactions when they did know they were working with the same prior collaborators. While the interactions in the data are only 10 minutes long each, their repeated nature may shed light on another research question:

Research Question 6 (RQ6) *How does the historic collaboration context influence team viability?*

3.2.4 Early warnings

When studying events with final outcomes, it can be helpful to understand whether that final result can be identified by some patterns or behaviors early in the process rather than after the outcome has already occurred. “Thin slicing” studies show how relatively small and relatively early interactions can be powerful predictors of long-term outcomes — famously by accurately predicting which marriages will fail years later [101, 54], or the grades at the end of a semester [123] — by making judgements based on only a few minutes of interactions. Understanding viability in the future, as opposed to after a team has already come to have low viability, would turn predicting viability into something strikingly useful for teams in practice. This goal motivates a final research question to study:

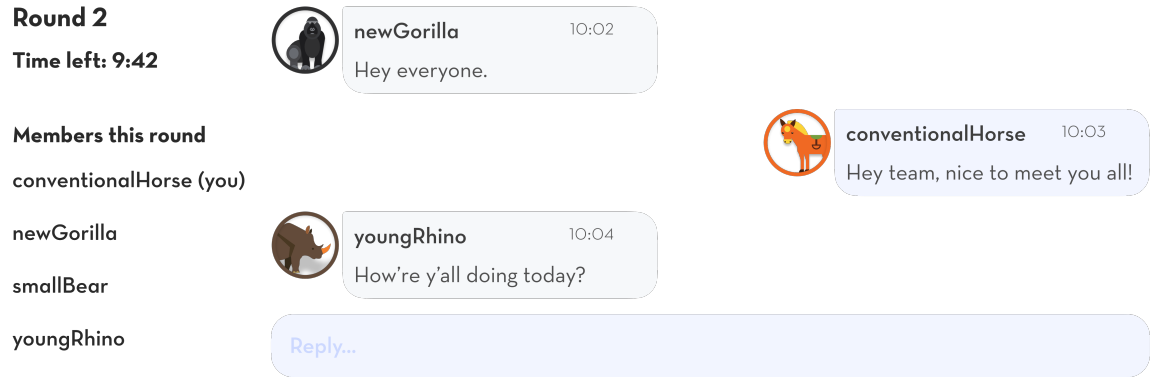


Figure 3.1: Participants collaborated in a chat room with anonymous usernames. Our dataset of interaction histories consists of all chat messages, who they were sent by, and a timestamp.

Research Question 7 (RQ7) *How little duration must be observed to make accurate predictions of team viability?*

In this section we have mapped out the challenges in classifying team viability and motivated a range of theoretically and practically interesting questions to try to answer. We next move in to explaining how we achieve the classification task.

3.3 Method

In this paper, we focus on differentiating high and low viability teams through chat transcripts of online team interaction. We operationize our goal as a binary classification task. Specifically, we derive both algorithmic and human-coded features from organizational behavior and behavioral science and apply to them to the interaction data from the teams' chat transcripts. The algorithmic features are drawn from a toolkit of modern natural language processing and computational social science tools; human-coded features are labeled by multiple independent workers on Amazon Mechanical Turk. Using these features, we then seek to classify high or low team viability using machine learning models.

In this section we will describe how the team interaction dataset was processed, how the features were encoded on the interaction data, and how the machine learning algorithms were set up to classify team viability.

3.3.1 Data processing

We work with data of teams' interactions gathered for a previous study that investigated team fracture and viability [244, 245]. In this dataset, participants recruited from Amazon Mechanical Turk were grouped into teams of 3–8 and remotely collaborated through a synchronous messaging

chat room to complete various tasks over several rounds of 10 minutes. The teams are all made up of only US workers who have completed at least 500 tasks on Mechanical Turk (48% female, and 38 years old on average). The identity of members shown to others was manipulated such that any one group of members would work together 4–6 times without knowing that they had collaborated previously. These data include all of the messages exchanged during the discourse associated with unique pseudonyms and the timestamp since the beginning of each interaction. In these interactions, three different types of tasks from McGrath’s task circumplex were used [231]: creative, intellectual, and cognitive conflict, although around 80% of the data utilize the creative task. After completion of each task, team members were asked to complete several surveys, including a 14-item viability instrument on a 5 point Likert scale [64].

As just mentioned, teams worked together multiple times in succession without knowing they were in fact interacting with the same collaborators [245, 244]. We achieved this effect through a two directional pseudonym masking technique designed to reset team dynamics between rounds — to participants, it appeared that they were starting a fresh collaboration with new team members. This was validated through an in-sample manipulation check to ensure that collaborators were truly not aware that they were working together repeatedly. Additionally, among the repeated interactions for each team, there was also one pair of interactions where the team members were knowingly *reconvened* with the same group by using consistent pseudonyms. In the prior experiments, this treatment was used to understand path dependence of viability in teams. In this study, we will leave aside data from these repeated teams for every part of our analysis which does not explicitly require reconvened teams, in order to avoid comparing teams with different lengths of interaction. We do take advantage of this unique aspect of the data to specifically make claims about the properties of viability signals in reconvening teams relating to RQ6. As far as we know, this is the only chat dataset affording comparative analysis across independent rounds and reconvening teams, which is why we selected it for this paper.

In total, we analyzed 669 independent chat interactions of teams, aggregating to a total of 40,024 messages. For each team, we summed each collaborator’s 14 viability scale items into a single individual score with a range of 14–70 and calculated the mean of all of the individual scores to determine a team viability score. Figure 3.2 illustrates the overall distribution of team viability: the median viability score of all teams is 45.5, the 10th percentile is 34.95, and the 90th percentile is 58.25.

3.3.2 Feature Collection

Computationally-derived Features

To study characteristic differences between high and low viability teams, we first derive features through algorithms via the series of chat messages in a multi-person team interaction. These features can be divided into four categories: work pattern, semantics, topic, and word choice, as motivated

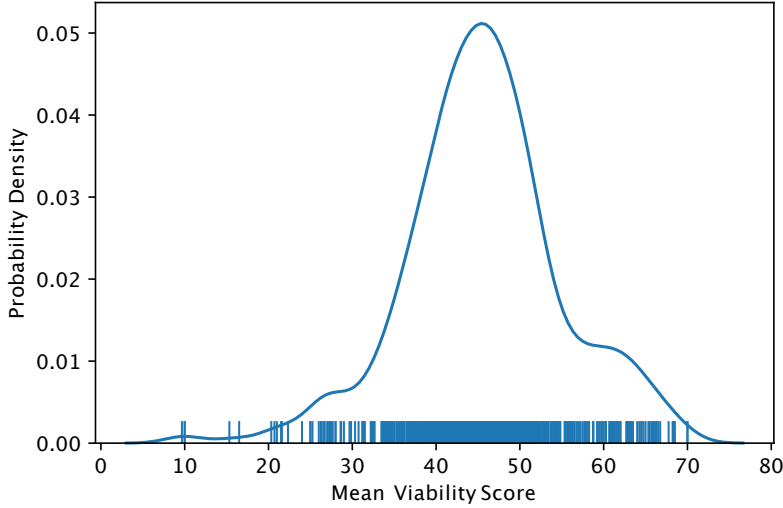


Figure 3.2: Distribution of average team viability scores across all teams in our dataset.

by prior literature discussed in Section 3.2.2. Because there are anywhere from 3 to 8 people in a group, we first calculate the features on the level of each individual team member from the entire chat interaction and consequently produce final team-level statistics of that feature (mean, standard deviation, minimum, maximum) that are used to model the chat discourse.

Table 3.1: Computationally-derived features associated with work pattern, semantics, topic, and word choice. Some features are multiplied by four, because the minimum, maximum, mean, and standard deviation were calculated for each.

Feature Group	Number	Specific Features
Work Pattern	2×4	Number of messages per person, number of words per person
Semantics	2×4	Subjectivity, polarity
Topic	1	TF-IDF cosine similarity
Word Choice	25	LIWC (anger, anxiety, sadness, articles, conjunctions, prepositions, inclusive, exclusive, quantifiers, certainty, discrepancies, negation, tentativeness, first person singular, first person plural, second person, indefinite pronouns, adverbs, social), argue, reference, readability

With work pattern features, we seek to capture higher level behavioral information from how team members are interacting. In task-oriented discussions, well-balanced participation among group members is a key factor present in productive teams [186]. We derive two features in this area: number of messages sent and number of words used by each team member. For each person, we count the number of separate messages he/she sent as well as the total number of words he/she used

Table 3.2: Computationally-derived features drawn from previous literature.

Feature	Measurement Method
Average message count, Talkative member message count, Laconic member message count, Variation in message counts, Average word count, Talkative member word count, Laconic member word count, Variation in word count [186]	Statistical measures on team members' chat messages
Average polarity score, Highest polarity individual score, Least polarity individual score, Variation in polarity scores, Average subjectivity score, Most subjective individual score, Least subjective individual score, Variation in subjectivity scores [184]	TextBlob sentiment analysis algorithms evaluated on team members' messages
Average reading level, Highest individual reading level, Lowest individual reading level, Variation in reading levels [208]	Dale-Chall readability score algorithm applied to team members' messages
Anger words, Anxiety words, Sadness words, Articles, Conjunctions, Prepositions, Inclusive language, Exclusive language, Certain language, Disagreement language, Negation language, Tentative language, First person singular, First person plural, Second person, Indefinite pronouns, Adverbs, Relationship-oriented language [65, 234]	LIWC lexical category counts
Collaborator language similarity [224]	Average cosine similarity of TF-IDF vectors based on unigrams and bigrams
Argument discourse markers [3, 9]	Counts of words from <i>ARGUE</i> corpus
Pseudonym use [220]	Counts of referring to others by pseudonym in the chat messages

across all messages for the entire chat discourse.

For semantic features, we applied text-based sentiment analysis as it has been widely used in predicting human behavior as in social media [184]. Specifically, we derive the polarity (whether language is positive or negative) and subjectivity (whether language is objective or subjective) of group members' messages using the natural language processing library TextBlob.²

For the topic features, we expanded upon previous work, which show that unigrams and bigrams are particularly effective for disagreement and argument detection [224]. Specifically, we create TF-IDF (term frequency-inverse document frequency) vectors based on unigrams and bigrams as a representation of topics in a person's chat messages. Preprocessing of the messages only involved removing accents to normalize to the Unicode standard, and converting all letters to lowercase;

²<https://textblob.readthedocs.io>

neither stop word removal or other token-level analysis like stemming or lemmatization were used. The average pairwise cosine similarity of these vectors is used as the feature to describe the alignment of topics in different team members' language.

For the word choice features, we applied dictionary-based text analysis, which has often been used to identify signals in language, for example, to study mental health using social media [65]. Specifically, we use LIWC (Linguistic Inquiry Word Count), an extensively human validated lexicon, to count the words that individuals used during the chat discourse that fall into different lexical categories [234]. For our task, we reduce the total category set to a subset of those that are cognitive- or function-oriented and those related to affect (see Table 3.1), culminating to 19 categories. Additionally, because we expect conversational conflict to influence team viability, we include as one feature the total count for the presence of any 17 discourse markers for recognizing disagreement from the *ARGUE* corpus [3, 9]. Prior work on chat-based team dialogues [220] motivated us to incorporate the number of references to other team members as another feature. As a higher level representation of word choice, we also calculate the minimum, maximum, mean, and standard deviation of the Dale-Chall readability scores for each members' chat contribution.

Human-labeled Features

Furthermore, we hypothesized that certain aspects of teamwork would be discernible to humans but not to algorithms. We have drawn 20 features from organizational behavior (Table 3.3) to be hand-labeled. We used the Amazon Mechanical Turk platform to recruit 366 human annotators who had completed at least 500 tasks to annotate teams' chats. For a given chat, the human annotators filled out a survey of 20 questions corresponding to the 20 features, answering each question using a 5-point Likert scale.

More specifically, our 366 human annotators or "reviewers" must have completed at least 1000 tasks and must be based in the USA. The recruiting HIT also excluded all reviewers who had participated in the original studies. Workers were paid at a rate of \$15 per hour, in accordance with existing literature [246]. We then built a platform that displays one chat per annotation task along with instructions given to the original participants in the task. With all pseudo-names of the participants replaced by a label of "Person_<number>" (e.g. Person1, Person2, ...) format, the reviewers were asked to rate their perception of the chat interaction on questions measuring selected features on a 5-point Likert scale and to predict whether a team would fracture. We included an attention check question asking for the number of participants in the chat and excluded responses that were not consistent with the truth from the analysis. The order of questions were randomized, and reviewers were allowed to participate in the task multiple times.

At least 3 reviewers annotated each chat, and the median response was selected as the feature value for analysis. The average standard deviation of scores across all raters and chats for a given feature was 1.08 prior to filtering. Responses that were more than 1.5 standard deviation away from

Table 3.3: Human-labeled features drawn from previous literature and the statements used to ask Amazon Mechanical Turk workers to rate them. The mean standard deviation ($\bar{\sigma}$) is for each feature across all included raters and across interactions is also shown.

Feature	Survey Question	$\bar{\sigma}$
Generating Many Ideas [187]	There were many ideas/suggestions/input generated from this group.	0.92
Reaching a Conclusion [83]	The group reached a decision in the end (there was a sense of consensus).	0.88
Progressing Slowly [25]	It took a long time for the team to reach a conclusion.	1.15
Thoughtful Response [159, 23]	The final answer of the team is reasonable and well-thought-out (e.g: a creative answer for ad writing).	0.90
Social Loafing [187]	At least one team member was not participating as much as the other team members.	1.15
Annoying Collaborators [137, 127]	At least one team member was annoying (used all caps, or sarcasm, or was obnoxious, etc).	1.32
Frustrated Collaborators [137, 127]	At least one team member expressed frustration/confusion.	1.27
Positive Interaction [24]	There were positive interactions (emojis, lol, great!, etc).	0.89
Starting Salutations [232, 102]	The team members greeted each other at the beginning of the interaction.	0.87
Ending Salutations [102]	The team members congratulated each other at the end of the interaction.	0.96
Sarcastic Interaction [180]	There were sarcastic comments.	1.28
Agreeable Interaction [252]	The group was agreeable.	0.89
Respectful Interaction [252]	The teammates were supportive and respectful as opposed to insulting.	0.85
Passive-aggressive Interaction [84]	Team members were passive-aggressive.	1.28
Dismissing Collaborators [86, 38]	At least one person in the team was dismissed at some point.	1.35
Punishing Collaborators [86, 38]	At least one person in the team was punished at some point.	1.35
Showing Embarrassment [86, 38]	At least one person in the team seemed embarrassed at some point.	1.31
Political Interaction [252, 130, 86, 13]	This interaction was Political as opposed to Non-political.	1.35
Playful Interaction [252, 130, 86, 13]	The conversation was Playful as opposed to ideologically charged.	0.96
Fact-driven Discussion [252, 130, 86, 13]	Ideas were Fact-based as opposed to Emotion-based.	0.99

the mean result were excluded prior to calculating the median. Mean standard deviation values for each feature across all interactions are shown in Table 3.3.

3.3.3 Machine learning models

We operationize our goal through a binary classification task to differentiate high and low viability teams. For our analysis, we divide all teams into two groups — either high viability or low viability — based on their viability score. Specifically, we set the cutoff viability score as a varying decision threshold, and classify all teams with viability score lower than the cutoff score as low viability teams and teams with viability score above the cutoff as high viability teams. For instance, if we set the cutoff threshold as 45.5 (the median viability score in our dataset), we divide all teams into two equal sized groups, where those teams with viability score above the median set as the high viability sample and those teams with viability lower than the median as the low viability sample. As such, we treat predicting team viability as a binary classification problem because we are interested in distinguishing teams with relatively higher viability from teams with relatively lower viability, rather than predicting the particular measure of viability.

We implemented 5 widely-used machine learning classification models from Python’s scikit-learn library [198], i.e., logistic regression, support vector classification (SVC), random forest classifier, multi-layer perceptron (MLP) classifier, and gradient boosting classifier, to train and classify low vs high team viability based on our derived features. We used the area under the curve of the receiver operating characteristics curve (AUC ROC) as a measure of prediction performance, since we formulated the viability differentiation as a binary classification problem and AUC is insensitive to an unbalanced testing set. Note that most experiments are done via 5-fold stratified cross validation, and each feature in the training and test sets are standardized.

3.4 Results

3.4.1 Differentiating team viability under different split

To answer RQ4, we evaluated the five models across nine different trials where we set the viability decision threshold at every ten percentile points, (e.g 10th percentile, 20th percentile, ... , 90th percentile). The models are able to differentiate higher from lower viability teams (Table 3.4).

Prediction Model	Data Partition Percentile								
	10	20	30	40	50	60	70	80	90
Logistic Regression	0.74	0.72	0.69	0.72	0.75	0.74	0.78	0.81	0.92
SVC	0.63	0.70	0.67	0.69	0.76	0.75	0.79	0.85	0.92
Random Forest	0.66	0.72	0.69	0.72	0.73	0.75	0.79	0.86	0.93
MLP Classifier	0.66	0.68	0.66	0.68	0.71	0.71	0.74	0.80	0.90
Gradient Boosting Classifier	0.67	0.70	0.67	0.72	0.75	0.73	0.76	0.86	0.93

Table 3.4: Viability classification performance (measured by ROC AUC) with different division of positive and negative samples using all features.

Prediction Model	Data Partition Percentile								
	10	20	30	40	50	60	70	80	90
Logistic Regression	0.77	0.76	0.73	0.75	0.76	0.76	0.78	0.82	0.94
SVC	0.63	0.74	0.72	0.73	0.76	0.75	0.79	0.84	0.93
Random Forest	0.70	0.70	0.69	0.73	0.72	0.73	0.77	0.84	0.92
MLP Classifier	0.69	0.71	0.71	0.71	0.73	0.74	0.75	0.82	0.92
Gradient Boosting Classifier	0.66	0.69	0.67	0.73	0.76	0.73	0.78	0.85	0.92

Table 3.5: Viability classification performance (measured by ROC AUC) with different division of positive and negative samples by computationally derived features.

Prediction Model	Data Partition Percentile								
	10	20	30	40	50	60	70	80	90
Logistic Regression	0.56	0.56	0.53	0.54	0.61	0.60	0.63	0.69	0.71
SVC	0.46	0.44	0.47	0.51	0.58	0.60	0.63	0.68	0.66
Random Forest	0.58	0.50	0.65	0.55	0.52	0.56	0.60	0.65	0.68
MLP Classifier	0.55	0.52	0.51	0.53	0.55	0.59	0.63	0.67	0.70
Gradient Boosting Classifier	0.57	0.48	0.53	0.55	0.5	0.56	0.59	0.67	0.70

Table 3.6: Viability classification performance (measured by ROC AUC) with different division of positive and negative samples by human labeled features.

Perhaps more importantly, our results suggest that there exist general mechanisms behind team viability vs. team interaction, independent of the team member composition.

When we consider the case of classifying a team’s viability relative to the median split, the best AUC score of the five models is 0.76. More interestingly, we find that as the data partition becomes such that we label fewer samples as high viability teams and focus on the teams with relatively higher viability scores, the classification accuracy becomes higher. For instance, when we label the top decile teams as high viability teams and designate the remaining 90% as low viability teams, we reach over 0.9 AUC, indicating there are salient differences between teams above the 90th percentile and the rest.

In the feature set we are using, in order to capture a wide variety of team interaction behaviors, some features were automatically developed by computational means, and those that could not be, due to the limitations of current algorithms, were labeled by humans. For the practicality of creating a tool for differentiating team viability, we further performed a preliminary ablative analysis of our features. We repeat the previous procedure separately for the computationally derived and human labeled features, which is shown in Table 3.5 and Table 3.6, respectively.

We see a noticeable difference in the performance of the two methods, but regardless, we witness a similar pattern of increasing accuracy as the threshold for dividing the data increases, which can be seen in the right-side plot of Figure 3.3. Across all decision thresholds, the classification accuracy of the computationally derived features is at least as good as the full features set, having an AUC

Features	Partition Percentile	F1 Score	Precision	Recall	ROC AUC
Computationally Derived	10	0.10	0.13	0.07	0.77
	50	0.67	0.70	0.64	0.76
	90	0.57	0.69	0.52	0.94
Human Labelled	10	0.03	0.20	0.02	0.56
	50	0.57	0.59	0.56	0.61
	90	0.08	0.50	0.05	0.71
All Features	10	0.20	0.32	0.15	0.74
	50	0.66	0.67	0.64	0.75
	90	0.56	0.62	0.52	0.92

Table 3.7: Additional metrics of viability classification for Logistic Regression when using a decision threshold at the 10th, 50th, and 90th percentile of viability score. Note that AUC is a better metric to evaluate viability classification performance in our case since the metric based on ranking, while F1 score, precision and recall are based on threshold to divide positive and negative labels. The high AUC for our model indicates the ability of our model to correctly rank high viability team ahead of low viability team, while the relatively low F1 score only reflects the cutoff threshold is not ideal.

of 0.76 of at the median split and 0.94 at the 90th percentile split as seen in Table 3.5. Using our human labeled features alone results in only an AUC of 0.61 at the median split and 0.71 at the 90th percentile split as seen in Table 3.6. This discrepancy could suggest that 1) there can be significant signal overlap between computationally derived and human labeled features, or 2) human labeled features may be noisier than computationally derived features. This result provides evidence that computationally derived features alone may be suited to further study.

Finally, to rule out the possibility that team performance may be a confounder, we carried out a robustness check by measuring the correlation between team performance (as measured by actual task performance, e.g. click through rate of advertisement text in the creative task) and team viability. In our dataset, 175 teams have team performance information. We observe weak correlation between the two ($R^2 = 0.10$). As another test, we add team performance as a feature to the classification model and compare it with the classification model without team performance on these 175 teams. We find no change in classification AUC: under 10% and 50% splits, the AUC is stable at 0.78, and under 90% split, the AUC is stable at 0.96 under both conditions, which further verifies that team viability cannot be explained by team performance.

For the rest of our analysis, we will focus on using our logistic regression model. It achieves the best results in most viability splits and as a model, it is computationally simpler and more interpretable than the other models we tested. Additionally, we report other performance metrics specifically for logistic regression under various conditions of decision threshold and used features in Table 3.7.

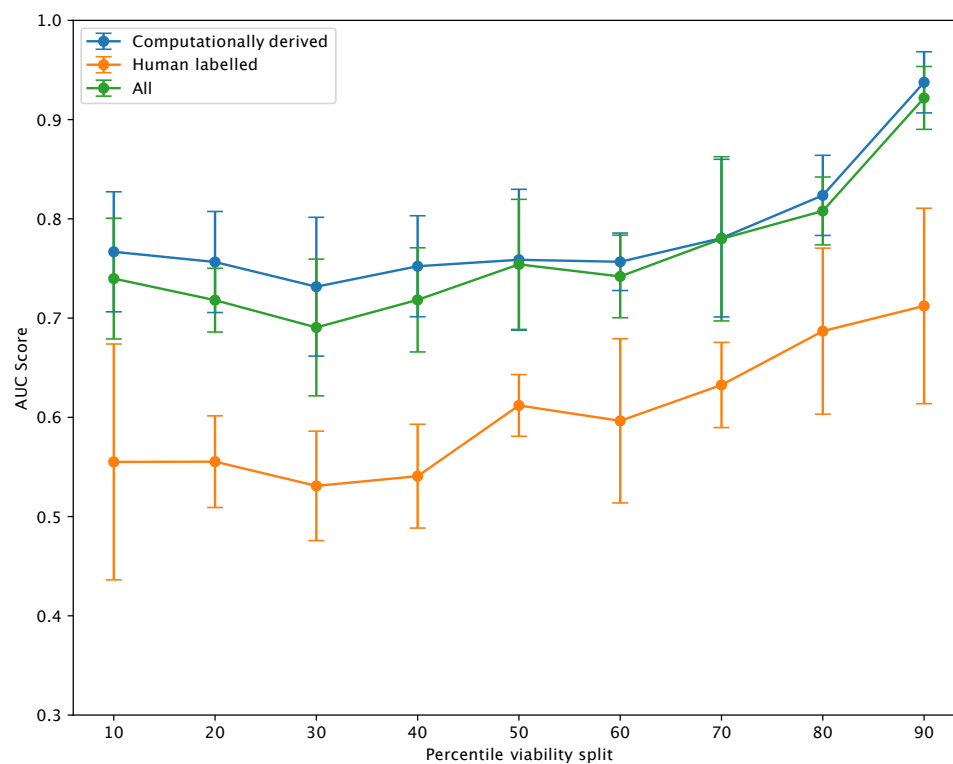


Figure 3.3: Performance of the logistic regression model in AUC score with computationally derived features, human labeled features, and all features across different viability percentile thresholds. Errors bars show the 95% confidence interval across 5 folds.

3.4.2 Feature Analysis

To answer RQ5, we proceed to investigate the role of different features in predicting team viability. Specifically, we run penalized logistic regression with L1 regularization (Lasso regression) for our model using threshold scores of 34.95 (bottom decile split), 45.5 (median split), and 58.25 (top decile split), and examine the coefficients of the features to understand whether certain features contribute positively or negatively to team viability. We choose penalized logistic regression since it helps prevent collinearity and sparsity with the effect that the coefficients' weights are moved to the most predictive features.

We present coefficients and their confidence intervals for all the significant features in Fig. 3.4. As can be observed, two features stand out as significant features across all three splits. Use of exclusive language (as defined by LIWC), such as 'but', 'either', 'except', and addressing each other in second person (e.g. 'you') are significantly and positively correlated with the teams with relatively higher viability across all three splits, which support findings that suggest teams whose members actively respond to and engage with the ideas of others are more likely to be viable [26]. In one of the top decile teams, the team members used exclusive language to consider a variety of different viewpoints and arguments when discussing their solution,

Person3: I believe it would be fair to allocate most, if not all of the funds to project 3.

Neither of the other two projects are vital, in my opinion.

Person4: great idea but sooo many people are hurting

Person2: #2 is the artsy option that we might get donations from the art community or art enthusiasts to help

On the other other hand, we observe that there are quite a few non-overlapping features that are significant when differentiating high and low viability teams at different thresholds, indicating that bottom decile viability teams, above and below median teams and top decile viability teams demonstrate different interaction behavior.

What distinguishes top decile teams from other teams? The use of exclusive language and second person are very strongly correlated with top decile teams, even more so than the other two splits for which these two features are also positive indicators. Other features such as articles and adverbs support this overarching trend of active engagement with other team members. As such, more active engagement appears to be an extremely important trend for top decile teams. A higher average reading level for the chat interaction is also positively correlated with membership in the top decile, suggesting that top decile teams often have the willingness and ability to communicate complex topics. In addition, the top decile has two interesting indicators which seem unintuitive upon first glance. The human-labeled presence of sarcastic interaction is positively correlated with viability scores above the 90th percentile. Upon further investigation, we found that while sarcasm has a negative connotation generally, the instances of human-labeled sarcasm which were present in

top decile team interactions were of the humorous variety.

Person2: 5) it's in the 120 range maybe? anyone live in Death Valley? ;)

In addition, the computationally-derived feature of Anxiety Words was also a positive indicator of top decile teams, but further investigation reveals that team members were using words which the algorithm identified as anxious in order to talk about their task, e.g., that "veterans were *struggling*".

What makes a team more likely to be a bottom decile team? The significant features that stand out make intuitive and theoretical sense. More exhibits of sadness, as well as the presence of a highly polarized individual — an individual who outputs very positive or very negative chat — are associated with the bottom decile of teams. These findings are consistent with the organizational behavior theory that displays of emotion, especially negative ones, will negatively impact the effectiveness of the team. [149] Interestingly, more use of first person singular terms (e.g., I, me, mine) and less use of first person plural (e.g., we, our, us) and second person words (e.g., you, your) is also associated with a higher chance of being in the bottom decile teams. This trend contributes to our conclusion that more active engagement with other team members correlates with higher viability, and less active engagement with other members correlates with lower viability. Finally, playful interactions were correlated with the bottom decile teams as well. Further investigation shows that overly playful teams had difficulty actually resolving the task at hand, ultimately leading to lower viability scores. For example, one team had a hard time deciding on a slogan for a chair after joking about a pun,

Person3: What about Bar Stool for American's Behind

Person4: that's America's Ass!

Person4: it would get meme points lol

Person3: A Stool from America's Ass

...

Person2: we have to decide ppl

When distinguishing teams above and below the median viability score, the general trends we discover in the more extreme splits hold true. More use of exclusive language and second person pronouns, which directly engage with other team members, are positive indicators of team viability greater than the median. The presence of argument discourse markers, and exhibitions of embarrassment, are correlated with team viability lower than the median. Two unusual features are specifically significant for the median viability split. Dismissing collaborators is a positive indicator for teams above median viability, which further corroborates the overarching trend that directly engaging with other team members — even when such engagement is active disagreement — is correlated with higher viability, since direct engagement allows team members to feel that the team actively communicates on issues. Finally, relationship-oriented language is interestingly correlated with teams below the median viability. We note that this LIWC category contained words that were colloquial and could easily be interpreted negatively, e.g. "give advice" and "bro". While these

words could be used positively in some situations, we expect that their negative interpretation may be disproportionately more impactful on team viability, while their positive interpretation may have little to no impact.

3.4.3 Differentiating team viability over time

To answer RQ6, we further investigate whether our model can be used to classify team viability over time. Specifically, we are curious about whether mechanisms behind team viability are time dependent or generalizable over time, i.e. is a team interaction - viability model learnt from team interactions at a certain stage applicable to classification of team viability at another stage?

The dataset we use is well-suited to study this research question since in the experiment design, people are asked to engage in multiple rounds of tasks (round 1 to round 4), and for each round they join newly-convened teams with teammates they do not recognize.

To shed light on this question, we ran a series of ablative experiments where for $j \in [1, 4]$, we trained a logistic regression model using team interactions from all rounds except round j and tested the model on round j , i.e. can we use interaction patterns mined from rounds other than j to predict team viability at round j ? Under this setting, we expect to observe rather consistent viability classification performance should the relationship between viability and team dynamics be static, otherwise there will be a significant performance drop since signals from round j are not encoded in the trained model.

In other words, if the behaviors that are associated with high or low viability are stable, we would expect the model trained on a team's first meeting to still function well on their fourth meeting; if the behaviors are not stable, we would expect the model to have weaker performance when run on a different meeting than its training set. We ran separate experiments for all features, computationally derived features and human labeled features with varying decision thresholds of viability scores. Additionally, we ensure there are equal numbers of training/testing samples (300 training, 60 testing) across trials to provide a fair comparison.

We present the results in Fig 3.5. As can be observed, in the case of the 10% and 50% split, there is not much deviation in the model's performance between different rounds used to train the model whether all features, computationally derived or human labeled features are used. Classification performance is stable around AUC of 0.8 for all features and algorithm derived features while around AUC of 0.65 for human labeled features. For the 90% viability split, we do observe a large drop from AUC close to 1.0 to AUC 0.8 for human labeled features at rounds 2 and 3 and for computationally derived features at round 4, yet there is no clear monotonic trend between performance and testing round indicating the variation is likely due to noise. Moreover, classification performance for all features remains relatively stable around 0.9 under the 90% split. Thus we conclude that our proposed model is rather robust regardless of the length of time team members have engaged in the work, which provides evidence that the relationship between viability and team interaction is static

over time.

3.4.4 Differentiating Initial and Reconvened Team Viability

As mentioned in Section 3.3.1, the dataset we used has a unique property of reconvened teams [244, 245] in addition to the independent rounds we have just discussed. In reconvened rounds, team members were aware that they were collaborating with the same team members for more than one separate round. These reconvened teams have not appeared in our training or testing sets so far to avoid any potential confounds, however, here we are interested to investigate whether our model will perform differently under the reconvened scenario, so as to answer RQ6 from another perspective. Specifically, we trained the viability classification model using all the masked interactions (300 interactions), and test on unmasked initial interaction (133 rounds) and reconvened interaction (132 rounds), for the three feature groups under the 10%, 50% and 90% division of viability. The result is reported in Figure 3.6.

We find that when using all features and computationally derived features, classifying team viability for reconvened interactions achieves much lower performance compared to classifying team viability for the initial interactions when we use the 10% and 50% division: for example, for the 10% split, computational features achieve 0.81 AUC when classifying the initial interaction but only 0.69 AUC when reconvened. Given prior analysis where we have shown that patterns behind team viability is rather consistent over time for 10% and 50% division, we conclude that the result indicates that computationally derived features are not so consistent at differentiating team viability in the reconvened condition, which may be a result of the model failing to capture signals embedded in the fact that team members know they are working with the same group of people for the reconvened rounds. Meanwhile, at the 90th percentile split, the viability classification performance remains relatively stable, indicating that the top decile viable teams have behavior patterns that remain consistent despite knowing they are working with the same group.

Interestingly, we find that for human labeled features, the viability classification performances for the initial and reconvened interactions are identical or even increase, indicating that human labeled features are perhaps more robust at classifying reconvened interactions.

3.4.5 Differentiating viability based on different amounts of interaction

Finally, to answer RQ7, we performed an additional analysis using our computationally derived features to test how little of the chat discourse is necessary to effectively differentiate teams' viabilities. Because we will need to consider smaller windows of the chat interactions, we can only apply the features that were derived computationally; the method with which we produced human labeled features was only performed once at the end of the whole chat interaction. We expect that the group interactions are dynamic and may change over the course of completing the task. For this study, since the chat histories we are analyzing are about 10 minutes long, we consider windows of

data at 10 second increments, totaling to 60 trials. We start by calculating our features on the first ten seconds and evaluating the performance of our model; next, we take the data within the first 20 second window, calculate features, and evaluate the AUC score; this continues until the full 10 minutes of data is used. All trials were performed only with our Logistic Regression model with three different viability threshold splits: 10%, 50%, and 90%.

In Figure 3.7, we see that when using the 10th and 50th percentile splits, the rate of improvement is gradual, and reaches consistent performance when at least 200 seconds of chat data is used. In these two cases, our approach obtains 95% of the full accuracy with at least 35% of the entire interaction. The performance trend when analyzing the top decile teams (90% split) is notably different; as the number of seconds of data increases, the AUC score rapidly increases and converges to steady performance as soon as when the training procedure considers at about 70 seconds of chat data. With only 12% of the chat interaction, we achieve 95% of the full accuracy when differentiating the top decile teams. This could suggest that the teams in the top decile of team viability exhibit some behavior earlier on that strongly contributes to predicting their final team viability. Regardless of the viability threshold used to divide the teams, we see that our model can consistently predict low v. high team viability much earlier in the process than at the full 10 minute mark.

To understand why this small window of chat data early in the collaboration can be predictive of the team viability by the end of the interaction, we compare the performance of consecutive disjoint windows of data to capture behaviors at different times in the interaction. Because the model achieved 95% of the full accuracy in about 200 seconds for the 10th and median viability split, we divide the ten minute interaction into three 200 second windows. Whereas the previous thin slicing analysis was done by considering cumulative windows of data, this time, none of the windows of data for each trial share any data. Looking at Figure 3.8, we see that the performance for all three viability splits remains relatively consistent over the three windows. This suggests that a small slice of data in the initial stages of the interaction is effective because by 200 seconds, team members start to exhibit consistent signals of behaviors that influence viability. Latter windows in the middle and end of the interaction continue to have the presence of these behaviors, hence the relatively consistent performance scores.

3.5 Discussion

3.5.1 Research Implications

The strong performance of our models in classifying teams by viability suggests that team viability largely is driven by underlying mechanisms that are common across different teams and repeated interactions of the same team. In particular, we find that work patterns and the language usage, which can be computationally derived, are effective in differentiating high and low viability teams

with high accuracy of $AUC = 0.76$ when equally splitting the data, and $AUC \geq 0.9$ when differentiating the top decile of high viability teams from the rest. Furthermore, the considerably high accuracy for classifying the teams with the top decile viability score and that only two features are shared as being significant across all decision thresholds of viability score suggest that the behavior of the most viable teams are notably different from that of the rest of the teams, with lowest to moderate viability. The relative decrease in the performance when classifying relative to the median or 10th percentile viability score implies that differentiating teams with an average level of viability and those with the lowest viability is more challenging. While all teams with the highest viability in our dataset are viable for the same reasons, indicated by the common signals of active contribution, equal division of labor, more accessible, objective, and positive language, less viable teams appear to reach that state for differing reasons, not as consistently as the former.

We also note that a majority of our significant features in all three classification viability thresholds (10th, 50th, 90th) were computationally derived, so they can be evaluated without exposing the interaction text to another party or person and remain accurate given a fraction of each interaction. On the other hand, human-labelled features were more robust against changes in context, suggested by the consistency in differentiating team viability in the initial interactions and the reconvened rounds. Although our features are not exhaustive by any means, we suggest that automated features, which offer a consistent comparison of all teams, identify distinctive characteristics of teams depending on their viability, while the human-labelled features reflect human judgment that can detect nuanced signals of team dynamics, such as sarcastic humor, more efficiently.

3.5.2 Design Implications

Our work also has several design implications for building computer-supported collaboration platforms. For instance, in online crowd-sourced collaboration systems, we show that it is possible to differentiate high and low viability teams with even a few instances of member interaction, providing opportunities for timely intervention if necessary. Designers could potentially leverage our detection pipelines to help teams monitor their viability and build intervention tools that respond to warning signals to promote a successful collaboration experience. To illustrate this, if the system recognizes signals associated with low team viability (e.g., overuse of negation, absence of second person pronouns), the system can remind the teams to take steps for positive behavior change.

Such systems can help improve the sustainability and longevity of teams in the workplace. Today, many teams rely on human leadership (i.e. a manager or a supervisor) to resolve team viability issues. However, such human figures can only detect and intervene after an incident has already occurred. Often times, after an incident like a heated dispute occurs, a team’s dynamics are beyond remedy. Meanwhile, the aforementioned systems can detect team viability issues in real time, allowing leadership to intervene before a major conflict erupts. This system affords teams a chance to resolve any growing issues and stay together before an irrevocable episode materializes.

The features used by these systems can also shed light on why a team’s viability is suffering and recommend tailored intervention depending on the factors for low viability. For example, if a team’s low viability can be explained by highly unequal participation, managers can subsequently address this specific issue and improve the team’s work patterns. On the other hand, if low viability of a team can be attributed to overuse of negation, a nudging reminder to be more positive may be more effective. It is important to note that a combination of behaviors may be affecting a team’s viability. Exploring different combinations may allow systems to learn the behavior patterns of a team and make more personalized recommendations for intervention. Furthermore, understanding how more viable teams behave can help provide areas for improvement for teams that are not necessarily considered not viable but simply want to improve how they work together. To illustrate, behavior associated with highest viability teams can be referenced when a team establishes a team norm.

Despite the previously mentioned beneficial use cases, these systems may also result in undesirable outcomes. To illustrate, a faulty prediction of the system can lead to the deterioration of viability going unnoticed until a major episode occurs or costly and unnecessary interventions. A more concerning implication of an automated system assessing viability is the possibility of the system adopting biases against certain communities based on distinctive language patterns, such as culturally-dependent colloquialism or accents. If the system is more sensitive to the language of certain groups of people, it may systematically and wrongly flag them as being problematic. Such an algorithmic bias would lead to discriminatory practices and inequitable treatment based on one’s cultural identity, which is both detrimental to team viability and unacceptable. Finally, the system may be hijacked for malicious or inappropriate uses. For example, the sheer amount of data that would be collected regarding individuals’ dialogue behavior raises concern for protecting one’s privacy against surveillance. Furthermore, the system could also be inappropriately applied to predict characteristics other than viability (e.g. performance), leading to flawed evaluations and perceptions of teams.

These issues motivate a variety of design principles and guidelines for usage in order to avoid unnecessary cost, inefficacy, injustices, and malicious usage. First, signals used to assess viability should be made public and should be readily audited. Further, indications of viability should not be made for individuals and attempts to measure individual viability potential should be avoided. Since viability is inherently a team property, extrapolation of assessment to individuals will be both flimsy and potentially damaging. Although it is tempting to develop recommendations with high precision, the trade-off between maximizing precision and minimizing potential abuse of targeted recommendations should be considered seriously. To resolve these issues in our own work, we are making available a version of our algorithm in which the features and their coefficients are transparently reported, and additionally, where a suitable amount of interaction material will be required to make assessments. Further discussion of ideas to respect privacy and ensure algorithmic fairness should be continued throughout the process of design, implementation, and deployment.

3.5.3 Limitations & Future Work

Considering the methods with which our dataset was gathered, there are a few limitations to the generalizability of the insights we draw from our analysis. First, the teams we analyze are relatively small in size and cannot represent the diversity and complexity in team roles and hierarchies in large organizations. Second, there may be important differences between remote collaboration and collocated work (even though the two forms demonstrate considerable characteristic similarities [45, 59]), particularly around finding common ground [190]. Prior research suggests that conflict of all types (task, affective, and process) will be detrimental to the performance of remote teams but may not have the same effects on traditional teams [111]. Thus, our approach may not fully capture interactions in collocated teams. Third, the chat interactions themselves are short in duration (10 mins for each round) and involve well-structured tasks. Coordination and problem defining in more complex, open-ended tasks may follow different dynamics than what participants did in our study, which could affect a team’s viability.

In future work, it would be interesting to extend this approach to real world cases like software engineering tasks, which is a prolonged collaboration and involves complex interactions between people and the work itself. This work focuses on remote synchronous collaboration, but whether the same insights apply to face-to-face interactions or what differences there may be are both useful areas to explore. Also, due to the size of the collected dataset, we are unable to do fine-grained analysis comparing our model under different task categories. In the future, it would be interesting to extend our analysis to different task categories.

Nevertheless, we believe our derived insights do generalize and can be of great value to understand team viability under a remote collaboration/interaction setting (which is becoming increasingly important as more collaborations are shifted online after the COVID-19 pandemic), especially for short-term team interactions where members do not know each other’s identities. Such flash interactions indeed happen often in online communities, e.g., in online forums, discussions under social media posts, and news reviews, and are also becoming common in crowdsourcing tasks. As such interactions are essentially similar in property to our collected data, we argue similar patterns are very likely to hold. Our findings and implications on team viability can help better understand user engagement in these settings.

Another possible line of analysis is delving into the cultural context of the teams, the human raters, and the computationally judged features. In the current work, all of the data are collected from US Amazon Mechanical Turk workers with moderate experience, and this sampling is likely to influence aspects of our result. Notably, the researchers attempted to code chats using the same prompts as the human raters did and found no notable difference in accuracy or inter rater reliability. Further, no stratification of the human rating data implicate sub populations within our analysis who systematically performed better. With this said, we expect these results may differ when other populations or cultural backgrounds are engaged. Finally, it would be useful to include

other factors of interpersonal communications (e.g., linguistic accommodation, between individual personality/dissimilarity, etc.) as future work to gain a deeper understanding of team viability.

3.6 Conclusion

Recognizing the importance of team viability — rather than performance — in effective collaboration, we study signals of team viability in text interaction data of remote teams. Our machine learning models using features motivated by previous literature can reliably differentiate teams that belong to the top decile, above median, and the bottom decile of viability scores. Further investigation of the models suggest that while the use of exclusive language and second person pronouns are associated with higher viability teams across thresholds, other behaviors are unique to distinguishing the top decile, such as higher average reading level, and the bottom decile, such as sadness and displays of extreme emotions. Furthermore, we demonstrate that the relationship between viability and team interaction is consistent over time, yet the pattern differs by de novo teams and reconvened teams. Finally, we find that signals from the initial moments of interaction are sufficient for successful prediction of the whole interaction, which suggests that viability can be evaluated and acted upon early. The strength of our models using automated features indicate opportunities for real time assessment of viability, which lends further space for intervention.

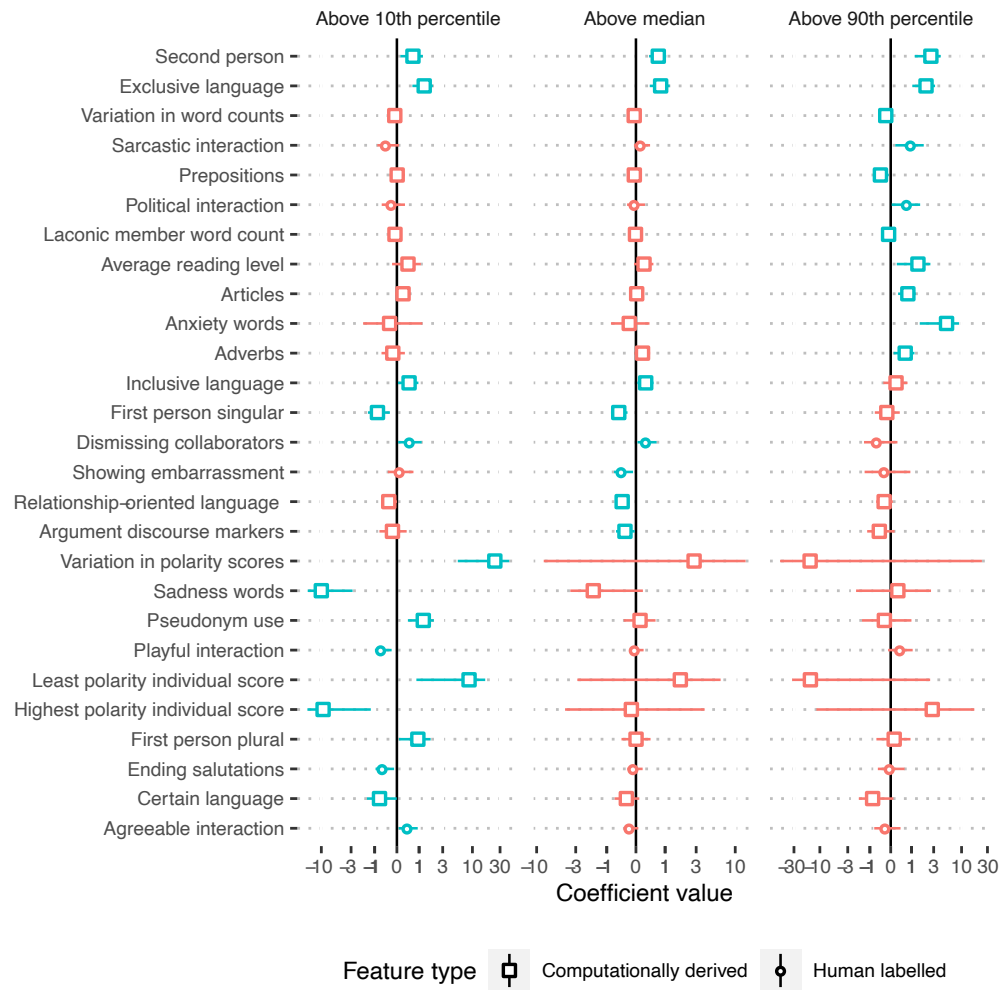


Figure 3.4: Logistic regression coefficients for features. Features with confidence interval distinguishable from zero (significant) are shown in blue, while features with confidence interval overlapping zero (insignificant) are shown in red. Features are ordered from most to least useful in classification, and they are clustered by viability split. Values are shown on a hyperbolic arcsine scale to cope with variance.

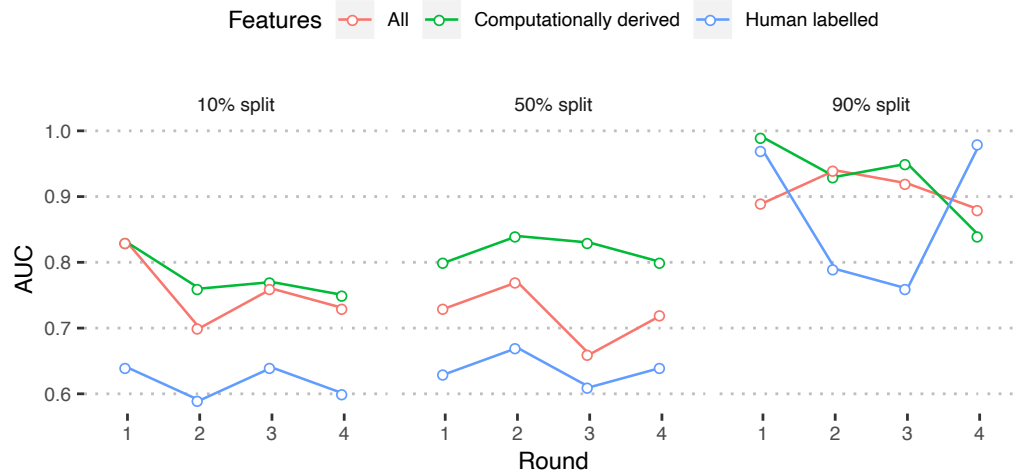


Figure 3.5: Predictive accuracy across rounds appears to not show any major trend. Notably, no confidence intervals are provided because these could not be calculated with cross validation — these results are exact given the data studied.

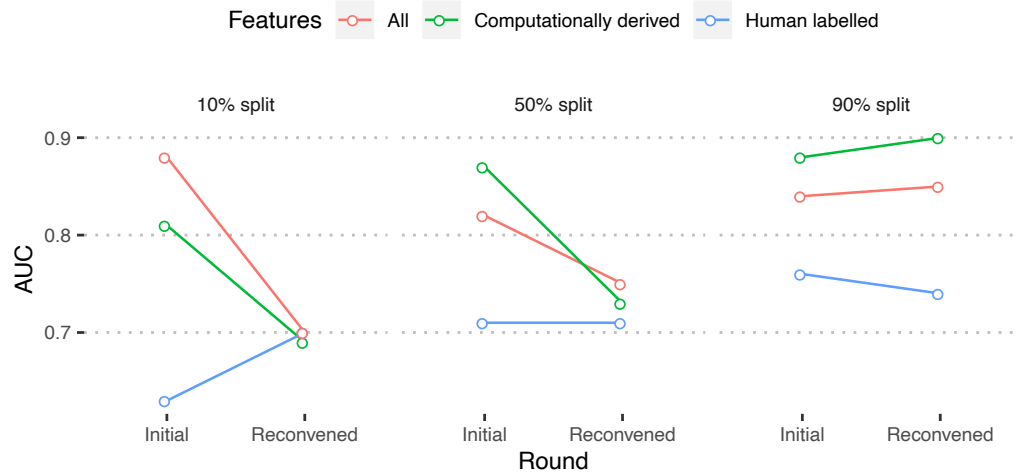


Figure 3.6: Predictive accuracy in reconvened rounds sees a substantial dip for algorithmic features in lower viability split cases, while human features appear less affected. Notably, no confidence intervals are provided because these could not be calculated with cross validation — these results are exact given the data studied.

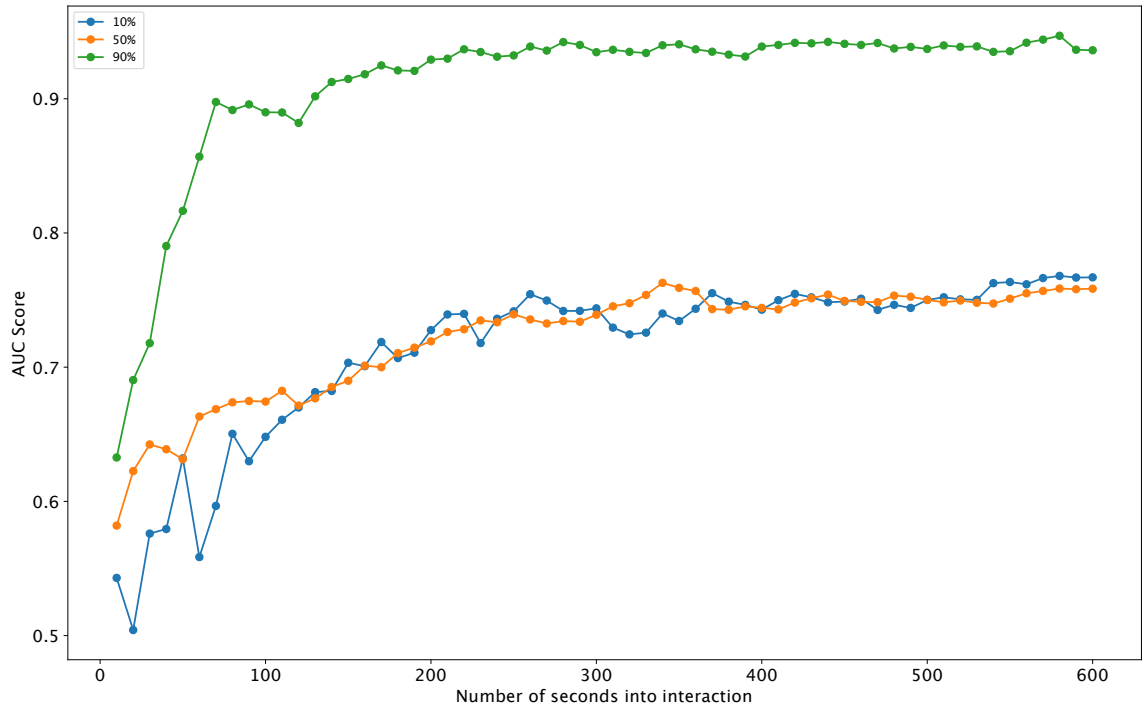


Figure 3.7: Results of training the Logistic Regression model with data from different numbers of seconds of the chat interaction when low v. high viability is split at the 10, 50, and 90 percentile viability.

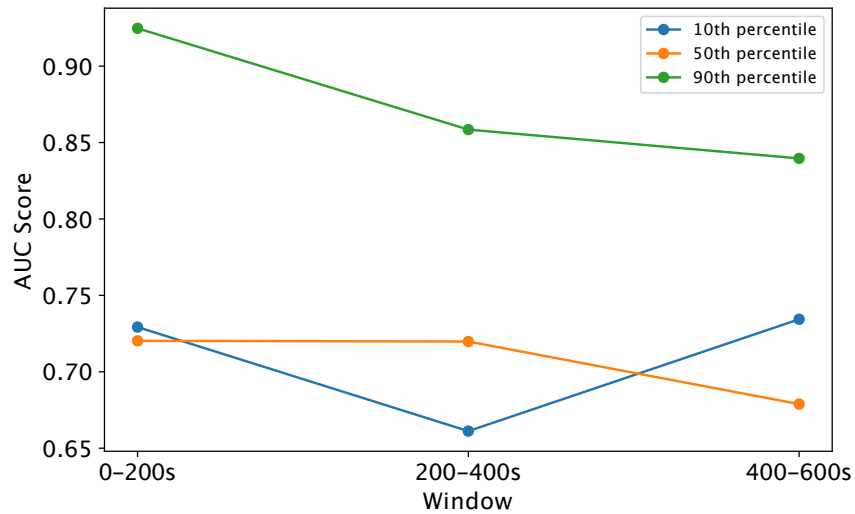


Figure 3.8: Performance of Logistic Regression model trained on disjoint windows of 200 seconds at different percentile splits (10th, 50th, 90th).

Chapter 4

Design and Evaluation of an AI tool for scientific feedback

In the previous chapter, we developed an AI method to classify and predict team viability, showing how *technical methods* can be used to further analyze, understand and even augment how people work. In this chapter, we discuss the approach of *design* in envisioning future workplace computing systems through prototyping, and evaluation. We developed a generative AI-based tool to provide scientific feedback in a peer review format to researchers, allowing them to receive timely feedback when needed. Through a mixed-method design that combines a retrospective evaluation and a prospective user study, we systematically evaluate the performance of the proposed feedback tool.

4.1 Introduction

While at Bell Laboratories in the 1940s, Claude Shannon embarked on developing a mathematical framework of information and communication [219]. Throughout this pursuit, he was faced with the challenge of naming his novel measure and considered terms such as ‘information’ and ‘uncertainty’. Shannon shared his work with John von Neumann, who quickly recognized the profound links between Shannon’s work and statistical mechanics, and proposed what later anchored modern information theory: ‘Information Entropy’ [236]. Scientific progress often rests on feedback and critique. Effective feedback among peer scientists not only elucidates and promotes the way new discoveries are made, interpreted, and communicated, but also catalyzes the emergence of new scientific paradigms by connecting individual insights, coordinating concurrent lines of thoughts, and stimulating constructive debates and disagreement [138].

However, the process of providing timely, comprehensive, and insightful feedback on scientific research is often laborious, resource-intensive, and complex [114]. This complexity is exacerbated

by the exponential growth in scholarly publications and the deepening specialization of scientific knowledge [200, 122]. Traditional avenues, such as peer review and conference discussions, exhibit constraints in scalability, expertise accessibility, and promptness. For instance, it has been estimated that peer review – one of the most major channels of scientific feedback – costs over 100M researcher hours and \$2.5B US dollars in a single year [5]. Yet at the same time, it has been increasingly challenging to secure enough qualified reviewers who can provide high-quality feedback given the rapid growth in the number of submissions [8, 33, 146, 133, 218]. For example, the number of submissions to the *ICLR* machine learning conference increased from 960 in 2018 to 4,966 in 2023.

While shortage of high-quality feedback presents a fundamental constraint on the sustainable growth of science overall, it also becomes a source of deepening global scientific inequities. Marginalized researchers, especially those from non-elite institutions or resource-limited regions, often face disproportionate challenges in accessing valuable feedback, perpetuating a cycle of systemic scientific inequities [36, 173].

Given these challenges, there is an urgent need for crafting scalable and efficient feedback mechanisms that can enrich and streamline the scientific feedback process. Adopting such advancements holds the promise of not just elevating the quality and scope of scientific research, given the concerning deceleration in scientific advancements [62, 34], but also of democratizing its access across the scientific community.

Large language models (LLMs) [40, 193, 191], especially those powered by Transformer-based architectures and pre-trained at immense scales, have opened up great potential in various applications [18, 148, 139, 235]. While LLMs have made remarkable strides in various domains, the promises and perils of leveraging LLMs for scientific feedback remain largely unknown. Despite recent attempts that explore the potential uses of such tools in areas such as automating paper screening [215], error identification [156], and checklist verification [207], we lack large-scale empirical evidence on whether and how LLMs may be used to facilitate scientific feedback and augment current academic practices.

In this work, we present the first large-scale systematic analysis characterizing the potential reliability and credibility of leveraging LLM for generating scientific feedback. Specifically, we developed a GPT-4 based scientific feedback generation pipeline that takes the raw PDF of a paper and produces structured feedback (Fig. 4.1a). The system is designed to generate constructive feedback across various key aspects, mirroring the review structure of leading interdisciplinary journals [181, 182] and conferences [209, 16, 15, 116, 185], including: 1) Significance and novelty, 2) Potential reasons for acceptance, 3) Potential reasons for rejection, and 4) Suggestions for improvement.

To characterize the informativeness of GPT-4 generated feedback, we conducted both a retrospective analysis and a prospective user study. In the retrospective analysis, we applied our pipeline on papers that had previously been assessed by human reviewers. We then compared the LLM feedback with the human feedback. We assessed the degree of overlap between key points raised by both

sources to gauge the effectiveness and reliability of LLM feedback. Furthermore, we compared the topic distributions of LLM feedback and human feedback. To enable such analysis, we curated two complementary datasets containing full-text of papers, their meta information, and associated peer reviews after 2022 ¹. The first dataset was sourced from *Nature* family journals, which are leading scientific journals covering multidisciplinary fields including biomedicine and basic sciences. Our second dataset was sourced from *ICLR* (International Conference on Learning Representations), a leading computer science venue on artificial intelligence. This dataset, although narrower in scope, includes complete reviews for both accepted and rejected papers. These two datasets allowed us to evaluate the performance of LLM in generating scientific feedback across different types of scientific writing (e.g. across fields).

For the prospective user study, we developed a survey in which researchers were invited to evaluate the quality of the feedback produced by our GPT-4 system on their authored papers. By analyzing researchers’ perspectives on the helpfulness, reliability, and potential limitations of LLM feedback, we can gauge the acceptability and utility of the proposed approach in the manuscript improvement process, and understand stakeholder’s subjective perceptions of the framework. Through recruitment over institute mailing lists, and contacting paper authors who put preprints on arXiv, we were able to collect survey responses from 308 researchers from 110 institutions in the field of AI and computational biology that come from diverse education status, experience, and institutes.

4.2 Methods

4.2.1 Overview of the *Nature* Family Journals Dataset

Several journals within the *Nature* group have adopted a transparent peer review policy, enabling authors to publish reviewers’ comments alongside the accepted papers [1]. For instance, by 2021, approximately 46% of *Nature* authors opted to make their reviewer discussions public [2]. Our dataset comprises papers from 15 *Nature* family journals, published between January 1, 2022, and June 17, 2023. We sourced papers from 15 *Nature* family journals, focusing on those published between January 1, 2022, and June 17, 2023. Within this period, our dataset includes 773 accepted papers from *Nature* with 2,324 reviews, 810 sampled accepted papers from *Nature* Communications with 2,250 reviews, and many others. In total, our dataset includes 3,096 accepted papers and 8,745 reviews (**Table 4.1**). The data were sourced directly from the *Nature* website (<https://nature.com/>).

¹Focusing data after 2022 avoids bias introduced by ‘testing on the training set’, since GPT-4, the LLM we used, is trained on data up to Sep 2021 [191].

4.2.2 Overview of the *ICLR* Dataset

The International Conference on Learning Representations (*ICLR*) is a leading publication venue in the machine learning field. *ICLR* implements an open review policy, making reviews for all papers accessible, including those for rejected papers. Accepted papers at *ICLR* are categorized into Oral presentations (top 5% of papers), Spotlight (top 25%), and Poster presentations. In 2022, *ICLR* received 3,407 submissions, which increased to 4,966 by 2023. Using a stratified sampling method, we included 55 Oral (with 200 reviews), 173 Spotlight (664 reviews), 197 Poster (752 reviews), 213 rejected (842 reviews), and 182 withdrawn (710 reviews) papers from 2022. For 2023, we included 90 Oral (317 reviews), 200 Spotlight (758 reviews), 200 Poster (760 reviews), 212 rejected (799 reviews), and 187 withdrawn (703 reviews) papers. The dataset comprises 1709 papers and 6,506 reviews in total (**Table 4.2**). The paper PDFs and corresponding reviews were retrieved using the OpenReview API (<https://docs.openreview.net/>).

4.2.3 Generating Scientific Feedbacks using LLM

We prototyped a pipeline to generate scientific feedback using OpenAI’s GPT-4 [191] (**Fig. 4.1a**). The system’s input was the academic paper in PDF format, which was then parsed with the machine-learning-based ScienceBeam PDF parser [88]. Given the token constraint of GPT-4, which allows 8,192 tokens for combined input and output, the initial 6,500 tokens of the extracted title, abstract, figure and table captions, and main text were utilized to construct the prompt for GPT-4 (**Fig. 4.11**). This token limit exceeds the 5,841.46-token average of *ICLR* papers and covers over half of the 12,444.06-token average for *Nature* family journal papers (**Table 4.4**). For clarity and simplicity, we instructed GPT-4 to generate a structured outline of scientific feedback. Following the reviewer report instructions from machine learning conferences [209, 16, 15, 116, 185] and *Nature* family journals [181, 182], we provided specific instructions to generate four feedback sections: significance and novelty, potential reasons for acceptance, potential reasons for rejection, suggestions for improvement (**Fig. 4.18**). The feedback for each paper was generated by GPT-4 in a single pass.

4.2.4 Retrospective Extraction and Matching of Comments from Scientific Feedback

To evaluate the overlap between LLM feedback and human feedback, we developed a two-stage comment matching pipeline (**Fig. 4.12**). In the first stage, we employed an extractive text summarization approach [158, 90, 176, 93]. Each feedback text, either from the LLM or a human, was processed by GPT-4 to extract a list of the points of comments raised in the text (see prompt in **Fig. 4.19**). The output was structured in a JSON (JavaScript Object Notation) format. Within this format, each JSON key assigns an ID to a specific point, while the corresponding value details the content of the point (**Fig. 4.19**). We focused on criticisms in the feedback, as they provide

direct feedback to help authors improve their papers [99]. The second stage focused on semantic text matching [78, 223, 37]. Here, we input both the JSON-formatted feedback from the LLM and the human into GPT-4. The LLM then generated another JSON output where each key identified a pair of matching point IDs and the associated value provided the explanation for the match. Given that our preliminary experiments showed GPT-4’s matching to be lenient, we introduced a similarity rating mechanism. In addition to identifying corresponding pairs of matched comments, GPT-4 was also tasked with self-assessing match similarities on a scale from 5 to 10 (**Fig. 4.20**). We observed that matches graded as “5. Somewhat Related” or “6. Moderately Related” introduced variability that did not always align with human evaluations. Therefore, we only retained matches ranked “7. Strongly Related” or above for subsequent analyses.

We validated our retrospective comment matching pipeline using human verification. In the extractive text summarization stage, we randomly selected 639 pieces of scientific feedback, including 150 from the LLM and 489 from human contributors. Two co-authors assessed each feedback and its corresponding list of extracted comments, identifying true positives (correctly extracted comments), false negatives (missed relevant comments), and false positives (incorrectly extracted or split comments). This process resulted in an F1 score of 0.968, with a precision of 0.977 and a recall of 0.960 (**Table 4.3a**), demonstrating the accuracy of the extractive summarization stage. For the semantic text matching stage, we sampled 760 pairs of scientific feedbacks: 332 comparing GPT to Human feedback and 428 comparing Human feedback. Each feedback pair was processed to enumerate all potential pairings of their extracted comment lists, resulting in 12,035 comment pairs. Three co-authors independently determined whether the comment pairs matched, without referencing the pipeline’s predictions. Comparing these annotations with pipeline outputs yielded an F1 score of 0.824, a recall of 0.878, and a precision of 0.777 (**Table 4.3b**). To assess inter-annotator agreement, we collected three annotations for 800 randomly selected comment pairs. Given the prevalence of non-matches, we employed stratified sampling, drawing 400 pairs identified as matches by the pipeline and 400 as non-matches. We then calculated pairwise agreement between annotations and the F1 score for each annotation against the majority consensus. The data showed 89.8% pairwise agreement and an F1 score of 88.7%, indicating the reliability of the semantic text matching stage.

4.2.5 Evaluating Specificity of LLM Feedback through Review Shuffling

To evaluate the specificity of the feedback generated by the LLM, we compared the overlap between human-authored feedback and shuffled LLM feedback. For papers published in the *Nature* journal family, the LLM-generated feedback for a given paper was randomly paired with human feedback for a different paper from the same journal and *Nature* root category. These categories included physical sciences, earth and environmental sciences, biological sciences, health sciences, and scientific community and society. If a paper was classified under multiple categories, the shuffle algorithm paired it with another paper that spanned the same categories. For the *ICLR* dataset, we compared

human feedback for a paper with LLM feedback for a different paper, randomly selected from the same conference year, either *ICLR* 2022 or *ICLR* 2023. This shuffling procedure was designed to test the null hypothesis: if LLM mostly produces generic feedback applicable to many papers, then there would be little drop in the pairwise overlap between LLM feedback and the comments from each individual reviewer after the shuffling.

4.2.6 Overlap Metrics for Retrospective Evaluations and Control

In the retrospective evaluation, we assessed the pairwise overlap of both GPT-4 vs. Human and Human vs. Human in terms of hit rate (**Fig. 4.2**). The hit rate, defined as the proportion of comments in set A that match those in set B , was calculated as follows:

$$\text{Hit Rate} = \frac{|A \cap B|}{|A|}$$

To facilitate a direct comparison between the hit rates of GPT-4 vs. Human and Human vs. Human, we controlled for the number of comments when measuring the hit rate for Human vs. Human. Specifically, we considered only the first N comments made by the first human (i.e., the human comments used as set A) for matching, where N is the number of comments made by GPT-4 for the same paper. The results, with and without this control, were largely similar across both the *ICLR* dataset for different decision outcomes (**Fig. 4.13,4.16**) and the *Nature* family journals dataset across different journals (**Fig. 4.14,4.15,4.16**). To examine the robustness of the results across different set overlap metrics, we also evaluated three additional metrics: the Szymkiewicz–Simpson overlap coefficient, the Jaccard index, and the Sørensen–Dice coefficient. These were calculated as follows:

$$\text{Szymkiewicz–Simpson Overlap Coefficient} = \frac{|A \cap B|}{\min(|A|, |B|)}$$

$$\text{Jaccard Index} = \frac{|A \cap B|}{|A \cup B|}$$

$$\text{Sørensen–Dice Coefficient} = \frac{2|A \cap B|}{|A| + |B|}$$

Results on these additional metrics suggest that our findings are robust on different set overlap metrics: the overlap GPT-4 vs. Human appears comparable to those of Human vs. Human both with and without control for the number of comments (**Fig. 4.6**).

4.2.7 Characterizing the comment aspects in human and LLM feedback

We curated an annotation schema of 11 key aspects to identify and measure the prevalence of these aspects in human and LLM feedback. This schema was developed with a focus on the *ICLR* dataset,

due to its specialized emphasis on Machine Learning. Each aspect was defined by its underlying emphasis, such as novelty, research implications, suggestions for additional experiments, and more. The selection of these 11 key aspects was based on a combination of the common schemes identified in the literature within the machine learning domain [32, 222, 132, 213], comments from machine learning researchers, and initial exploration by the annotators. From the *ICLR* dataset, a random sample of 500 papers was selected to ensure a broad yet manageable representation. Using our extractive text summarization pipeline, we extracted lists of comments from both the LLM and human feedback for each paper. Each comment was then annotated according to our predefined schema, identifying any of the 11 aspects it represented (**Table 4.5, 4.6, 4.7**). To ensure annotation reliability, two researchers with a background in machine learning performed the annotations.

4.2.8 Prospective User Study and Survey

We conduct a prospective user study to further validate the effectiveness of leveraging LLMs to generate scientific feedback. To facilitate our user study, we launched an online Gradio demo [4] of the aforementioned generation pipeline, accessible at a public URL (**Fig. 4.17**). Users are prompted to upload a research paper in its original PDF format, after which the system delivers the feedback comments to user’s email. We ask users to only upload papers published after 9/2021 to ensure the papers are never seen by GPT-4 during training (the cutoff date of GPT-4 training corpora is 9/2021). We have also incorporated an ethics statement to discourage the direct use of LLM content for any review-related tasks. After the feedback is generated and sent to users, users are asked to fill a 6-page survey (Figure 4.4), which includes 1) author background information, 2) review situation in author’s area, 3) general impression of LLM feedback, 4) detailed evaluation of LLM feedback, 5) comparison with human peer review (if available), and 6) additional questions and feedback, which systematically investigates human evaluations of different aspects of LLM reviews.

We carefully designed the user study instructions and questions by following the standards and best practices in human computer interaction (HCI) user research [194, 131]. Following prior HCI user study designs [192, 148, 147], our survey first focus on high-level impression (e.g. the overall ‘helpfulness’ of the system) then dive into more specific and detailed evaluations. We adopted the commonly used 5 point Likert Scale in the survey as it allows for a wider range of responses, making it suitable for exploring subtler differences in opinions or attitudes than 3 point Likert Scale [194, 77, 214, 228, 148]. The phrasing of our survey questions followed the phrasings used in previous successful HCI studies to reduce potential bias [194, 131, 148]. Coauthors with expertise in HCI and user studies drafted the survey. Then all the authors discussed and iterated on the questions and descriptions to make sure our questions are objective and unbiased. After we designed initial version of the user study survey, interface and instructions, we conducted a pilot study with 15 participants to further validate the design. We first asked participants to complete the full user study on their own following the study’s instructions, then interviewed them to discuss whether there are any unclear

parts in the study. We walked through instructions, questions and scales in detail, and noted any part participants reported confusion or have misunderstanding. We then iterated on the design and tested with the participants until no more issues were reported.

We further validated the survey design by quantifying the inter-rater reliability of the survey responses on the LLM generated scientific feedback. Specifically, three coauthors on this work chose four recent papers (after 2022) that they are familiar with, and then independently completed the survey for GPT-4’s feedback on each paper. On the paper specific survey questions, the responses of the three coauthors achieved average Krippendorff’s Alpha of 0.76, showing consistency among raters’ evaluations on the same paper.

In our formal survey data collection, we recruited the participants by (1) circulating the survey invitation through relevant institute mailing lists, and 2) contacting all authors affiliated with US academic institutions who have published at least one preprint on arXiv in the field of computer science and computational biology during January to March, 2023, provided we can find an associated email address within the first three pages of their arXiv PDF preprints. We acknowledge the potential for self-selection bias among our study participants, a common issue encountered in many online surveys and HCI studies [31]. However, we believe the participants in our study still constitute a relevant and meaningful cohort, since researchers who opted into our study likely represent those who are curious about and interested in the capabilities of LLMs to improve their papers, which are also more likely to become potential early adopters of these tools. This population is broadly consistent with the typical target of HCI user study research, which predominantly focuses on prospective users of new tools [194, 131].

We calculated Cronbach Alpha, a widely used measure of internal consistency, on survey questions related to the ‘helpfulness’ of LLM generated scientific feedback. This metric quantifies how well the survey items measure the same concept or construct. The Cronbach Alpha score is 0.81, supporting the internal consistency of our survey responses.

The study has been approved by Stanford University’s Institutional Review Board.

4.3 Results: Retrospective Evaluation

To evaluate the quality of LLM feedback retrospectively, we systematically assess the content overlap between human feedback given to submitted manuscripts and the LLM feedback using two large-scale datasets. The first dataset, sourced from *Nature* family journals, includes 8,745 comments from human reviewers for 3,096 accepted papers across 15 *Nature* family journals, including *Nature*, *Nature Biomedical Engineering*, *Nature Human Behaviour*, and *Nature Communications* (**Table 4.1**). The second dataset comprises 6,505 comments from human reviewers for 1,709 papers from the International Conference on Learning Representations (*ICLR*), a leading venue for artificial intelligence research in computer science (**Table 4.2**). These two datasets complement each other: the

first dataset (*Nature* portfolio journals) spans a broad range of prominent journals across various scientific disciplines and impact levels, thereby capturing both the universality and variations in human-based scientific feedback. The second dataset (*ICLR*) provides an in-depth perspective of scientific feedback within leading venues of a rapidly evolving field – machine learning. Importantly, this second dataset includes expert feedback on both accepted and rejected papers.

We developed a retrospective comment matching pipeline to evaluate the overlap between feedback from LLM and human reviewers (**Fig. 4.1b**). The pipeline first performs extractive text summarization [158, 90, 176, 93] to extract the comments from both LLM and human-written feedback. It then applies semantic text matching [78, 223, 37] to identify shared comments between the two feedback sources. We validated the pipeline’s accuracy through human verification, yielding an F1 score of 96.8% for extraction (**Table 4.3a**) and 82.4% for matching (**Table 4.3b**). As additional validation, four human annotators independently assessed LLM’s feedback and human reviews for 400 *Nature* portfolio papers. They also found the overlap between LLM’s feedback and human reviews (28% hit rate) to be higher than the overlap between two human reviews (25% hit rate).

4.3.1 LLM feedback significantly overlaps with human-generated feedback

We began by examining the overlap between LLM feedback and human feedback on *Nature* family journal data (**Table 4.1**). More than half (57.55%) of the comments raised by GPT-4 were raised by at least one human reviewer (**Fig. 4.5a**). This suggests a considerable overlap between LLM feedback and human feedback, indicating potential accuracy and usefulness of the system. When comparing LLM feedback with comments from each individual reviewer, approximately one third (30.85%) of GPT-4 raised comments overlapped with comments from an individual reviewer (**Fig. 4.2a**). The degree of overlap between two human reviewers was similar (28.58%), after controlling for the number of comments.

This indicates that the overlap between LLM feedback and human feedback is comparable to the overlap observed between two human reviewers. We further stratified these overlap results by academic journals (**Fig. 4.2c**). While the degree of overlap between LLM feedback and human comments varied across different academic journals within the *Nature* family — from 15.58% in *Nature Communications Materials* to 39.16% in *Nature* — the overlap between LLM feedback and human feedback comments largely mirrored the overlap found between two human reviewers. The robustness of the finding further indicates that scientific feedback generated from LLM is similar to what researchers could get from peer reviewers. As additional sensitivity experiments, we found that the overlap analysis were consistent across other statistical metrics including Szymkiewicz–Simpson overlap coefficient, Jaccard index and Sørensen–Dice coefficient (**Fig. 4.6**). For a subset of 408 *Nature* family publications, we also obtained associated Research Square preprints. The GPT-4’s feedback on these preprints also significantly overlaps the reviewers’ comments—55.4% of the points

raised by GPT-4 were raised by at least one human reviewer (**Fig. 4.7**).

In parallel experiments, we investigated the comment overlap between LLM feedback and human feedback on *ICLR* papers data (**Table 4.2**), and the results were largely similar. A majority (77.18%) of the comments raised by GPT-4 were also raised by at least one human reviewer (**Fig. 4.5b**), indicating considerable overlap between LLM feedback and human feedback. When comparing LLM feedback with comments from each individual reviewer, more than one third (39.23%) of GPT-4 raised comments overlapped with comments from an individual reviewer (**Fig. 4.2b**). The overlap between two human reviewers was similar (35.25%), after controlling for the number of comments (**Fig. 4.6**). We further stratified these overlap results by the decision outcomes of the papers (**Fig. 4.2d**). Similar to results over *Nature* family journals, we found that the overlap between LLM feedback and human feedback comments largely mirrored the overlap found between two human reviewers.

In addition, as *ICLR* dataset includes both accepted and rejected papers, we conducted stratification analysis and found a correlation between worse acceptance decisions and larger overlap in *ICLR* papers. Specifically, papers accepted with oral presentations (representing the top 5% of accepted papers) have an average overlap of 30.63% between LLM feedback and human feedback comments. The average overlap increases to 32.12% for papers accepted with a spotlight presentation (the top 25% of accepted papers), while rejected papers bear the highest average overlap at 47.09%. A similar trend was observed in the overlap between two human reviewers: 23.54% for papers accepted with oral presentations (top 5% accepted papers), 24.52% for papers accepted with spotlight presentations (top 25% accepted papers), and 43.80% for rejected papers. This suggests that rejected papers may have more apparent issues or flaws that both human reviewers and LLMs can consistently identify. Additionally, the increased overlap between LLM feedback and actual human reviewer feedback for rejected papers indicates that LLM feedback could be particularly constructive and formative for papers that require more substantial revisions to be accepted. Indeed, by raising these concerns earlier in the scientific process before review, these papers and the science they report may be improved.

Since our primary analyses focus on GPT-4, we wanted to assess how consistent is GPT-4 feedback over time. Accordingly, we evaluated the March 2023 and June 2023 checkpoints of GPT-4 on *Nature* family and *ICLR* papers, and found that the two GPT-4 checkpoints produced very consistent feedback (**Fig. 4.8**). We also evaluated the ability of two state-of-the-art open-source LLMs: Llama 2 (70 billion parameters) and Falcon (40 billion parameters), to provide feedback. Both open-source LLMs perform significantly worse than GPT-4 (**Fig. 4.8**).

4.3.2 LLM could generate non-generic feedbacks.

Is it possible that LLM merely generates generic feedback applicable to multiple papers? A potential null model is that LLM mostly produces generic feedback applicable to many papers. To test this

hypothesis, we performed a shuffling experiment aimed at verifying the specificity and relevance of LLM generated feedback. For each paper in the *Nature* family journal data, the LLM feedback was shuffled for papers from the same journal and within the same *Nature* category. If the LLM were producing only generic feedback, we would observe no decrease in the pairwise overlap between shuffled LLM feedback and human feedback. In contrast, the pairwise overlap significantly decreased from 30.85% to 0.43% after shuffling (**Fig. 4.2a**). A similar drop from 39.23% to 3.91% was observed on *ICLR* (**Fig. 4.2b**). These results suggest that LLM feedback is paper-specific.

4.3.3 LLM is consistent with humans on major comments

What characteristics do LLMs’ comments exhibit? What are the distinctive features of the human comments that align with LLMs’? Here we evaluate the unique characteristics of comments generated by LLMs. Our analysis revealed that comments identified by multiple human reviewers are more likely to be echoed by LLMs. For instance, in the *Nature* family journal data (**Fig. 4.2e**), a comment raised by a single human reviewer had an 11.39% chance of being identified by LLMs. This probability increased to 20.67% for comments raised by two reviewers, and further to 31.67% for comments raised by three or more reviewers. A similar trend was observed in the *ICLR* data (**Fig. 4.2f**), where the likelihood of LLMs identifying a comment increased from 15.39% for a single reviewer to 26.21% for two reviewers, and 39.33% for three or more reviewers. These findings suggest that LLMs are more likely to identify common issues or flaws that are consistently recognized by multiple human reviewers, compared to specific comments raised by a single reviewer. This alignment of LLM with human perspectives indicates its ability to identify what is generally considered as major or significant issues.

We further examined the likelihood of LLM comments overlapping with human feedback based on their position in the sequence, as earlier comments in human feedback (e.g. “concern 1”) may represent more significant issues. To this end, we divided each human reviewer’s comment sequence into four quarters within the *Nature* journal data (**Fig. 4.2g**). Our findings suggest that comments raised in the first quarter of the review text are most likely (21.23%) to overlap with LLM comments, with subsequent quarters revealing decreasing likelihoods (16.74% for the second quarter). Similar trends were observed in the *ICLR* papers data, where earlier comments in the sequence showed a higher probability of overlap with LLM comments (**Fig. 4.2h**). These findings further support that LLM tends to align with human perspectives on what is generally considered as major or significant issues.

4.3.4 LLM feedback emphasizes certain aspects more than humans

We next analyzed whether certain aspects of feedback are more/less likely to be raised by the LLM and human reviewers. We focus on *ICLR* for this analysis, as it’s more homogeneous than *Nature* family journals, making it easier to categorize the main aspects of review. Drawing on

existing research in peer review literature within the machine learning domain [32, 222, 132, 213], we developed a schema comprising 11 distinct aspects of comments. We then performed human annotation on a randomly sampled subset (see Methods section).

Fig. 4.3 presents the relative frequency of each of the 11 aspects of comments raised by humans and LLM. LLM comments on the implications of research 7.27 times more frequently than humans do. Conversely, LLM is 10.69 times less likely to comment on novelty than humans are. While both LLM and humans often suggest additional experiments, their focuses differ: humans are 6.71 times more likely than LLM to request more ablation experiments, whereas LLM is 2.19 times more likely than humans to request experiments on more datasets. These findings suggest that the emphasis put on certain aspects of comments varies between LLMs and human reviewers. This variation highlights the potential advantages that a human-AI collaboration could provide. Rather than having LLM fully automate the scientific feedback process, humans can raise important points that LLM may overlook. Similarly, LLM could supplement human feedback by providing more comprehensive comments.

4.4 Results: Prospective User Study and Survey

Taken together, our retrospective evaluations above suggest that LLMs can generate scientific feedback that focuses on similar aspects as human reviewers. Yet, consistency in concerns is only one of the many factors contributing to the utility of scientific feedback. Recent studies in human-AI interaction have also identified additional factors that individuals may consider when evaluating and adopting AI-based tools [148], prompting us to ask: how do scientific researchers respond to feedback generated by LLMs? Thus, we launched a survey study on 308 researchers from 110 institutions who opted in to receive LLM-generated scientific feedback on their own papers, and were asked to evaluate its utility and performance. While our sampling approach is subject to biases of self-selection, the data can provide valuable insights and subjective perspectives from researchers complementing our retrospective analysis [175, 211]. The results from the user study are illustrated in **Fig. 4.4**.

Our user study provides additional evidence that is largely consistent with retrospective evaluations. First, the user study survey results corroborate the findings from the retrospective evaluation on significant overlaps between LLM feedback and human feedback: more than 70% of participants think there is at least “partial alignment” between LLM feedback and what they think/would expect on the significant points and issues with their paper, and 35% of participants think the alignment is considerable or substantial (**Fig. 4.4b**). Second, the survey study further corroborates the findings from the automated evaluation on the ability of the language model to generate non-generic feedback: 32.9% of participants think our system-generated feedback is “less specific than many, but more specific than some peer reviewers”, while 17.3% and 14% think it is “about as specific as

peer reviewers", or "more specific than many peer reviewers", further corroborating that LLMs can generate non-generic reviews (**Fig. 4.4d**).

4.4.1 Researchers find LLM feedback helpful

Participants were surveyed about the extent to which they found the LLM feedback helpful in improving their work or understanding of a subject. The majority responded positively, with over 50.3% considering the feedback to be helpful, and 7.1% considering it to be very helpful (**Fig. 4.4a**). When compared with human feedback, while 17.5% of participants considered it to be inferior to human feedback, 41.9% considered it to be less helpful than many, but more helpful than some human feedback. Additionally, 20.1% considered it to be about the same level of helpfulness as human feedback, and 20.4% considered it to be even more helpful than human feedback (**Fig. 4.4c**). Our evaluation also revealed that the perceptions of alignment and helpfulness were consistent across various demographic groups. Individuals from different educational backgrounds, ranging from undergraduate to postgraduate levels, found the feedback equally helpful and aligned with human feedback. Similarly, whether an experienced or a novice researcher, participants across the spectrum of publishing and reviewing experience reported similar levels of satisfaction and utility from the LLM based feedback, indicating that LLM based feedback tools could potentially be helpful to a diverse range of population (**Fig. 4.9, 4.10**).

In line with the helpfulness of the system, 50.5% of survey participants further expressed their willingness to reuse the system (**Fig. 4.4g**). The participants expressed optimism about the potential improvements that continued use of the system could bring to the traditional human feedback process (**Fig. 4.4e, f**). They believe that the LLM technology can further refine the quality of reviews and possibly introduce new capabilities. Interestingly, the evaluation also revealed that participants believe authors are more likely to benefit from LLM based feedback than other stakeholders such as reviewers, and area chairs (**Fig. 4.4h**). Many participants envisioned a timely feedback tool for authors to receive comments on their papers in a timely manner, e.g. one participant wrote, "The review took five minutes and was of a reasonably high quality. This can tremendously help authors to receive a fast turnaround feedback and help in polishing their submissions." Another participant wrote, "After writing a paper or a review, GPT could help me gain another perspective to re-check the paper."

4.4.2 LLM could generate novel feedback not mentioned by humans.

Beyond generating feedback that aligns with humans, our results also suggest that LLM could potentially generate useful feedback that has not been mentioned by humans, e.g., 65.3% of participants think at least to some extent LLM feedback offers perspectives that have been overlooked or underemphasized by humans. Several participants mentioned that:

- "It consists more points, covering aspects which human may forget to think about."

- “It actually highlighted a few limitations which human reviewers didn’t point out to, but as authors we were aware of it and were expecting it. But this GPT figured out some of them, so that’s interesting.”
- “The GPT-generated review suggested me to do visualization to make a more concrete case for interpretability. It also asked to address data privacy issues. Both are important, and human reviewers missed this point.”

4.4.3 Limitations of LLM feedback

Study participants also discussed limitations of the current system. The most important limitation is its ability to generate specific and actionable feedback, e.g.

- “Potential Reasons are too vague and not domain specific.”
- “GPT cannot provide specific technical areas for improvement, making it potentially difficult to improve the paper.”
- “The reviews crucially lacked much in-depth critique of model architecture and design, something actual reviewers would be able to comment on given their likely considerable experience in fields closely related to the focus of the paper.”

As such, one future direction to improve the LLM based scientific feedback system is to nudge the system towards generating more concrete and actionable feedback, e.g. through pointing to specific missing work, experiments to add. As one participant nicely summarized:

- “(large language model generated) reviews were less about the content and more about the testing regime as well as less ML details-focused, but this is okay as it still gave relevant and actionable advice on areas of improvement in terms of paper layout and presenting results. GPT-generated reviews are especially useful here when less-experience authors may leave out details on implementation and construction or forget to thoroughly explain testing regime by providing pointers on areas to polish the paper in, potentially decreasing the number of review cycles before publication.”

4.5 Discussion and Conclusion

In this study, we characterized the usefulness and reliability of LLM in scientific evaluation by building and evaluating an LLM-based scientific feedback generation framework. Through a combination of retrospective (comparing LLM feedback with human feedback from the peer review process) and prospective evaluation design (user study with researchers), we have seen a substantial level of overlap and positive user perceptions regarding the usefulness of LLM feedback. Furthermore, in evaluating user perceptions, we found that a majority of participants regarded LLM feedback as useful in the manuscript improvement process, and sometimes LLM could bring up novel points not covered by humans. The positive feedback from users highlights the potential value and utility

of leveraging LLM feedback as a valuable resource for authors seeking constructive feedback and suggestions for enhancing their manuscripts. This could be especially helpful for researchers who lack access to timely quality feedback mechanisms, e.g., researchers from traditionally underprivileged regions who may not have resources to access conferences, or even peer review (their works are much more likely than those of “mainstream” researchers to get desk rejected by journals and thus seldom go through the peer review process [173]). For others, the framework could be used as a mechanism for authors to self-check and improve their work in a timely manner, especially in an age of exponentially growing scientific papers and increasing challenges to secure timely and quality peer reviewer feedback. Our analysis suggests that people from diverse educational backgrounds and publishing experience can find the LLM scientific feedback generation framework useful (**Supp. Fig. 4.9,4.10**).

Despite the potential of LLMs in providing timely and helpful scientific feedback, it is important to note that expert human feedback will still be the cornerstone of rigorous scientific evaluation. As demonstrated in our findings, our analysis reveals limitations of the framework, e.g., LLM is biased towards certain aspects of scientific feedback (e.g., “add experiments on more datasets”), and sometimes feels “generic” to the authors (while participants also indicate that quite often human reviewers are “generic”). While comparable and even better than some reviewers, the current LLM feedback cannot substitute specific and thoughtful human feedback by domain experts.

It is also important to note the potential misuse of LLM for scientific feedback. We argue that LLM feedback should be primarily used by researchers identify areas of improvements in their manuscripts prior to official submission. It is important that expert human reviewers should deeply engage with the manuscripts and provide independent assessment without relying on LLM feedback. Automatically generating reviews without thoroughly reading the manuscript would undermine the rigorous evaluation process that forms the bedrock of scientific progress.

More broadly, our study contributes to the recent discussions on the impacts of LLM and generative AI on existing work practices. Researchers have discussed the potential of LLM to improve productivity [188, 199], creativity [91], and facilitate scientific discovery [242]. We envision that LLM and generative AI, if deployed responsibly, could also potentially bring a paradigm change to how researchers conduct research, collaborate, and provide evaluations, influencing the way science and technology advance. Our work brings a preliminary investigation into such potentials through a concrete prototype for scientific feedback.

There are several limitations to our study that are important to highlight. First, our results are based on one specific instantiation of scientific feedback from LLM, i.e., our framework is based on the GPT-4 model, enabled by a specific prompt. While we have spent significant efforts in improving the performance of our GPT-4 feedback pipeline (and achieved reasonable utility), the results should be interpreted as a lower bound, rather than an upper bound, on the potential of leveraging LLMs for scientific feedback. Moreover, our system only leverages zero-shot learning of

GPT-4 without fine-tuning on additional datasets. Further, the architecture and prompt used in our study only represent one of the many possible forms of LLM-based scientific feedback. Aside from exploring other LLMs and conducting more sophisticated prompt engineering, future work could incorporate labeled datasets of “high quality scientific feedback” to further fine-tune the LLM, or prompt LLM to leverage tools (e.g., fact check API) so that the feedback could be more detailed and method-specific. Nevertheless, our proposed framework proves to be helpful and aligns well with comments brought up by human reviewers, demonstrating the potential of incorporating LLM in the scientific evaluation process, echoing prior works arguing for AI in the scientific process [242]. Our retrospective evaluation used *Nature* family data and *ICLR* data. While these data cover a range of scientific fields, including biology, computer science, etc., and the *ICLR* dataset includes both accepted and rejected papers, all studied papers are targeted at top venues in English. Future work should further evaluate the framework with more coverage. Our user study is limited in coverage of participant population and suffer from a self-selection issue. For example, we focused on U.S. based researchers, and those who opted in to receive LLM generated scientific feedback may not fully represent either their own scientific fields or researchers generally (see **Methods** for a more extended discussion). Current results are based on responses from researchers in machine learning and computational biology, and while we aim to ensure representativeness of researchers by reaching out to randomly sampled authors who upload preprints to arXiv in recent months, participants who opt into the study are likely to be interested and familiar with LLM or AI in general. Finally, the version of the GPT-4 model we utilized for our primary experiments does not possess the capability to understand or interpret visual data such as tables, graphs, and figures, which are integral components of scientific literature. As we wrapped out our study, OpenAI released GPT-4V(ision). In a pilot analysis, we applied GPT-4V on 500 *Nature* family papers, now including all the figures, and evaluated it using the same framework as above. Adding the vision capability further increased the overlap between AI generated reviews and human reviews (hit rate of 43%) compared to 31% without vision). This finding suggests that a promising direction of future work is to leverage visual-language models to provide more comprehensive feedback on manuscript drafts.

One other direction for future work is to explore the extent to which the proposed approach can help identify and correct errors in scientific papers. This would involve artificially introducing various types of errors, including typos, mistakes in data analyses, and errors in mathematical equations. By evaluating whether LLM-generated feedback can effectively detect and rectify these errors, we can gain further insights into the system’s ability to improve the overall accuracy and quality of scientific manuscripts. Furthermore, investigating the limitations and challenges associated with error detection and correction by LLM is crucial. This includes understanding the types of errors that may be more challenging for the model to detect and correct, as well as evaluating the potential impact of false positives or false negatives in the generated feedback. Such insights can inform the development of more robust and accurate AI-assisted review systems. Additionally, we intend to

broaden the scope of evaluated scientific papers to include manuscripts written in languages other than English, or by authors for whom English is not their first language, in order to understand whether LLMs can provide useful feedback for such papers.

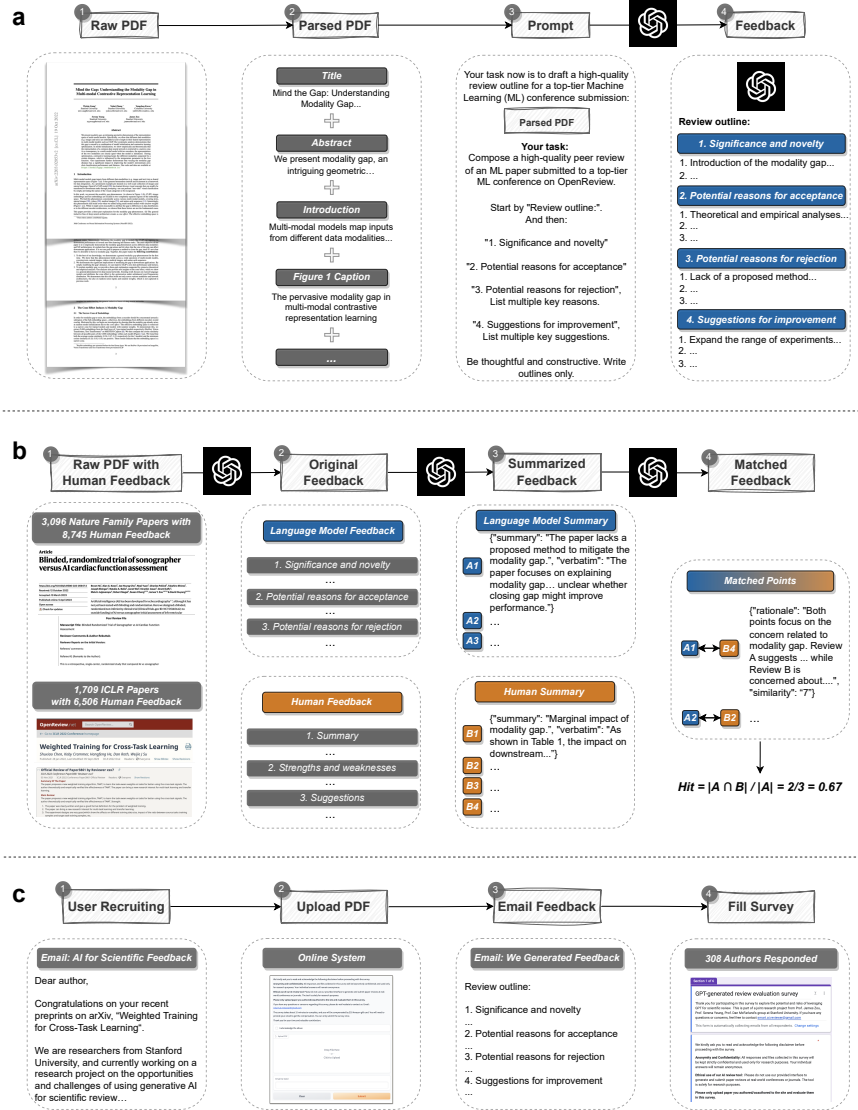


Figure 4.1: **Characterizing the capability of LLM in providing helpful feedback to researchers.** **a**, Pipeline for generating LLM scientific feedback using GPT-4. Given a PDF, we parse and extract the paper’s title, abstract, figure and table captions, and main text to construct the prompt. We then prompt GPT-4 to provide structured comments with four sections, following the feedback structure of leading interdisciplinary journals and conferences: significance and novelty, potential reasons for acceptance, potential reasons for rejection, and suggestions for improvement. **b**, Retrospective analysis of LLM feedback on 3,096 *Nature* family papers and 1,709 *ICLR* papers. We systematically compare LLM feedback with human feedback using a two-stage comment matching pipeline. The pipeline first performs extractive text summarization to extract the points of comments raised in LLM and human-written feedback respectively, and then performs semantic text matching to match the points of shared comments between LLM and human feedback. **c**, Prospective user study survey with 308 researchers from 110 US institutions in the field of AI and computational biology. Each researcher uploaded a paper they authored, and filled out a survey on the LLM feedback generated for them.

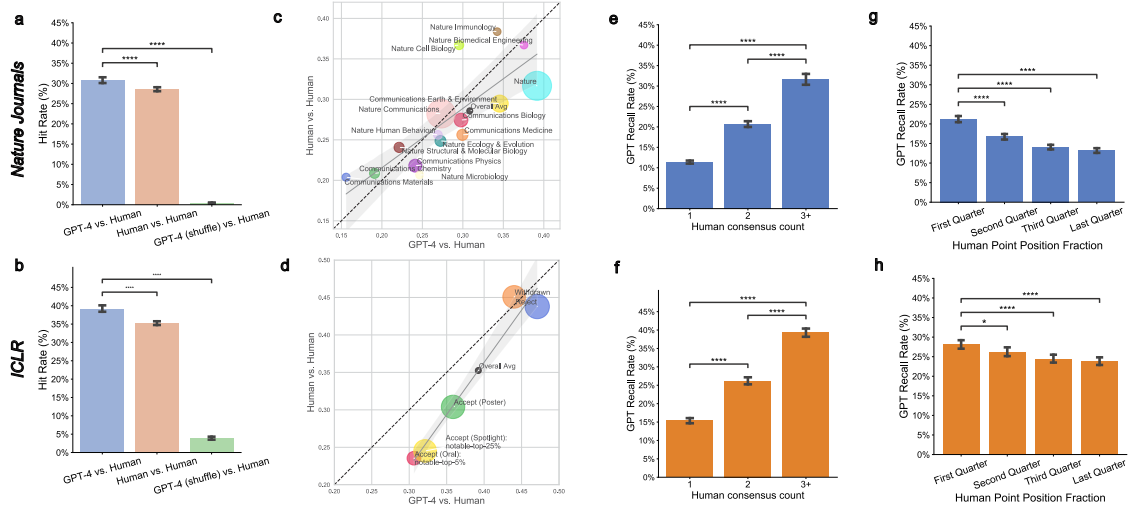


Figure 4.2: **Retrospective analysis of LLM and human scientific feedback.** **a**, Retrospective overlap analysis between feedback from the LLM versus individual human reviewers on papers submitted to *Nature* Family Journals. Approximately one third (30.85%) of GPT-4 raised comments overlap with the comments from an individual reviewer (hit rate). “GPT-4 (shuffle)” indicates feedback from GPT-4 for another randomly chosen paper from the same journal and category. As a null model, if LLM mostly produces generic feedback applicable to many papers, then there would be little drop in the pairwise overlap between LLM feedback and the comments from each individual reviewer after the shuffling. In contrast, the hit rate drops substantially from 57.55% to 1.13% after shuffling, indicating that the LLM feedback is paper-specific. **b**, In the International Conference on Learning Representations (*ICLR*), more than one third (39.23%) of GPT-4 raised comments overlap with the comments from an individual reviewer. The shuffling experiment shows a similar result, indicating that the LLM feedback is paper-specific. **c-d**, The overlap between LLM feedback and human feedback appears comparable to the overlap observed between two human reviewers across *Nature* family journals (**c**) ($r = 0.80$, $P = 3.69 \times 10^{-4}$) and across *ICLR* decision outcomes (**d**) ($r = 0.98$, $P = 3.28 \times 10^{-3}$). **e-f**, Comments raised by multiple human reviewers are disproportionately more likely to be hit by GPT-4 on *Nature* Family Journals (**e**) and *ICLR* (**f**). The X-axis indicates the number of reviewers raising the comment. The Y-axis indicates the likelihood that a human reviewer comment matches a GPT-4 comment (GPT-4 recall rate). **g-h**, Comments presented at the beginning of a reviewer’s feedback are more likely to be identified by GPT-4 on *Nature* Family Journals (**g**) and *ICLR* (**h**). The X-axis indicates a comment’s position in the sequence of comments raised by the human reviewer. Error bars represent 95% confidence intervals. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$.

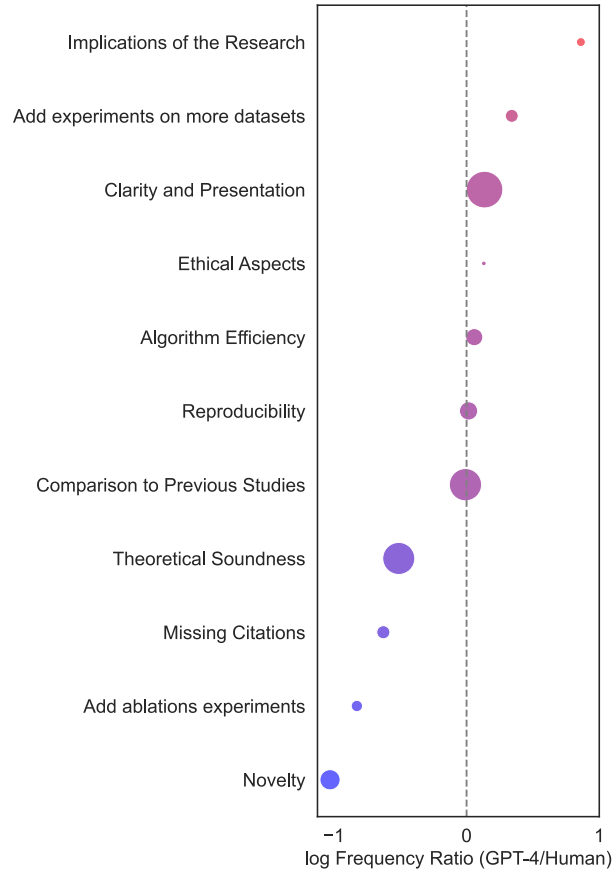


Figure 4.3: **LLM based feedback emphasizes certain aspects more than humans.** LLM comments on the implications of research 7.27 times more frequently than human reviewers. Conversely, LLM is 10.69 times less likely to comment on novelty compared to human reviewers. While both LLM and humans often suggest additional experiments, their focuses differ: human reviewers are 6.71 times more likely than LLM to request additional ablation experiments, whereas LLM is 2.19 times more likely than humans to request experiments on more datasets. Circle size indicates the prevalence of each aspect in human feedback.

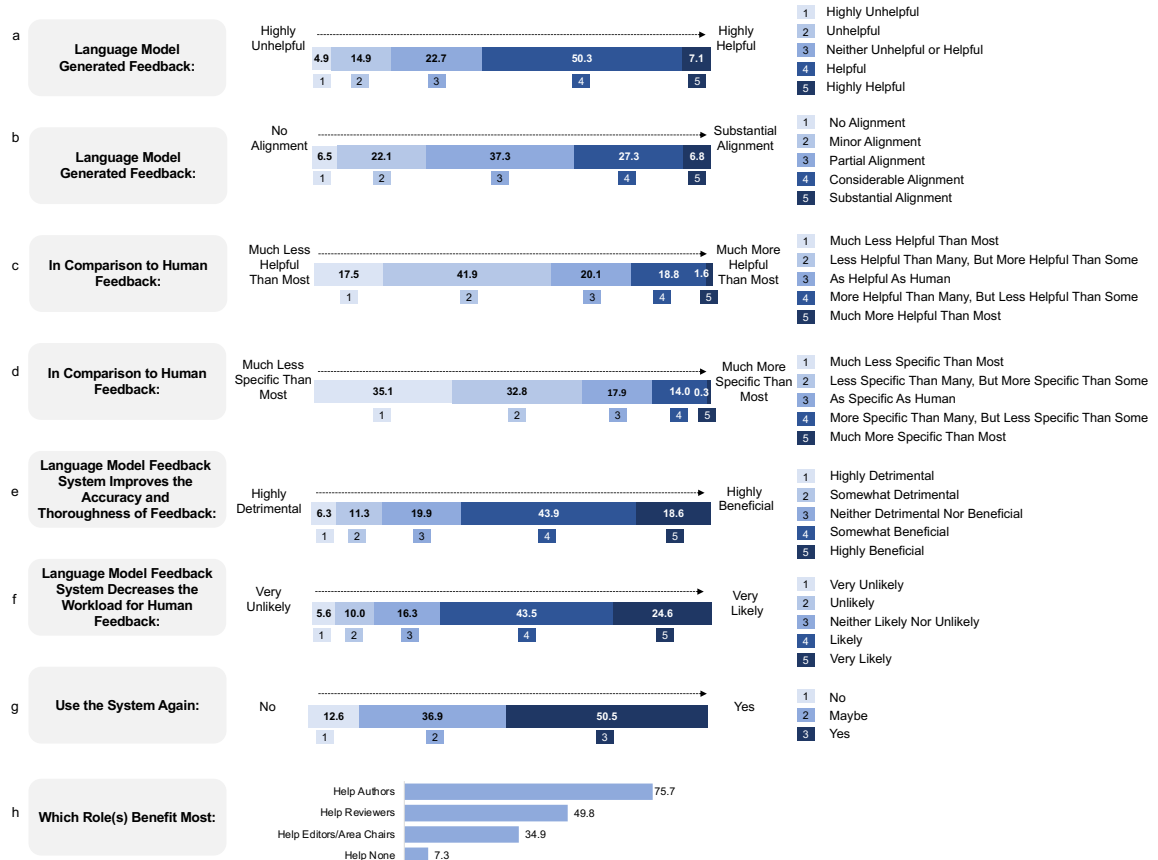


Figure 4.4: **Human study of LLM and human review feedback** ($n = 308$). **a-b**, LLM generated feedback is generally helpful and has substantial overlaps with actual feedback from human reviewers. **c-d**, Compared to human feedback, LLM feedback is slightly less helpful and less specific. **e-f**, Users generally believe that the LLM feedback system can improve the accuracy and thoroughness of reviews, and reduce the workload of reviewers. **g**, Most users intend to use or potentially use the LLM feedback system again. **h**, Users believe that the LLM feedback system mostly helps authors, followed by reviewers and editors / area chairs. Numbers are percentages (%).

Table 4.1: Summary of papers and their associated reviews sampled from 15 *Nature* family journals.

Journal	Count	
	Papers	Reviews
<i>Nature</i>	773	2324
<i>Nature Communications</i>	810	2250
<i>Communications Earth & Environment</i>	299	807
<i>Communications Biology</i>	200	526
<i>Communications Physics</i>	174	464
<i>Communications Medicine</i>	123	343
<i>Nature Ecology & Evolution</i>	113	332
<i>Nature Structural & Molecular Biology</i>	110	290
<i>Communications Chemistry</i>	101	279
<i>Nature Cell Biology</i>	78	233
<i>Nature Human Behaviour</i>	72	211
<i>Communications Materials</i>	67	181
<i>Nature Immunology</i>	62	165
<i>Nature Microbiology</i>	57	174
<i>Nature Biomedical Engineering</i>	57	166
Total	3096	8745

Table 4.2: Summary of *ICLR* papers and their associated reviews sampled from the years 2022 and 2023, grouped by decision.

Decision Category	<i>ICLR 2022</i>		<i>ICLR 2023</i>	
	# of Papers	# of Reviews	# of Papers	# of Reviews
Accept (Oral) - notable-top-5%	55	200	90	317
Accept (Spotlight) - notable-top-25%	173	664	200	758
Accept (Poster)	197	752	200	760
Reject after author rebuttal	213	842	212	799
Withdrawn after reviews	182	710	187	703
Total	820	3168	889	3337

Table 4.3: **Results of human verification on the retrospective comment extraction and matching pipeline.** **a**, Human verification for the extractive summarization stage. We randomly sampled 639 scientific feedbacks. Human annotators reported the numbers of true positives (correctly extracted comments), false negatives (overlooked relevant comments), and false positives (incorrectly extracted or split comments). Results suggest high accuracy of the extractive summarization stage. **b**, Human verification for the semantic matching stage. We randomly sampled 12,035 pairs of extracted comments. Human annotators annotated each pair of extracted comments, determining whether the two comments are semantically matched in content. Results suggest high accuracy of the semantic text matching stage.

Extractive Summarization		Semantic Matching		
Extracted Comments	Count	Human Matching	Predicted Matching	
			Matched	Not Matched
TP (True Positives)	2634	Matched	685	95
FN (False Negatives)	110	Not Matched	197	11058
FP (False Positives)	63			
Precision	0.977	Precision	0.777	
Recall	0.960	Recall	0.878	
F1 Score	0.968	F1 Score	0.824	

Table 4.4: Mean token lengths of papers and human reviews in the two datasets.

Dataset	Paper (Mean Token Length)	Human Review (Mean Token Length)
<i>ICLR</i>	5,841.46	671.53
<i>Nature</i> Family Journals	12,444.06	1,337.93

Table 4.5: Example comments extracted from LLM and human feedback on *ICLR* by human coding

Human Coding	Source	Comment
Clarity and Presentation	Human	The writing is hard to follow. Since this paper introduces multiple new concepts, it was hard for me to understand...
Clarity and Presentation	GPT-4	The paper is highly technical and may be difficult to understand for readers who are not familiar with the field. The authors could provide a more detailed explanation of the IB principle...
Comparison to Previous Studies	Human	The comparisons are flawed. In particular the 'label consistency' and 'class-center consistency' losses are disjoint with the GNN methodology, and a fairer comparison would be with GNNs that also use these two losses...
Comparison to Previous Studies	GPT-4	The paper lacks a thorough comparison with existing methods. While the authors have compared their approach with a few baselines, a more comprehensive comparison with...
Theoretical Soundness	Human	IMHO, the theoretical proof is relatively trivial. The final conclusion is if the similarity is proper, the predicted action is accurate. Since the model is actually learning the proper similarity, this is equivalent to saying if the model h is well trained, the output is accurate. This is obviously true.
Theoretical Soundness	GPT-4	The authors should provide more details on the theoretical analysis of the connections between message passing and consistency constraints to make it more understandable for readers...
Novelty	Human	A major concern of the paper is about the model's novelty. The reviewer has doubts on the argument that a new combination of existing techniques (BN, LSTM, S&E, skip connection) for the task of video prediction is significant enough to publish in <i>ICLR</i> ...
Novelty	GPT-4	The paper could be more explicit in explaining the novelty of the proposed method compared to existing techniques. It is not entirely clear how the proposed method differs from other methods that use projection features...
Reproducibility	Human	As for reproducibility, although the authors provide a reproducibility statement, it is still better to provide codes for other readers to implement the experiments...
Reproducibility	GPT-4	Reproducibility: The authors should provide more details about the experimental setup to ensure reproducibility...

Table 4.6: Example comments extracted from LLM and human feedback on *ICLR* by human coding

Human Coding	Source	Comment
Add ablations experiments	Human	It’s best to make clearer how much of the observed performance gains in experiments is from the proposed learning confidence-conditioned values, vs from some of the empirical/engineering decisions in Sec 5 practical algorithms (IQN, approx inverse visitation). Perhaps if there’s a simpler (maybe tabular?) problem that doesn’t have these deep RL complexities that would strengthen the results and help the reader understand the contribution better - and help to show what the learned Q-values are actually doing...
Add ablations experiments	GPT-4	The paper does not provide a clear explanation of how the proposed auxiliary tasks contribute to the improved performance of the model. The authors should provide a more detailed analysis of the impact of these tasks on the model’s performance. It would also be helpful to see a comparison of the model’s performance with and without these tasks...
Implications of the Research	Human	One aspect I would have liked to see more of is how the findings of this submission influence future work on Gaussian processes and other Bayesian models? Are there any specific takeaways which might guide the design of models yielding improved estimates with better uncertainty calibration?
Implications of the Research	GPT-4	The paper could provide more information on the potential real-world implications of membership leakage in PLMs. For example: The authors could discuss the potential impact on users’ privacy and data security. The authors could explore the potential legal and ethical implications of such privacy leakage...
Ethical Aspects	Human	My main concern: I did not find IRB approval information on the human experiment. If there is, it should be mentioned, if not the authors should explain why it is not necessary in this case (and should be validated with the conference chairs). Also, the details of the experiment and instructions to the demonstrators should accompany the paper.
Ethical Aspects	GPT-4	The paper does not discuss the ethical implications of their research. While the authors’ intention is to improve the security of federated learning systems, their research could potentially be misused by malicious actors...

Table 4.7: Example comments extracted from LLM and human feedback on *ICLR* by human coding

Human Coding	Source	Comment
Add ablations experiments	Human	It’s best to make clearer how much of the observed performance gains in experiments is from the proposed learning confidence-conditioned values, vs from some of the empirical/engineering decisions in Sec 5 practical algorithms (IQN, approx inverse visitation). Perhaps if there’s a simpler (maybe tabular?) problem that doesn’t have these deep RL complexities that would strengthen the results and help the reader understand the contribution better - and help to show what the learned Q-values are actually doing...
Add ablations experiments	GPT-4	The paper does not provide a clear explanation of how the proposed auxiliary tasks contribute to the improved performance of the model. The authors should provide a more detailed analysis of the impact of these tasks on the model’s performance. It would also be helpful to see a comparison of the model’s performance with and without these tasks...
Implications of the Research	Human	One aspect I would have liked to see more of is how the findings of this submission influence future work on Gaussian processes and other Bayesian models? Are there any specific takeaways which might guide the design of models yielding improved estimates with better uncertainty calibration?
Implications of the Research	GPT-4	The paper could provide more information on the potential real-world implications of membership leakage in PLMs. For example: The authors could discuss the potential impact on users’ privacy and data security. The authors could explore the potential legal and ethical implications of such privacy leakage...
Ethical Aspects	Human	My main concern: I did not find IRB approval information on the human experiment. If there is, it should be mentioned, if not the authors should explain why it is not necessary in this case (and should be validated with the conference chairs). Also, the details of the experiment and instructions to the demonstrators should accompany the paper.
Ethical Aspects	GPT-4	The paper does not discuss the ethical implications of their research. While the authors’ intention is to improve the security of federated learning systems, their research could potentially be misused by malicious actors...

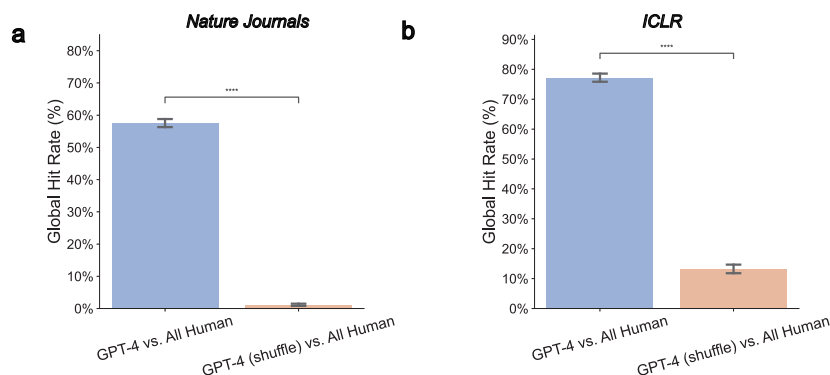


Figure 4.5: **Fraction of GPT-4 comments that overlap with comments raised by at least one human reviewer.** **a**, In the *Nature* family journal data, 57.55% of the comments made by GPT-4 overlap with comments made by at least one human reviewer, suggesting a significant overlap between LLM feedback and human feedback. “GPT-4 (shuffle)” refers to feedback from GPT-4 for a randomly selected manuscript from the same journal and category. As a null model, if LLM mostly produces generic feedback applicable to many papers, then there would be little drop in the global hit rate between LLM feedback and the comments from each individual reviewer after the shuffling. However, the global hit rate decreases markedly from 57.55% to 1.13% after shuffling, indicating that the LLM feedback is paper-specific. **b**, The results are similar with *ICLR* data. 77.18% of the comments made by GPT-4 overlap with comments made by at least one human reviewer. The shuffling result also suggests that LLM feedback is paper-specific. Error bars represent 95% confidence intervals. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$.

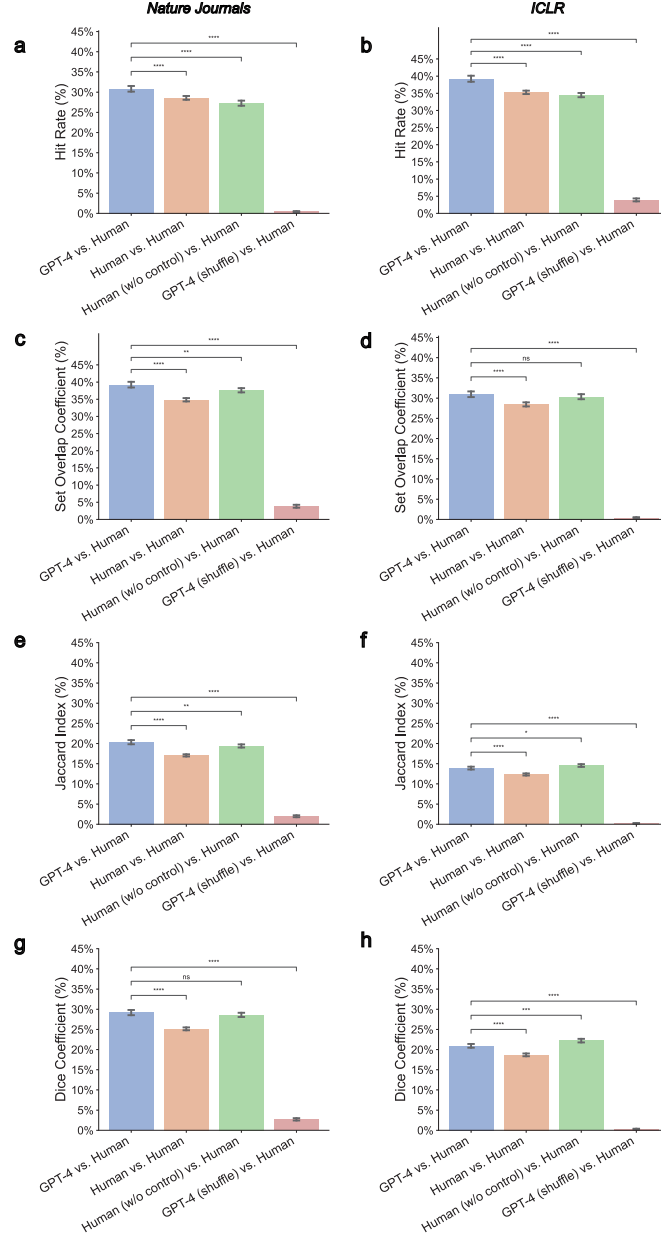


Figure 4.6: **Retrospective evaluation using alternative set overlap metrics for robustness check.** (a, b) Hit rate. (c, d) Szymkiewicz–Simpson overlap coefficient. (e, f) Jaccard index. (g, h) Sørensen–Dice coefficient. Additional metrics results suggest the overlap between GPT-4 and Human is comparable to Human vs. Human overlap, suggesting the robustness of our findings across different set overlap metrics. Error bars represent 95% confidence intervals. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$.

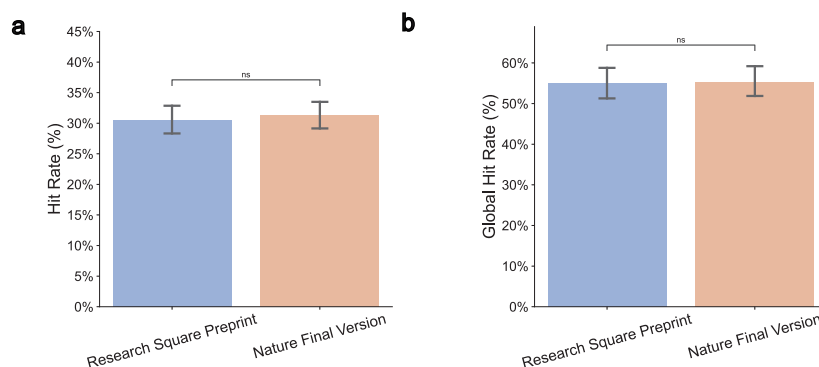


Figure 4.7: **Assessing GPT-4’s Feedback on Research Square Preprints from the *Nature* Family of Publications.** This analysis focuses on 408 preprints from the Nature family of publications available on Research Square. We selected only the preprints posted within four week of the official “Received Date” indicated in the final publication to ensure that the preprint matches the version sent for review. These 408 preprints include, for example, 103 papers eventually published in *Nature Communications*, 72 papers published in *Nature*. **(a)** Overlap analysis between GPT-4’s feedback on these preprints and individual human reviews. Over 30% of GPT-4 raised comments n these preprints overlap with the comments from an individual reviewer (hit rate). **(b)** Over half (55.4%) of the suggestions made by GPT-4 on these preprints were mentioned by at least one human reviewer. This is referred to as the global hit rate, which is the percentage of GPT-4 comments that overlap with comments raised by at least one human reviewer. Error bars represent 95% confidence intervals.

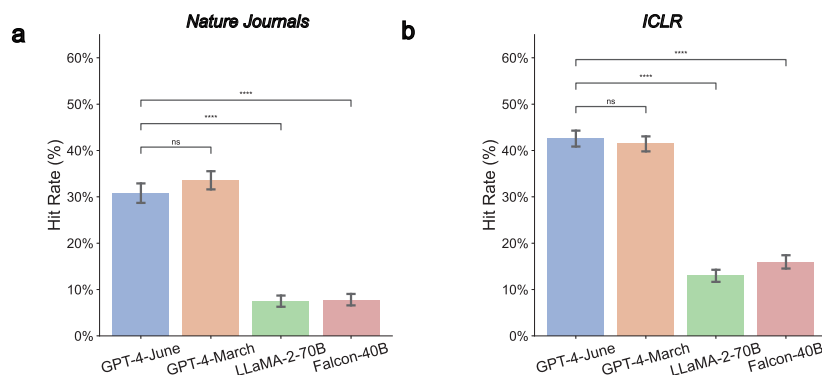


Figure 4.8: **Assessing variability in GPT-4’s feedback and feedback of open-source LLMs.** Overlap analysis of the March 2023 and June 2023 checkpoints of GPT-4, along with two state-of-the-art open-source language models: Llama 2 (70 billion parameters) and Falcon (40 billion parameters) to provide feedback. We evaluated their performance on two datasets: **(a)** 500 randomly selected papers from the *Nature* family of journals and **(b)** 500 randomly selected papers from the *ICLR* conference. The two versions of GPT-4 have very similar overlap with human reviews: for *Nature* family, 33.6% overlap with human reviews in March and 30.8% in June; for *ICLR*, 41.5% overlap with human reviews in March and 42.6% in June. In contrast, both open-source models, Llama 2 and Falcon, perform significantly worse than GPT-4. For *Nature* family papers, the overlap between Llama 2 and human reviews is just 7.5%, and for Falcon, it’s 7.8%. For *ICLR* papers, the overlap is slightly higher, with Llama 2 at 13.0% and Falcon at 15.9%. Error bars represent 95% confidence intervals. *P < 0.05, **P < 0.01, ***P < 0.001, and ****P < 0.0001.

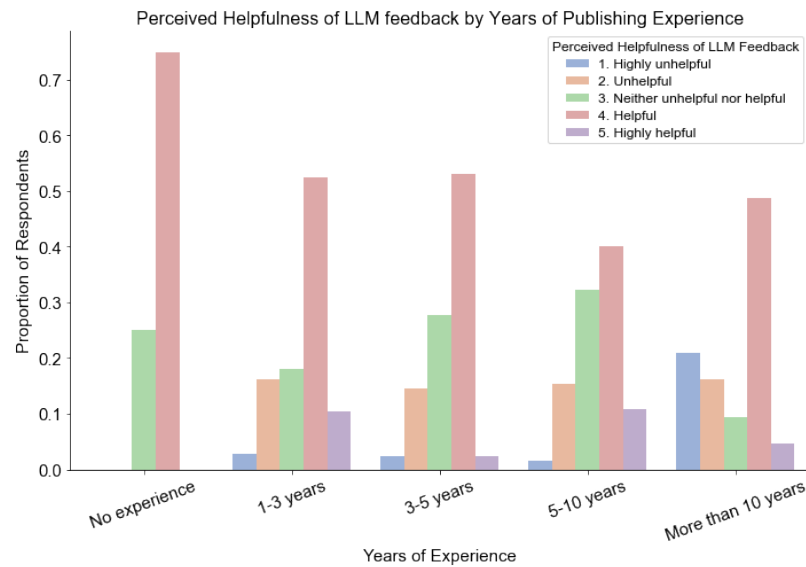


Figure 4.9: LLM-based scientific feedback is considered helpful among participants with varying publishing experience.

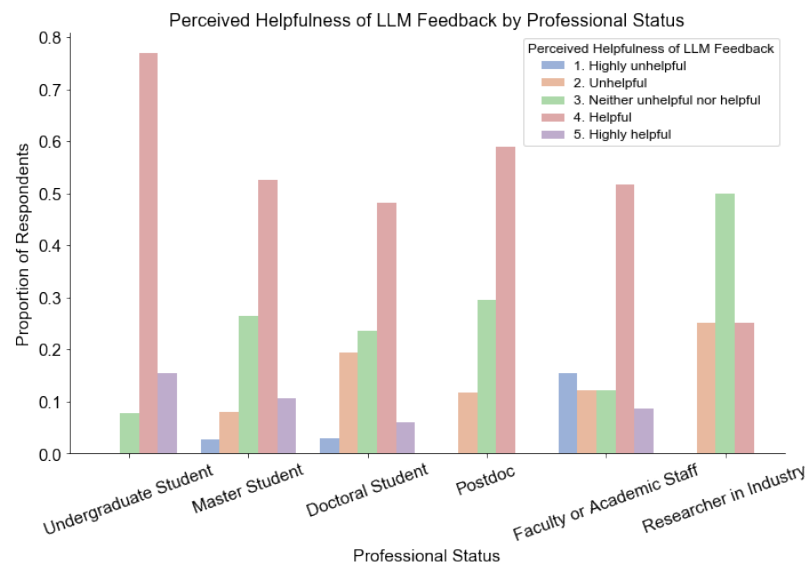


Figure 4.10: LLM-based scientific feedback is considered helpful among participants of different professional statuses.

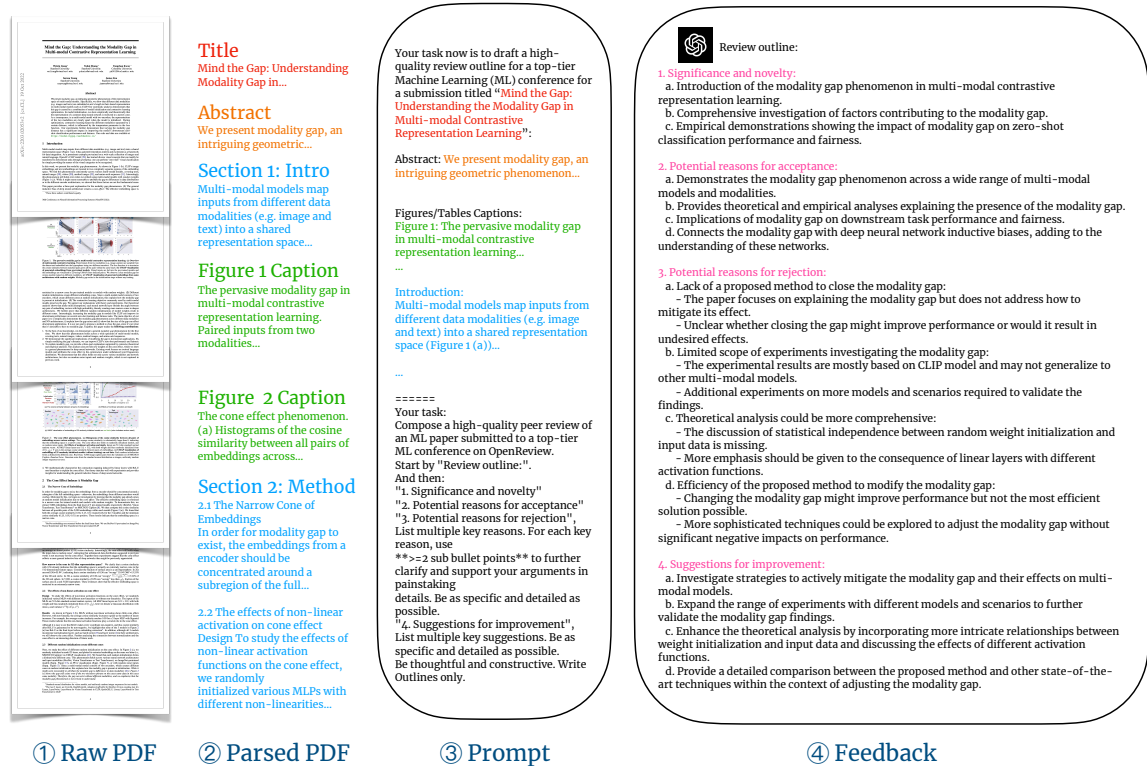


Figure 4.11: **Schematic of the LLM scientific feedback generation system.** Manuscript text, including figure captions, is extracted from the manuscript PDFs and integrated into a prompt for LLM GPT-4, which then generates feedback. The generated feedback provides structured comments in four sections: significance and novelty, potential reasons for acceptance, potential reasons for rejection, and suggestions for improvement. In the example, GPT-4 raised a comment that the paper reported a modality gap phenomenon but did not propose methods to close the gap or demonstrate the benefits of doing so.

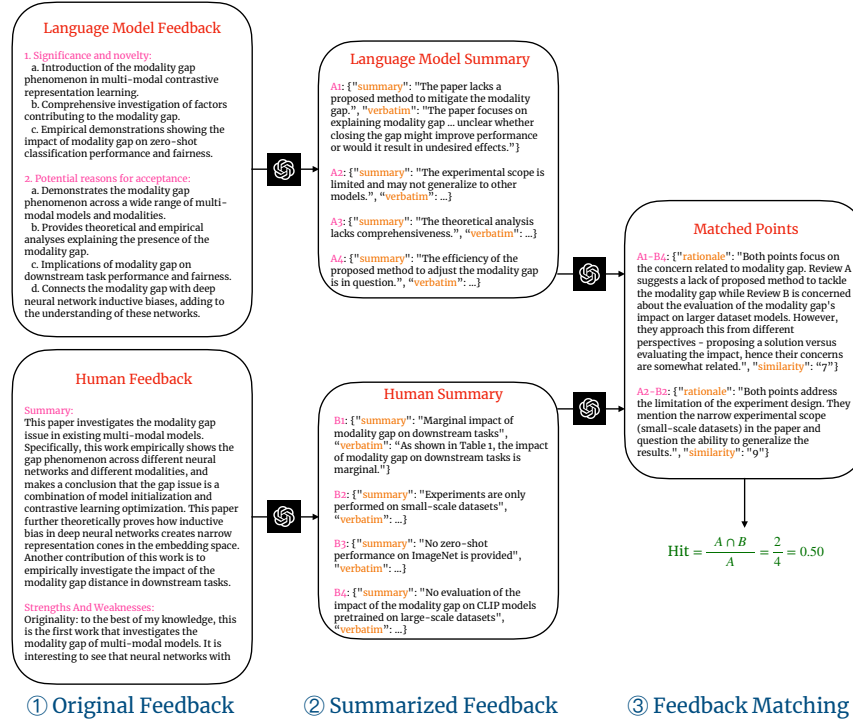


Figure 4.12: **Workflow of the retrospective comment matching pipeline for scientific feedback texts.** **a**, This two-stage pipeline compares comments raised in LLM generated feedback with those from human reviewers. **b**, Extraction: Utilizing LLM's capabilities for information extraction, key comments are extracted from both LLM generated and human-written reviews. **c**, Matching: LLM is used for semantic similarity analysis, where comments from LLM and human feedback are matched. For each paired comment, a similarity rating and justifications are provided. A similarity threshold ≥ 7 is set to filter out weakly-matched comments. This threshold is chosen based on human validations of the matching stage.

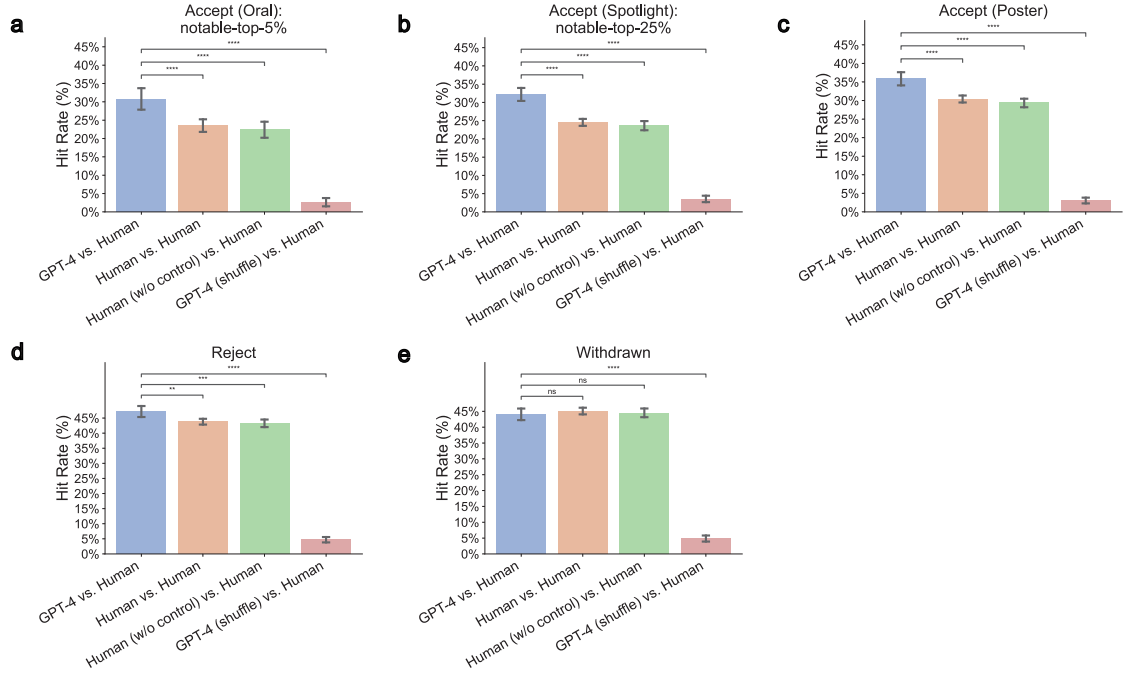


Figure 4.13: **Robustness check on controlling the number of comments for hit rate measurement in *ICLR* data.** Consider two sets of comments A and B . The hit rate refers to the percentage of comments in set A that match those in set B . To enable a more direct comparison between the hit rates of GPT-4 vs. Human and Human vs. Human, we have controlled for the number of comments when calculating the hit rate for Human vs. Human. Specifically, we chose only the first N comments made by the first reviewer (i.e., the comments used as set A) for matching, where N corresponds to the number of comments made by GPT-4 for the same paper. We also included Human (w/o control) vs. Human, where we matched all human comments without limiting the number of comments. **a**, Hit rate comparison for *ICLR* papers that were accepted with oral presentations (top 5% accepted papers), **b**, papers accepted with spotlights (top 25% of accepted papers), **c**, papers accepted for poster presentations, **d**, rejected papers, **e**, papers withdrawn post-review. This figure indicates that results with and without the control for the number of comments are largely similar. The overlap between LLM feedback and human feedback appears comparable to the overlap observed between two human reviewers. Error bars represent 95% confidence intervals. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$.

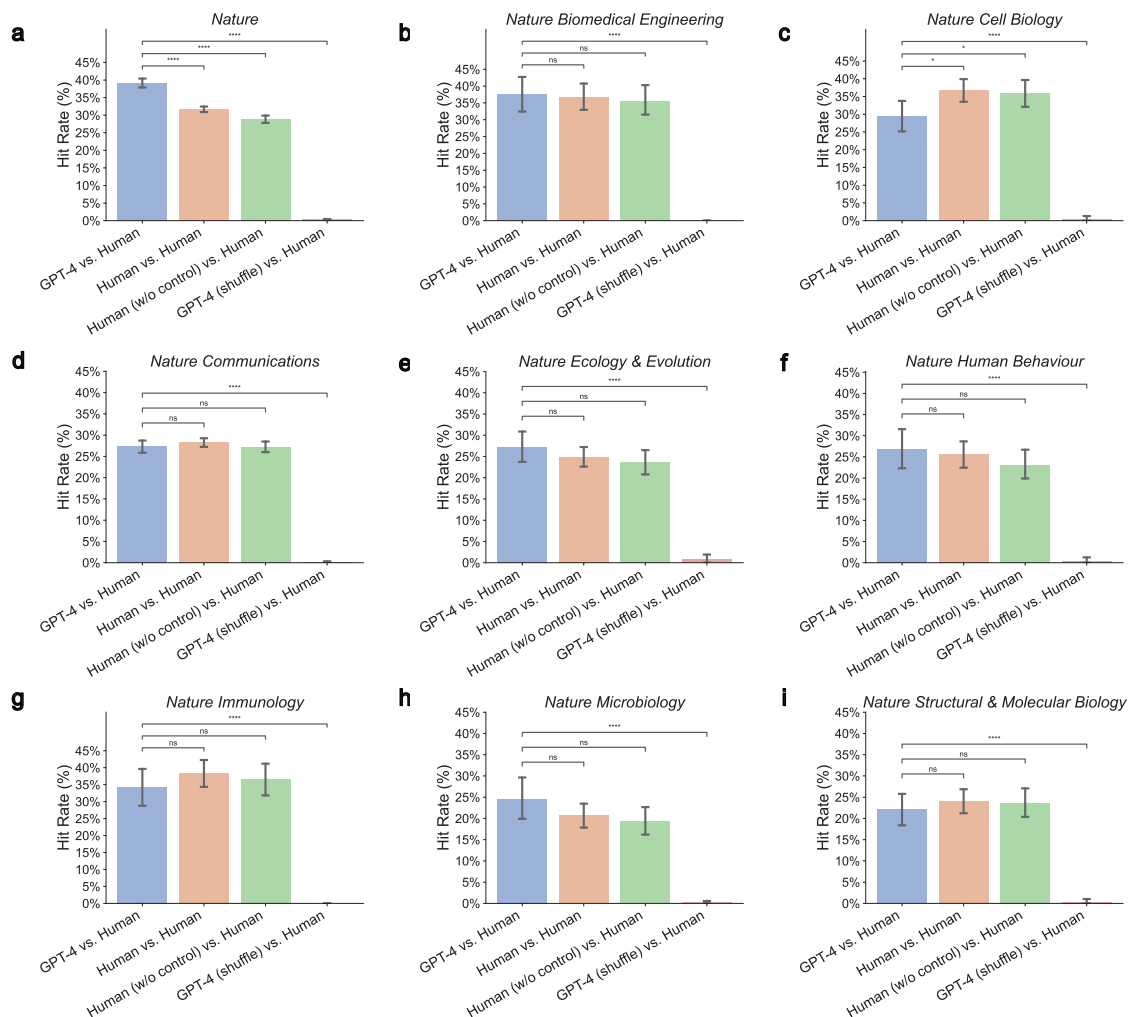


Figure 4.14: **Robustness check on controlling for the number of comments when measuring overlap in *Nature* family journal data (1/2).** Results are stratified by journals: (a) *Nature*, (b) *Nature Biomedical Engineering*, (c) *Nature Cell Biology*, (d) *Nature Ecology & Evolution*, (e) *Nature Human Behaviour*, (f) *Nature Communications*, (g) *Nature Immunology*, (h) *Nature Microbiology*, and (i) *Nature Structural & Molecular Biology*. This figure indicates that results with and without the control for the number of comments are largely similar. The overlap between LLM feedback and human feedback seems comparable to the overlap observed between two human reviewers. Results for additional *Nature* family journals are shown in **Supp. Fig. 4.15**. Error bars represent 95% confidence intervals. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$.

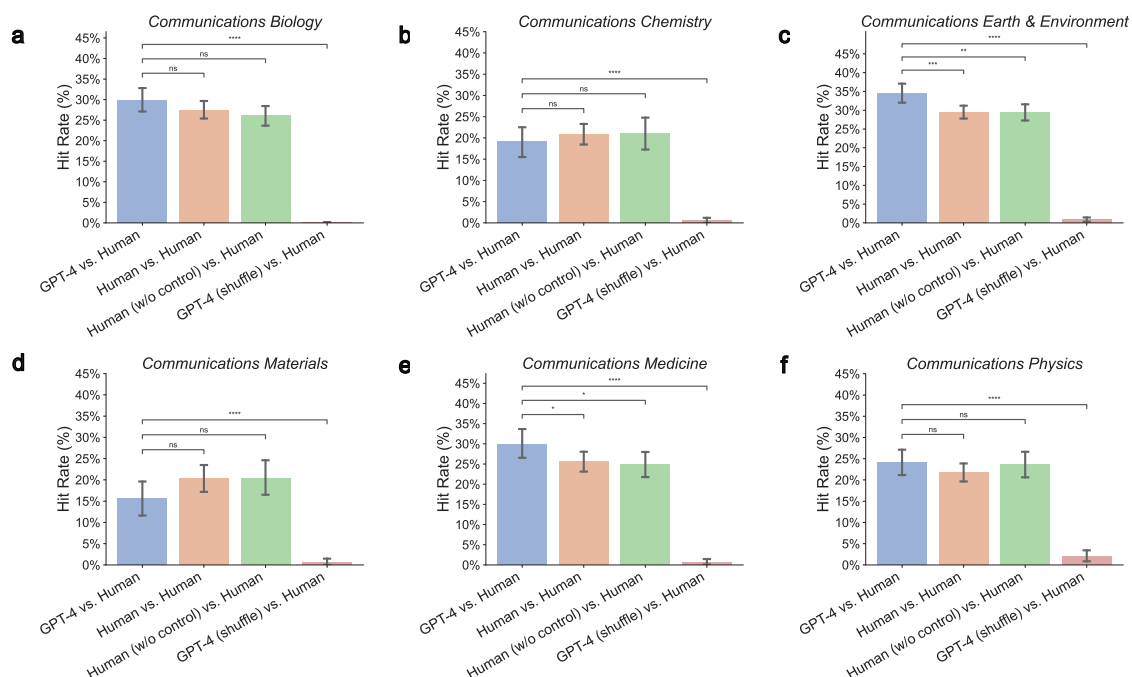


Figure 4.15: **Robustness check on controlling for the number of comments in measuring hit rates in *Nature* family journal data (2/2).** Results are stratified by journals (continuing **Supp. Fig. 4.14**): (a) *Communications Biology*, (b) *Communications Chemistry*, (c) *Communications Earth & Environment*, (d) *Communications Materials*, (e) *Communications Medicine*, and (f) *Communications Physics*. This figure indicates that results with and without the control for the number of comments are largely similar. The overlap between LLM feedback and human feedback seems comparable to the overlap observed between two human reviewers. Error bars represent 95% confidence intervals. *P < 0.05, **P < 0.01, ***P < 0.001, and ****P < 0.0001.

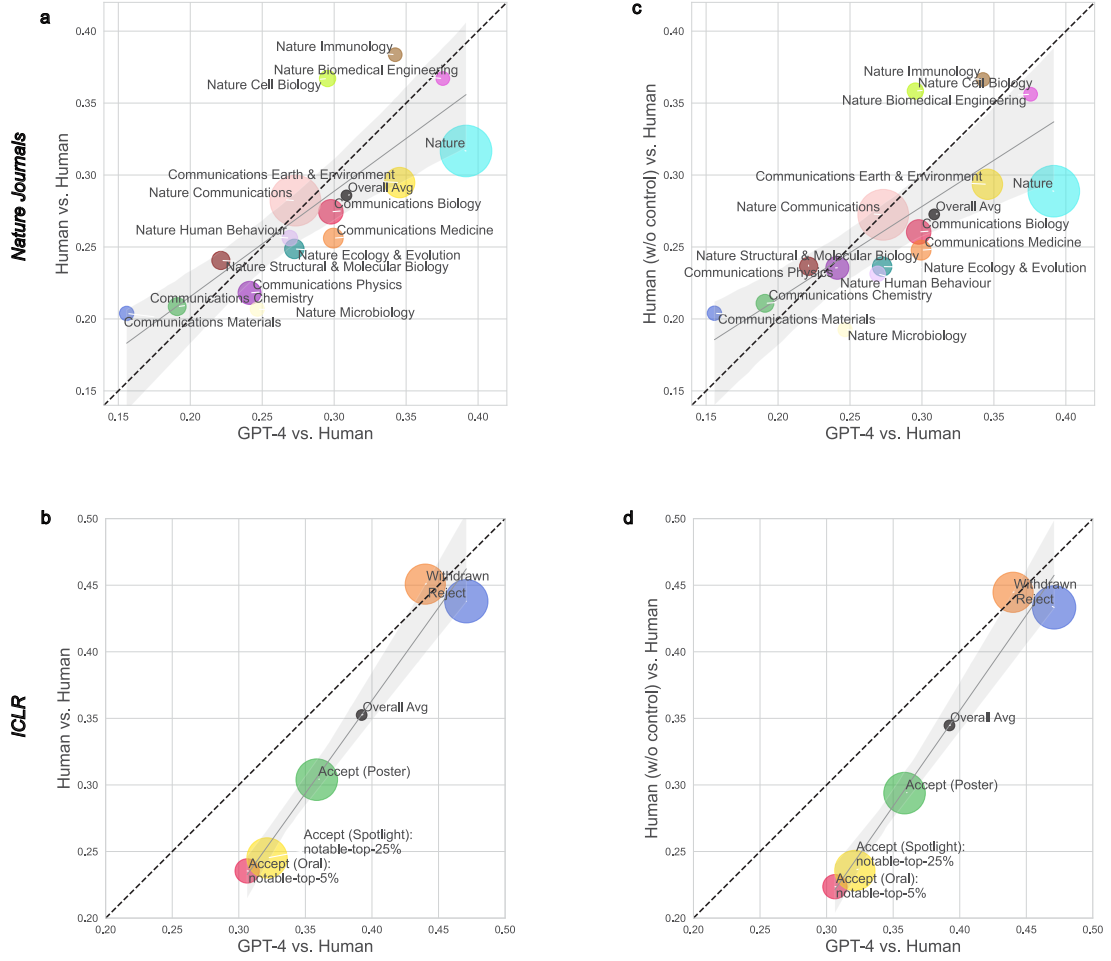


Figure 4.16: **Robustness check on controlling for the number of comments in the correlation of hit rates.** (a) Hit rates across various *Nature* family journals, controlling for the number of comments ($r = 0.80$, $P = 3.69 \times 10^{-4}$). (b) Hit rates across *ICLR* papers with different decision outcomes, controlling for the number of comments ($r = 0.98$, $P = 3.28 \times 10^{-3}$). (c) Hit rates across various *Nature* family journals without control for the number of comments ($r = 0.75$, $P = 1.37 \times 10^{-3}$). (d) Hit rates across *ICLR* papers with different decision outcomes without control for the number of comments ($r = 0.98$, $P = 2.94 \times 10^{-3}$). Human (w/o control) represents matching with all human comments without controlling for the number of comments. Circle size indicates sample size.

We kindly ask you to read and acknowledge the following disclaimer before proceeding with the survey.

Anonymity and Confidentiality: All responses and files collected in this survey will be kept strictly confidential and used only for research purposes. Your individual answers will remain anonymous.

Ethical use of our AI review tool: Please do not use our provided interface to generate and submit paper reviews at real-world conferences or journals. The tool is solely for research purposes.

Please only upload paper you authored/coauthored to the site and evaluate them in this survey.

If you have any questions or concerns regarding this survey, please do not hesitate to contact us. Email: smart.ai.reviewer@gmail.com

The survey takes about 15 minutes to complete, and you will be compensated by \$20 Amazon gift card. You will need to provide your email to get the compensation. You can only submit the survey once.

Thank you for your time and valuable contribution.

☒ I acknowledge the above

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Clear
Submit

Confirmation

Your draft is received. We just sent you an email confirmation. We will send you the review once it is ready. The process will take some time, approximately 5 mins, so feel free to check back after some time. The generated review may go into your spam box.

Flag

Figure 4.17: Web interface for our prospective user study designed to characterize the capability of LLM in providing helpful scientific feedback. Users are instructed to upload a research paper they authored in PDF format, following which LLM feedback is sent to their email. Users were guided to upload only papers published after GPT-4’s last training cut-off in September 2021. An ethics statement has been included to discourage the direct application of LLM content for review-related tasks.

```

Your task now is to draft a high-quality review outline for a Nature family journal for a
submission titled <Title>:

'''
<Paper_content>
'''

=====

Your task:
Compose a high-quality peer review of a paper submitted to a Nature family journal.

Start by "Review outline:".
And then:
"1. Significance and novelty"
"2. Potential reasons for acceptance"
"3. Potential reasons for rejection", List multiple key reasons. For each key reason, use
    **>=2 sub bullet points** to further clarify and support your arguments in painstaking
    details. Be as specific and detailed as possible.
"4. Suggestions for improvement", List multiple key suggestions. Be as specific and
    detailed as possible.

Be thoughtful and constructive. Write Outlines only.

```

Figure 4.18: Prompt template employed with GPT-4 for generating scientific feedback on papers from the *Nature* journal family dataset. <Paper_content> denotes text extracted from the paper, including the paper’s abstract, figure and table captions, and other main text sections. For clarity and succinctness, GPT-4 was directed to formulate a structured outline of scientific feedback. GPT-4 was requested to generate four feedback sections: significance and novelty, potential reasons for acceptance, potential reasons for rejection, and suggestions for improvement. The feedback was generated by GPT-4 in a single pass.

```

Your goal is to identify the key concerns raised in the review, focusing only on potential
    reasons for rejection.

Please provide your analysis in JSON format, including a concise summary, and the exact
    wording from the review.

Submission Title: <Title>

====Review:
'''
<Review_Text>
'''

====
Example JSON format:
{{
  "1": {"summary": "<your concise summary>", "verbatim": "<concise, copy the exact
    wording in the review>"}},
  "2": ...
}}

Analyze the review and provide the key concerns in the format specified above. Ignore
    minor issues like typos and clarifications. Output only json.

```

Figure 4.19: Prompt template employed with GPT-4 for extractive text summarization of comments in LLM and human feedback. The output is structured in JSON (JavaScript Object Notation) format, where each JSON key assigns an ID to a specific point, and the corresponding value provides the content of the point.

```

Your task is to carefully analyze and accurately match the key concerns raised in two
reviews, ensuring a strong correspondence between the matched points. Examine the
verbatim closely.

====Review A:
'''
<JSON extracted comments for Review A from previous step>
'''

====Review B:
'''
<JSON extracted comments for Review B from previous step>
'''

Please follow the example JSON format below for matching points. For instance, if point 1
from review A is nearly identical to point 2 from review B, it should look like this:

{{
"A1-B2": {"rationale": "<explain why A1 and B2 are nearly identical>", "similarity":
        "<5-10, only an integer>"}},
...
}}

Note that you should only match points with a significant degree of similarity in their
concerns. Refrain from matching points with only superficial similarities or weak
connections. For each matched pair, rate the similarity on a scale of 5-10.

5. Somewhat Related: Points address similar themes but from different angles.
6. Moderately Related: Points share a common theme but with different perspectives or
   suggestions.
7. Strongly Related: Points are largely aligned but differ in some details or nuances.
8. Very Strongly Related: Points offer similar suggestions or concerns, with slight
   differences.
9. Almost Identical: Points are nearly the same, with minor differences in wording or
   presentation.
10. Identical: Points are exactly the same in terms of concerns, suggestions, or praises.

If no match is found, output an empty JSON object. Provide your output as JSON only.

```

Figure 4.20: Prompt template employed with GPT-4 for semantic text matching to match the points of shared comments between two feedback. The input comprises two lists of comments in JSON format obtained from the preceding step. GPT-4 was then directed to identify common points between the two lists, and generate a new JSON where each key corresponds to a pair of matching point IDs, and the associated value provides the rationale for the match.

Chapter 5

Discussion, Conclusion and Future Work

Throughout the proceeding chapters, we demonstrated through three focal studies, how *empirical evidence*, *technical methods* and *design* can be utilized to enhance the study of computer-mediated work.

To conclude, this chapter will review the main contributions of the thesis and consider important avenues for future work.

5.1 Summaries of Contributions

Chapter 2 showcases the use of empirical evidence to analyze multitasking during remote meetings through a comprehensive study. It leverages a large telemetry dataset and a diary study of Microsoft employees in the U.S., recording activities during video conferences, emails, and file edits from February to May 2020. The analysis reveals that meeting characteristics like size, length, and timing significantly influence multitasking, which can have both positive and negative effects on productivity and well-being. The chapter suggests best practices for remote meetings, such as avoiding important sessions in the morning, and design implications for tools that support beneficial multitasking.

Chapter 3 introduces an AI approach to assess team viability, a crucial element for effective teamwork. By analyzing 669 online teams' 10-minute text conversations, the study identifies teams with high and low viability using a lasso regression model that achieves 0.74–0.92 AUC ROC accuracy. Key predictors include exclusive language and second-person pronouns, indicating that active engagement is vital for team success. It also finds that only about 70 seconds of conversation is needed to predict team interaction viability, enabling teams to monitor and visualize their viability in real time.

Chapter 4 explores the design and evaluation of a large language model (LLM) system to generate peer-review-like feedback, aiming to innovate how researchers receive feedback. Despite the challenges posed by the rapid increase in scholarly output and specialization, this system, based on GPT-4, provides feedback on research papers. It compares favorably with human reviewers in tests against feedback from major journals and a user study with researchers in AI and computational biology. While GPT-4 generally provides helpful feedback, it tends to focus on certain aspects and lacks in-depth critique, highlighting that LLM and human feedback can complement each other. This suggests the potential of LLM feedback, particularly when expert input is unavailable or during the early stages of research.

Broadly, this thesis makes contributions in the areas of social computing, design, and management.

- **Social computing.** The current dissertation contributes to the literature on computer supportive cooperative work (CSCW). The dissertation studies how recent computing technologies, such as video conferencing, and generative AI, introduces new challenges and potentials at the workplace. The dissertation further measures and studies how work practices have been impacted by these computing technologies.
- **Design.** The dissertation also contributes to novel design to improve workplace experience. The dissertation provides both concrete design prototypes (e.g. generative AI based tool to generate scientific feedback), algorithms (e.g. AI algorithm to predict team viability), and design implications (feature designs for people to focus on remote meetings).
- **Management.** Finally, the dissertation also contributes to the field of management. On one hand, the dissertation contributes novel empirical evidence to organization behavior literature, including how workers collaborate over video conferencing, and explore mechanisms behind team viability. On the other hand, the dissertation also contributes to managerial best practices guidelines, e.g. discussions on best practices for running effective remote meetings. Some managerial insights have been featured and discussed in news¹, and blog posts².

5.2 Discussion

In this section, I further discuss some lessons I learnt through the three studies.

5.2.1 Ethical considerations for building workplace computing systems

Recent advancements and widespread adoption of workplace computing systems have transformed modern workplaces. However, alongside their benefits, these systems have also sometimes led to

¹<https://www.wired.com/story/stop-looking-your-email-youre-video/>

²<https://slack.com/blog/collaboration/sharpen-your-focus-identifying-bad-habits>

unsatisfactory experiences and even heightened tensions among workers [124]. A primary source of these issues often lies in the foundational design of these systems. As we integrate increasingly powerful technologies, such as generative AI, into the workplace, it becomes crucial to consider the ethical dimensions of these tools, particularly what objectives they are set to optimize and who ultimately benefits from their deployment.

What objectives is the tool optimizing for?

Traditionally, the primary goal of many workplace applications has been to enhance worker productivity [85, 14, 250]. However, as we develop the next generation of workplace computing systems, it is essential to recognize that boosting productivity is merely one of several goals that a well-rounded workplace should strive for. Modern designs should also prioritize fostering creativity, improving employee well-being, and enhancing engagement [76, 147]. Design and research in these areas, unfortunately, is not as robust as studies focused on productivity, indicating a gap in our understanding of how diverse objectives can be balanced in technological solutions.

While striving for productivity, it's important to acknowledge that it might conflict with other valuable objectives [105]. For instance, an overemphasis on productivity can lead to a 'local minimum' scenario—a state where workers may become less creative and innovation stagnates. This could ultimately be detrimental to the long-term success of an organization.

Who is benefiting from the tool?

Equally important is the consideration of who benefits from workplace computing systems [100]. Often, these systems are perceived to favor management, leading to their characterization as 'bossware.' This raises critical ethical questions about the balance of power in workplace technologies and the potential for these tools to overlook the needs and voices of the average worker [126, 60].

To address these issues, it is vital to incorporate inclusive design practices that consider the perspectives and needs of all stakeholders [63, 58]. This includes presenting technology in a way that is transparent and accountable to the end users, not just the managers or implementers. Ensuring that worker voices are heard and valued in the deployment of workplace computing systems is fundamental to foster a supportive and innovative working environment.

5.2.2 Do we really need more computing tools at the workplace?

While reflecting on the ethical design principles of workplace software is essential, it is equally important to consider the actual need for additional technological tools [109]. The key is not merely to increase the quantity of tools but to enhance the quality of those we choose to implement. We need good tools, not just more tools.

The scattering of worker attention

The proliferation of technology in the workplace, while beneficial in many ways, has also led to a scattering of people's attention [162]. The constant bombardment of notifications, messages, and digital interruptions fractures concentration and can diminish overall productivity and well-being. This phenomenon calls for a critical reassessment of the types of technologies deployed in the workplace. It is essential to question whether each new tool introduced is truly necessary or if it merely adds to the clutter, complicating tasks and reducing efficiency. Effective workplace technology should simplify workflows, not complicate them, ensuring that employees can focus on tasks that require deeper cognitive engagement.

(Over)Reliance on technology?

The increasing reliance on technology also raises significant concerns about human behavior and capabilities in the workplace. As organizations integrate more technological tools, there is a growing dependency on these systems, often at the expense of human skills and problem-solving abilities. Not only does this change introduce new societal issues, such as the ownership of ideas and determining who should be responsible for problematic outcomes [151, 153, 154], but also creates a dependency that poses significant risks. Should these systems fail, employees and the organization as a whole may find themselves ill-equipped to adapt or revert to manual processes.

This scenario highlights the need for a balanced approach where technology complements human skills rather than replacing them. Employees should be encouraged to develop skills that remain relevant and useful, even in the absence of advanced technological aids. Training programs should focus not only on how to use new technologies but also on nurturing critical thinking, creativity, and problem-solving skills that technology cannot replicate.

5.2.3 The need for more interdisciplinary research bridging computer science and social science

Finally, it's also important to note computing systems is only part of the solution to a better work experience. Traditionally, the computer science communities have primarily focused on technological aspects, whereas social scientists have concentrated on the human side, exploring the implications for human behavior and social interaction [145, 61]. It is essential to bridge these distinct focuses to foster a more integrated understanding of how managerial strategies and computing applications can effectively coexist and enhance the workplace.

The field of Human-Computer Interaction (HCI) and Computer-Supported Cooperative Work (CSCW) have provided valuable works moving forward in this direction, yet much of research in the community relies on small-scale empirical evidence, often collected from controlled environments or specific settings, which challenges the generalizability of the findings. There is a notable gap in

research scale and application that needs addressing to extend these insights to broader, more diverse workplace contexts [49]. There is also more opportunities for HCI and CSCW to better incorporate theories from the social science literature [107, 74].

5.3 Future directions

Finally, I would like to end with discussions on some potential future directions scholars could pursue relevant to this dissertation.

5.3.1 Evaluating workplace computing systems in the wild

First, more work is needed to evaluate computing systems in the wild [56, 210, 152, 237]. Given the application nature of workplace computing systems, researchers should seek to derive more empirical evidence (ideally at scale) how actual users/workers interact with such systems, so as to better understand the systems' values and areas for improvement. While there has been rich literature on workplace computing systems in HCI and CSCW, many of them are based on small-scale user study through interview or survey, and often conducted in lab based setting rather than in-situ, thus the findings may be not easily generalizable to a different setting. By collecting larger-scale evidence in-situ, and comparing trends over different platforms, it would be easier to derive more generalizable conclusions that would more likely to apply to different scenarios [250].

A possible way moving forwards on this direction is through research collaboration with industry partners. Through my PhD, I have seen increasing interest from industry in working together with researchers on this angle [49, 46, 47], and have engaged in a number of collaborations with industries [48, 45, 44, 50, 251]. I strongly encourage future researchers to take advantage of this opportunity by engaging in conversations with practitioners.

5.3.2 Understanding human-AI interaction guidelines in the era of generative AI

Secondly, specifically to AI, I think it is important to delve into best practices/guidelines to integrate AI into workplace computing systems [12]. While many organizations are excited about incorporating generative AI as part of their work flow/workplace experience, the specific use case and the design (e.g. how AI should be embedded in the work process remains largely an open question today). For instance, lots of industrial efforts have been put into creating ChatGPT like chatbot in various scenarios at the workplace, yet chatbot interfaces may not be the best UI for these scenarios thus many of the attempts could potentially fail (e.g. we get some initial qualitative evidence suggesting that sometimes a dashboard may be more useful than a chatbot interface as it would provide stakeholders more agency and control over decision making). Thus, I think it critical

to study how generative AI can effectively work together with human workers to achieve best user experience and performance. For instance, what should be agency level of AI for a specific workplace use case? How much data should AI analyze that people are okay with? What action should AI make for that particular scenario?

As an example, my recent work have initiated a large-scale survey study to understand how individuals from various backgrounds view the trade-offs associated with AI-enhanced teamwork in different collaboration areas, including freelance work, online crowd work, and traditional employment. Drawing from the taxonomy of team processes [168] – (1) Coordinating and Planning; (2) Working on Tasks; and (3) Managing Relationships, we measure and compare each population’s desired levels of autonomy, privacy, and accuracy for using AI in each collaboration domain.

Of course, a survey would just be a preliminary step towards this goal. Future work should further strengthen these initial human-AI interaction guidelines through user study and field experiments in specific grounded scenarios, e.g., by comparing user reactions and other outcome metrics on various conditions AI presents to them.

5.3.3 Workplace Human behavior simulation through generative AI

Finally, I believe there is great potential to leverage generative AI, especially large language model to better understand How people work. to simulate human behavior, e.g. social interactions. AI agents based on large language models have demonstrated the ability to simulate realistic human behavior and interactions [196]. I envision that these techniques hold significant potential for social scientists and policy makers by serving as a testing ground for evaluating theories, strategies and policies, which advances classical agent based modeling/simulation approaches. For instance, within AI simulated environments, researchers can create simulacra that replicate real-world workplace scenarios, integrating factors such as communication patterns, decision-making processes, collaboration dynamics, and organizational structures. AI agents can role-play, mimicking employees, managers of different personalities/backgrounds/work patterns [53, 52, 249], allowing researchers to observe how various policies play out in different simulated contexts. As such, researchers can systematically vary inputs to analyze the outcomes and consequences of different strategies. For instance, exploring the effects of implementing flexible work arrangements, introducing new performance evaluation methods, or altering team compositions [253]. By observing the simulated interactions, researchers can identify potential challenges, opportunities, and unintended consequences that may arise from management policies. I believe there are lots of future work in need for achieving this vision, including building the technical foundations that enable such large-scale simulations, and exploring potential applications through collaborations with domain experts.

In conclusion, this dissertation has placed a significant emphasis on the design and evaluation of workplace computing systems by combining technical approach from computer science, and theoretical foundations from social science. I hope the outlined future directions offer a starting point

for both researchers and practitioners to push the boundaries of this field further, and collectively build a better future of work.

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